TCP/IP: Concepts and IP Routing

Modernized 2026 edition (US English) | Rodolfo Nützmann

Based on the "TCP/IP Concepts" course presented at Technology Week, March 1999

Course agenda

Module 1: General routing concepts

- Heterogeneity and the components of an internetwork
- Bridges and Layer 2 switches versus routers
- Routable and nonroutable protocols (historical view)
- Network-layer versus data-link addressing
- Encapsulation and packet transmission
- Static and dynamic routing
- Distance vector: metrics, slow convergence and its fixes
- Link state: principles, advantages and protocols
- Autonomous systems

Module 2: TCP/IP

- TCP/IP stack, IPv4 datagram, protocol and port numbers
- IPv4 addressing: classes (legacy), CIDR, subnetting and VLSM
- Host behavior: route determination and ARP (local, remote, proxy)
- Complete transfer example across a routed internetwork
- RIP, EIGRP, administrative distance and redistribution
- OSPF in detail; IS-IS; BGP-4
- ACLs, broadcasts, secondary addresses, unnumbered interfaces
- IPv6: header, addressing, ND, routing, transition
- IP multicast (IGMP, PIM), RSVP, MPLS and VRRP

From 1999 to 2026: what changed in this course

Topic	In 1999	In 2026
LAN links	Ethernet 10/100, Token Ring, FDDI	Ethernet 1G to 400G, Wi-Fi 6/7; Token Ring and FDDI extinct
Network protocols	TCP/IP, IPX, AppleTalk, DECnet, SNA	IP only (IPv4 and IPv6); the rest is legacy
IPv4 addressing	Classes A/B/C, contiguous subnets	CIDR/VLSM (RFC 4632); IPv4 exhaustion (IANA 2011, LACNIC 2020); NAT and CGNAT
IPv6	"IPng" in draft (RFC 1883)	RFC 8200 (2017); > 50% of Google access (Mar 2026); Brazil ≈ 59% (APNIC, Oct 2026)
IGP	RIP, IGRP, OSPF	OSPFv2/v3, IS-IS, EIGRP (RFC 7868); RIP only in small networks
EGP	EGP on the way out, BGP-4 arriving	BGP-4 (RFC 4271) universal; 32-bit ASNs (RFC 6793)
Multicast	DVMRP, MOSPF, MBone	PIM-SM/SSM (RFC 7761, RFC 4607), IGMPv3/MLDv2
Gateway redundancy	VRRP just published (RFC 2338)	VRRPv3 (RFC 9568) for IPv4 and IPv6; HSRP
Layer 2 interconnection	Bridges, STP 802.1D	Multilayer switches, RSTP/MSTP, 802.1Q VLANs, EVPN

MODULE 1

General routing concepts

Internetworking, bridges and switches, routers, logical addressing, static and dynamic routing, distance vector, link state and autonomous systems

Module 1 objectives

1. **Internetworking vocabulary**
   Recognize the basic internetworking terms and concepts: LAN and WAN links, interconnection devices, broadcast and collision domains.

2. **Bridges and switches versus routers**
   Compare the operation of bridges/Layer 2 switches and routers, and understand Layer 2 and Layer 3 switching.

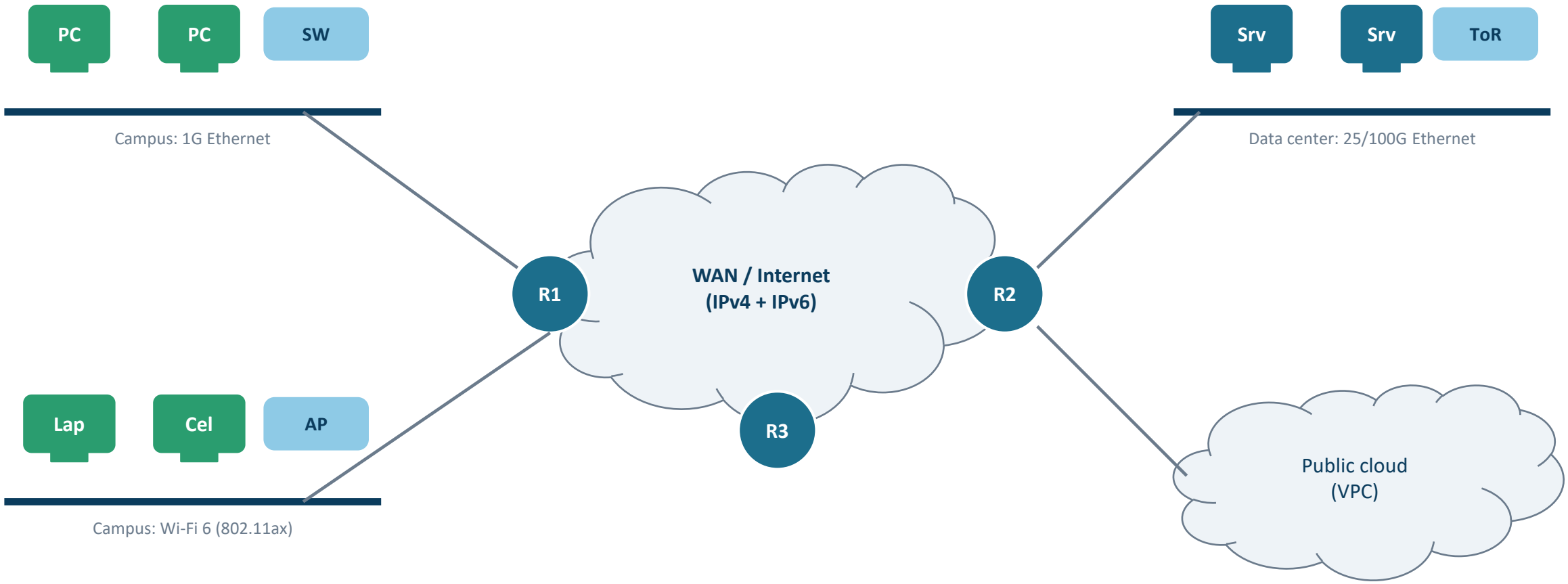
3. **Routable protocols**
   Describe the properties of a routable protocol and put nonroutable protocols in historical context.

4. **Distance vector**
   Explain the principle of operation, the inherent instability (slow convergence) and the techniques that correct it.

5. **Link state**
   Explain SPF (Dijkstra) algorithms, their advantages over distance vector and the protocols that implement them.

6. **Autonomous systems**
   Define autonomous system and distinguish interior from exterior routing protocols.

Heterogeneity: the reason internetworks exist



In 1999 the challenge was making TCP/IP and NetWare coexist on Token Ring and Ethernet; in 2026 the challenge is connecting campus, Wi-Fi, data center and cloud under a single protocol: IP.

Components of an internetwork

LAN links

- IEEE 802.3 Ethernet: 1G, 10G, 25G, 100G, 400G
- IEEE 802.11 Wi-Fi (ax, be)
- One broadcast domain per VLAN
- **Legacy:** Token Ring, FDDI, ARCnet

WAN links

- Leased circuits, dark fiber, Metro Ethernet
- Carrier MPLS (L3VPN, L2VPN)
- Public Internet with tunnels (IPsec, SD-WAN)
- 4G/5G, LEO satellite
- **Legacy:** serial lines, Frame Relay, X.25, ATM

Interconnection devices

- Bridge / Layer 2 switch (forwards by MAC)
- Router (forwards by IP address)
- Layer 3 switch (routing in ASIC)
- **Legacy:** repeaters and hubs (Layer 1)

TCP/IP internetworks are connectionless: every packet is forwarded independently, hop by hop.

Interconnection devices and the OSI model

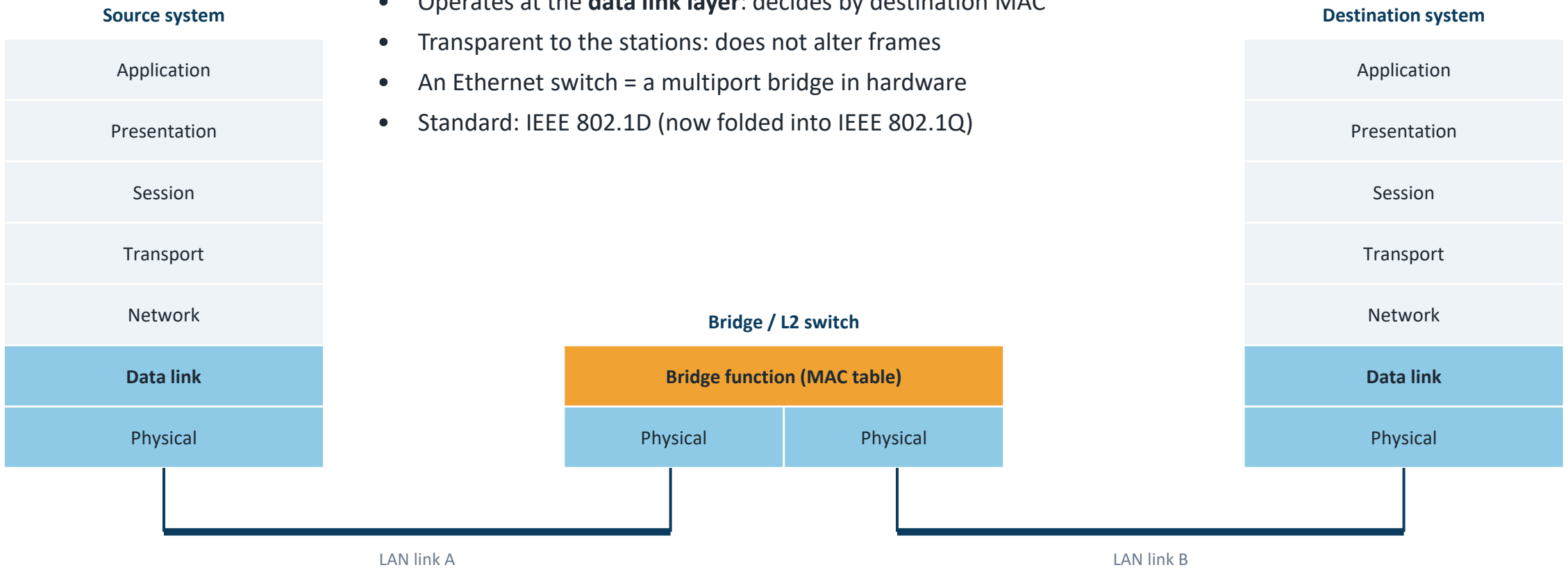
Hub / repeater	Bridge / L2 switch	Router	L3 switch
7 Application	7 Application	7 Application	7 Application
6 Presentation	6 Presentation	6 Presentation	6 Presentation
5 Session	5 Session	5 Session	5 Session
4 Transport	4 Transport	4 Transport	4 Transport
3 Network	3 Network	**3 Network**	**3 Network**
2 Data link	**2 Data link**	2 Data link	2 Data link
1 Physical	1 Physical	1 Physical	1 Physical

LEGACY

The highlighted layer is the one that decides forwarding: destination MAC (Layer 2) or destination IP address (Layer 3). An L3 switch routes between VLANs with the same logic as a router, but in ASIC.

Bridges and Layer 2 switches

- Operates at the **data link layer**: decides by destination MAC
- Transparent to the stations: does not alter frames
- An Ethernet switch = a multiport bridge in hardware
- Standard: IEEE 802.1D (now folded into IEEE 802.1Q)



Frame flow through a bridge or switch

1. **Learn**
   Record the source MAC (SA) and the arrival port in the MAC table; refresh the aging timer.

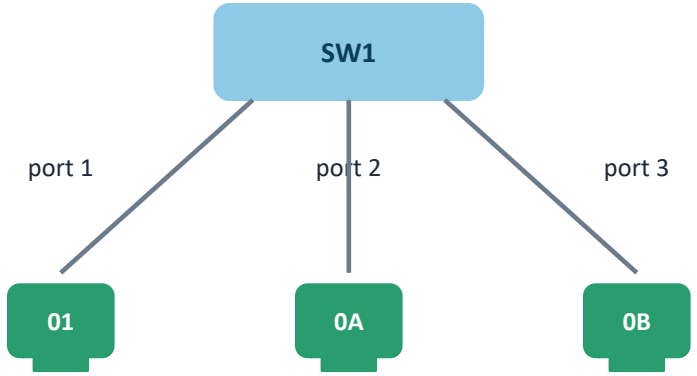
2. **Look up**
   Search for the destination MAC (DA). Found on another port: forward. Found on the same port: filter (discard).

3. **Flood**
   Unknown DA, broadcast or multicast: send the frame out every port except the one it arrived on (flooding).

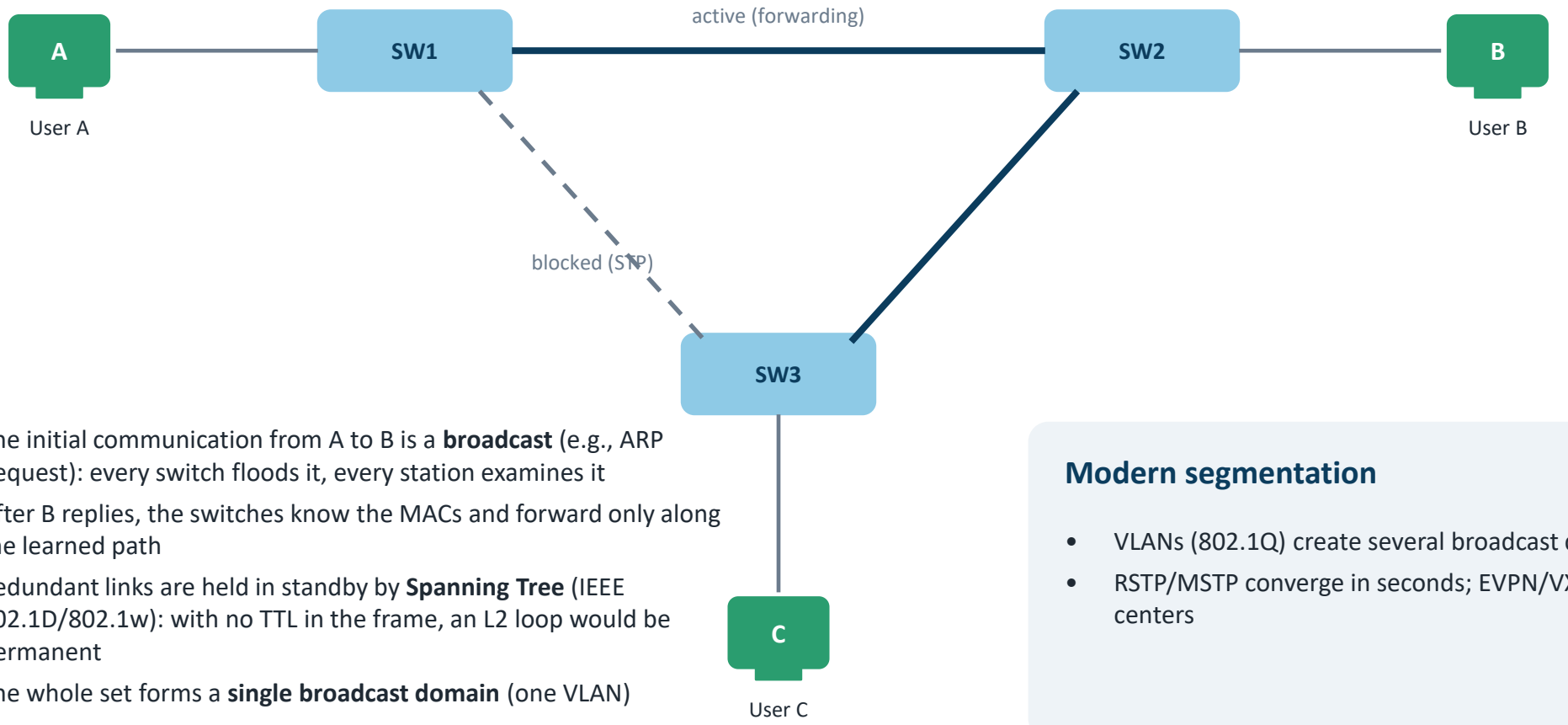
4. **Age out**
   Entries without traffic are removed after the aging time (default 300 s), letting stations move between ports.

MAC	Port	Age (s)
00-00-5E-00-53-01	1	12
00-00-5E-00-53-02	1	40
00-00-5E-00-53-0A	2	3
00-00-5E-00-53-0B	3	275

MAC table (SAT / CAM) of a switch



Switched network: broadcast domain and Spanning Tree



- The initial communication from A to B is a **broadcast** (e.g., ARP Request): every switch floods it, every station examines it
- After B replies, the switches know the MACs and forward only along the learned path
- Redundant links are held in standby by **Spanning Tree** (IEEE 802.1D/802.1w): with no TTL in the frame, an L2 loop would be permanent
- The whole set forms a **single broadcast domain** (one VLAN)

Modern segmentation

- VLANs (802.1Q) create several broadcast domains on the same switch
- RSTP/MSTP converge in seconds; EVPN/VXLAN replaces STP in data centers

Layer 2 switching: advantages and limitations

Advantages

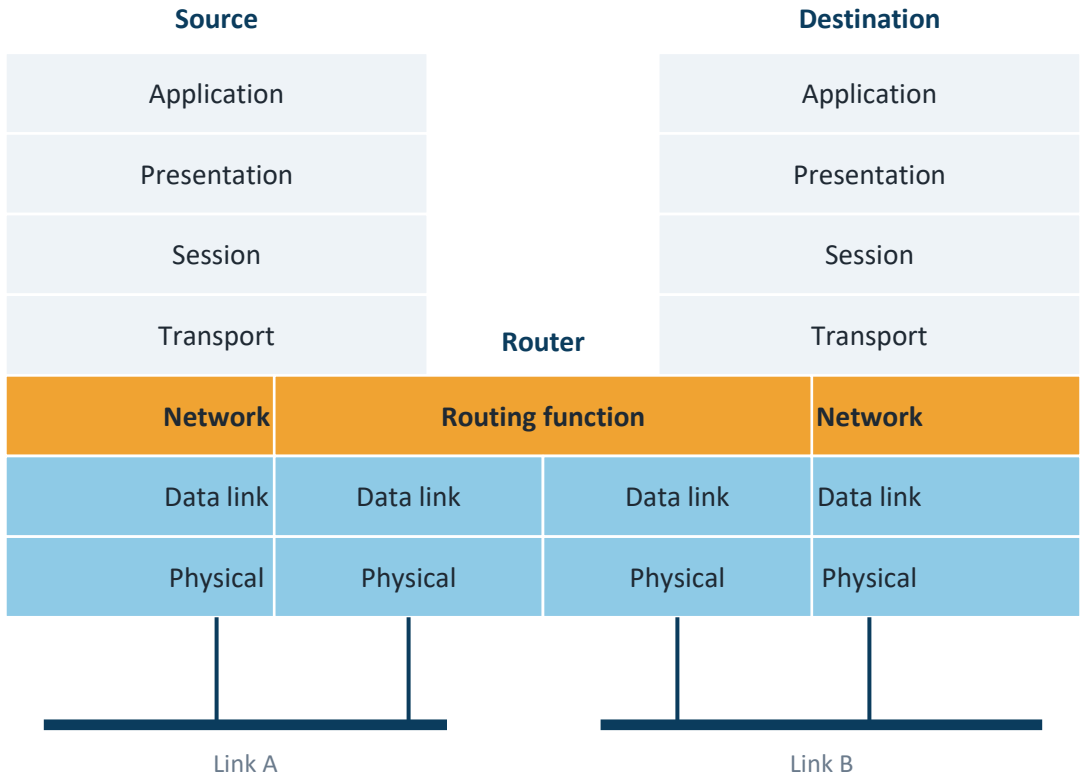
- **Cost:** lower price per port than routers
- **Performance:** does not parse the network-layer header; line-rate forwarding in ASIC
- **Plug and play:** learns addresses by itself; little configuration
- **Protocol independence:** carries any Layer 2 payload (IP, ARP, LLDP, legacy protocols)

Limitations

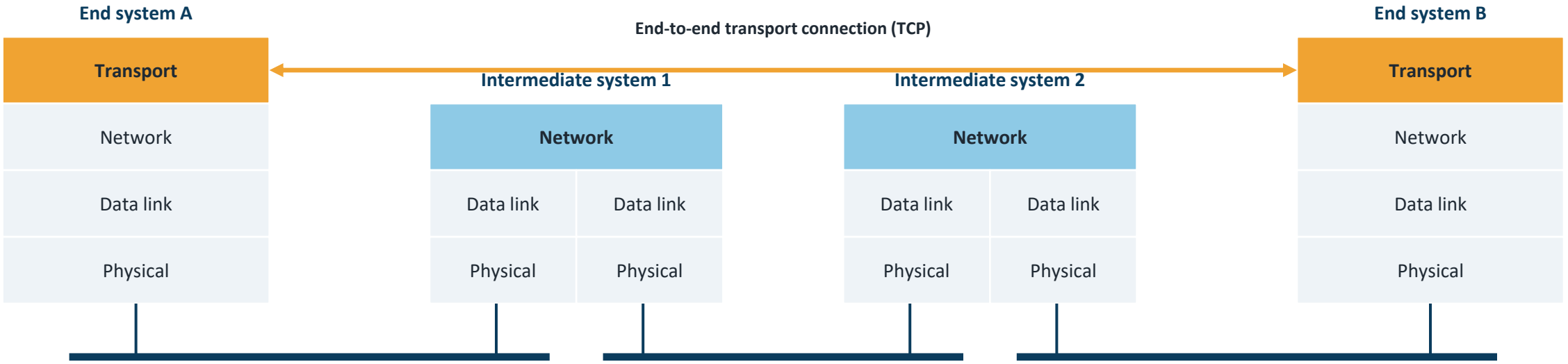
- **Broadcast:** no protection against broadcast storms; the failure domain grows with the network
- **Topology:** uses only a subset of the links (spanning tree); classic STP converges in 30 to 50 s (RSTP: seconds)
- **MTU:** does not fragment; frames larger than the MTU of the egress link are dropped
- **No congestion feedback** to the stations (Ethernet frame flow is not end to end)
- **Scale:** 802.1D recommended a maximum diameter of 7 bridges; today the limit is a design choice (size of the broadcast domain)

Routers

- Operate at the **network layer** of the OSI model (the Internet layer of TCP/IP)
- Forward packets from end station to end station, allowing **multiple paths** between them (ECMP)
- Routing decision based on the **network address** (IP), not the data-link (MAC) address
- Bound broadcast domains: traffic isolation and the natural place for **policy and security**
- Take part in a **distributed algorithm** (routing protocol) to choose the optimal path
- Rewrite the data-link header at every hop and decrement the TTL (IPv4) / Hop Limit (IPv6)

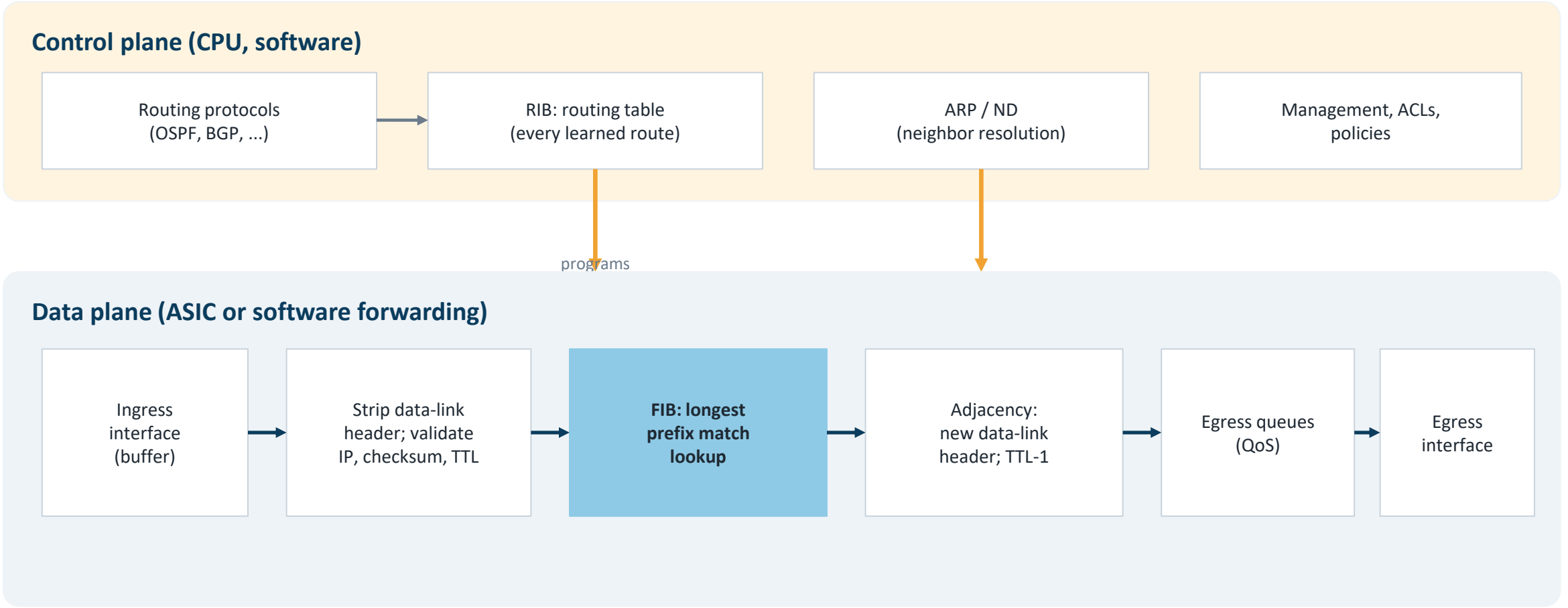


End systems and intermediate systems



- **End systems** (hosts) implement every layer; the transport connection is end to end
- **Intermediate systems** (routers) implement up to the network layer and treat every packet independently
- Reliability, ordering and flow control belong to TCP at the endpoints; the router may drop packets

Data flow inside a router



RIB (Routing Information Base) = what the router knows; FIB (Forwarding Information Base) = what it uses to forward. Packets addressed to the router itself go up to the control plane.

Routers: advantages and limitations

Advantages

- **Broadcast isolation:** every interface (or VLAN) is a separate broadcast domain
- **Path selection:** uses the best physical path available; load-balances across equal-cost paths (ECMP)
- **Flexibility:** any topology, including redundant meshes with no blocked links
- **Practically unlimited scale:** the Internet is a routed network
- **Policy point:** ACLs, QoS, NAT, tunnels

Limitations

- **Protocol dependence:** only forwards what it is configured to route (today: IPv4 and IPv6)
- **Throughput:** in 1999, software routing was slower than bridging; in 2026, L3 switches and ASIC routers forward at line rate
- **Configuration complexity:** addressing, protocols, policies; hosts need an address, mask and gateway (automated by DHCP/SLAAC)
- **Cost:** more complex and more expensive devices per port

LEGACY

Routable and nonroutable protocols

Routable protocols

- Use a layered structure close to the OSI model, with a **network layer** and a hierarchical logical address (network + node)
- Designed for internetworks with several LAN/WAN links joined by routers
- **In use (2026):** IPv4 (RFC 791) and IPv6 (RFC 8200)
- **Legacy:** XNS, Novell IPX, AppleTalk, DECnet Phase IV and DECnet/OSI, Banyan VINES, ISO CLNP, Xerox PUP

Nonroutable protocols

- Do not implement a network layer, or use a flat addressing scheme (no 'network' part)
- Designed for a single link or an extended LAN (with bridges)
- **Legacy:** NetBIOS/NetBEUI, SNA (subarea and APPN), DEC LAT, DEC MOP
- **Still today, by design:** ARP, STP, LLDP and other Layer 2 protocols stay confined to the link

How to 'route' the nonroutable: encapsulation over IP. NetBIOS moved to TCP/UDP (RFC 1001/1002, ports 137 to 139); SNA was carried by DLSw (RFC 2166) and later by Enterprise Extender; in general, tunnels and Layer 2 VPNs (EVPN/VXLAN, L2TPv3) extend a link across a routed network.

Bottom line for 2026: 'routable protocol' means IP. The criterion is still useful to explain why ARP, STP and LLDP do not cross routers.

Addressing: network layer versus data link layer

Data-link address (MAC)

- 48 bits (EUI-48): 24-bit manufacturer OUI + 24-bit serial
- Flat: says nothing about where the interface is in the internetwork
- Scope: valid only within the link (broadcast domain)
- Example (RFC 9542 documentation range): **00-00-5E-00-53-1A**

Network address (IP)

- Logical and hierarchical: **network (prefix) + node (host)**
- Scope: unique across the whole internetwork (the Internet, when public)
- Assigned by the administrator, DHCP (IPv4/IPv6) or SLAAC (IPv6)
- Examples: **172.16.20.10/24** (IPv4), **2001:db8:20::10/64** (IPv6)

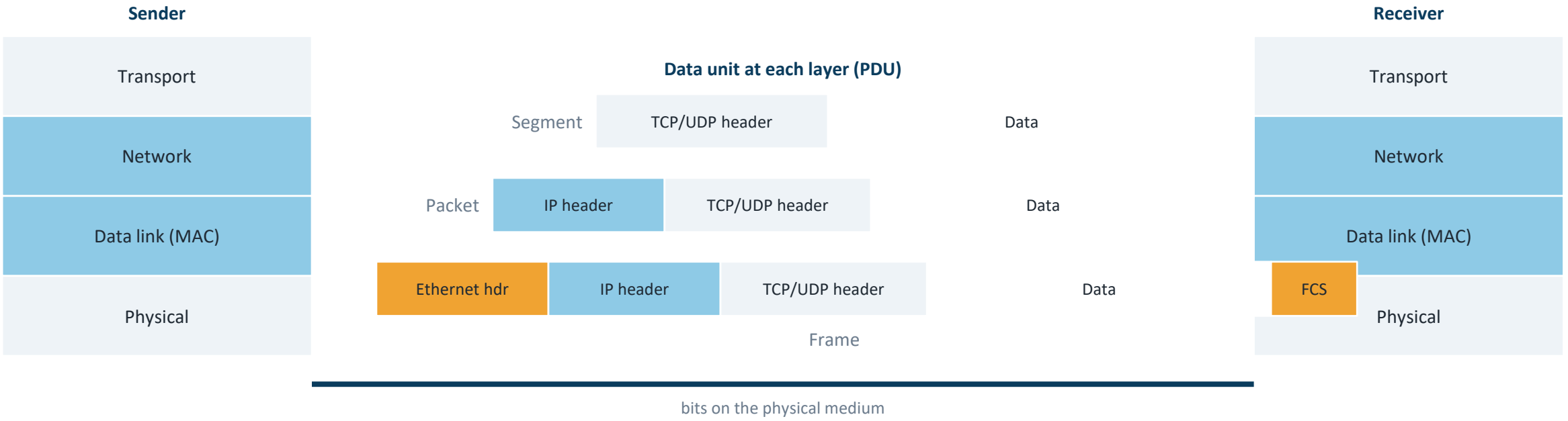
General structure of a network-layer address



IPv4: 172.16.20 | .10 IPv6: 2001:db8:20:: | :10 (the boundary is set by the mask or the prefix length)

In 1999 every routable protocol had its own scheme (IPX: 32-bit network + node = MAC; AppleTalk: 16-bit network + 8-bit node). Today, only IPv4 and IPv6.

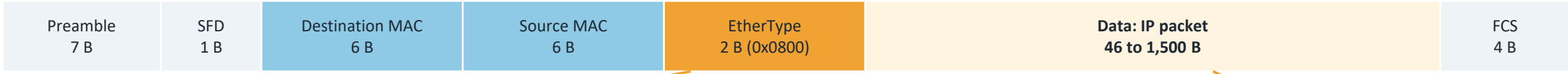
Packet transmission: down the stack



- Each layer adds its header on the way out and removes it on the way in (**encapsulation**)
- The frame's **EtherType** (0x0800 IPv4, 0x86DD IPv6, 0x0806 ARP) identifies the network protocol
- Legacy: the IEEE 802.3 format with LLC/SNAP (LLC-PDU) was common in 1999; today IP over Ethernet uses Ethernet II; Wi-Fi still uses LLC/SNAP

Encapsulation: an IPv4 packet inside an Ethernet frame

Ethernet II frame (IEEE 802.3 with EtherType)



IPv4 packet (datagram), RFC 791

bit 0 bit 31

Version	IHL	DSCP	ECN	Total length
Identification			Flags	Fragment offset
Time to live (TTL)		Protocol		Header checksum
Source IP address				
Destination IP address				
Options (rare)				Padding
Transport-layer data (TCP/UDP segment), up to 65,515 bytes				

Minimum IPv4 header: 20 bytes (IHL = 5). Ethernet MTU of 1,500 bytes: larger packets are fragmented (IPv4) or dropped with ICMP 'Packet Too Big' (IPv6).

Routing algorithms: how routers learn about networks

Manual configuration: static routes

- The administrator enters prefix, mask and next hop
- Default route (0.0.0.0/0 or ::/0) as 'everything else'
- Predictable, no control traffic, no CPU
- Does not react to failures (unless tracked with IP SLA/BFD)

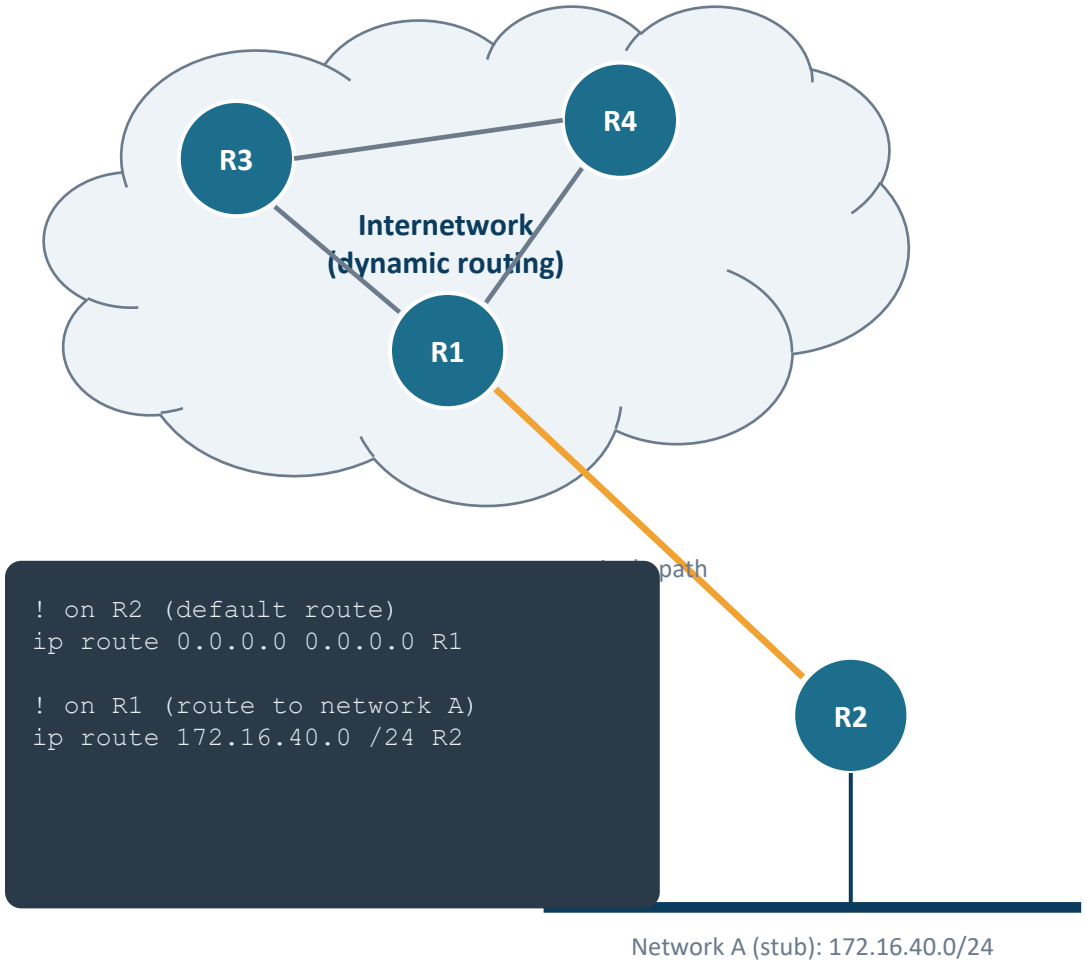
Automatic distribution: dynamic routing

- Routers exchange information and compute paths
- **Distance vector:** RIP, EIGRP (advanced distance vector)
- **Link state:** OSPF, IS-IS
- **Path vector:** BGP (between autonomous systems)
- Reacts to failures and topology changes (convergence)

Rule of thumb: static at the edges (stubs and default), dynamic in the core. The final routing table combines both sources, arbitrated by administrative distance (Module 2).

Static routing

- All routing information is **precomputed** and supplied by manual configuration
- Must be recomputed and reapplied on **every topology change**
- Unsuitable for large, dynamic internetworks
- Ideal for **stub** networks (a single path) and for the **default route**
- No protocol overhead; fully predictable behavior



Distributed adaptive (dynamic) routing

Distance vector

- 'Routing by rumor': each router advertises to its neighbors the distance (metric) to each destination
- Bellman-Ford algorithm
- Simple; converges more slowly; subject to transient loops
- RIP (RFC 2453), EIGRP (RFC 7868, 'advanced')

Link state

- Each router floods a description of its links (LSA); everyone builds the same map
- Dijkstra's SPF algorithm
- Converges fast; scales with areas; more CPU and memory
- OSPF (RFC 2328 / 5340), IS-IS (RFC 1195)

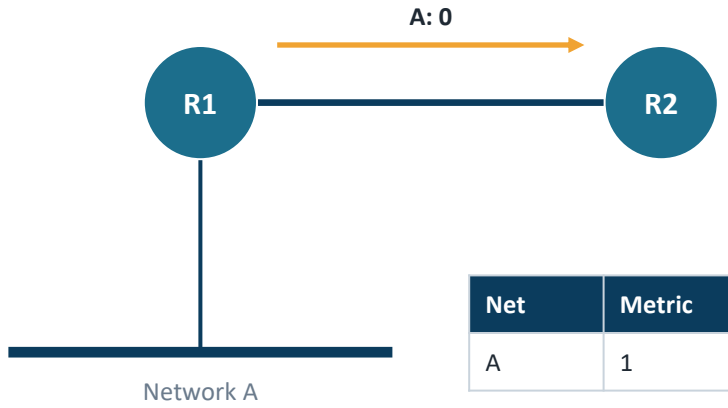
Path vector

- Advertises the full path (list of ASs) to each destination; avoids loops by inspecting the path
- Policy based, not metric based only
- Between autonomous systems (and inside large data centers)
- BGP-4 (RFC 4271)

Routers sense the local environment, exchange information with each other and jointly run a distributed algorithm to determine the optimal path.

Distance-vector routing

- Each router keeps the **distance** (metric) from itself to every possible destination and the **vector** (outgoing neighbor)
- Learns the topology by exchanging **complete tables** with direct neighbors at periodic intervals
- On receiving a neighbor's update, it updates the table when:
  - the destination **was unknown**
  - the advertised route is **shorter** than the known one
  - the current route **already goes through that neighbor** and the advertised distance changed (accepted even if worse)
- Based on the **Bellman-Ford** algorithm; no view of the full map ('routing by rumor')

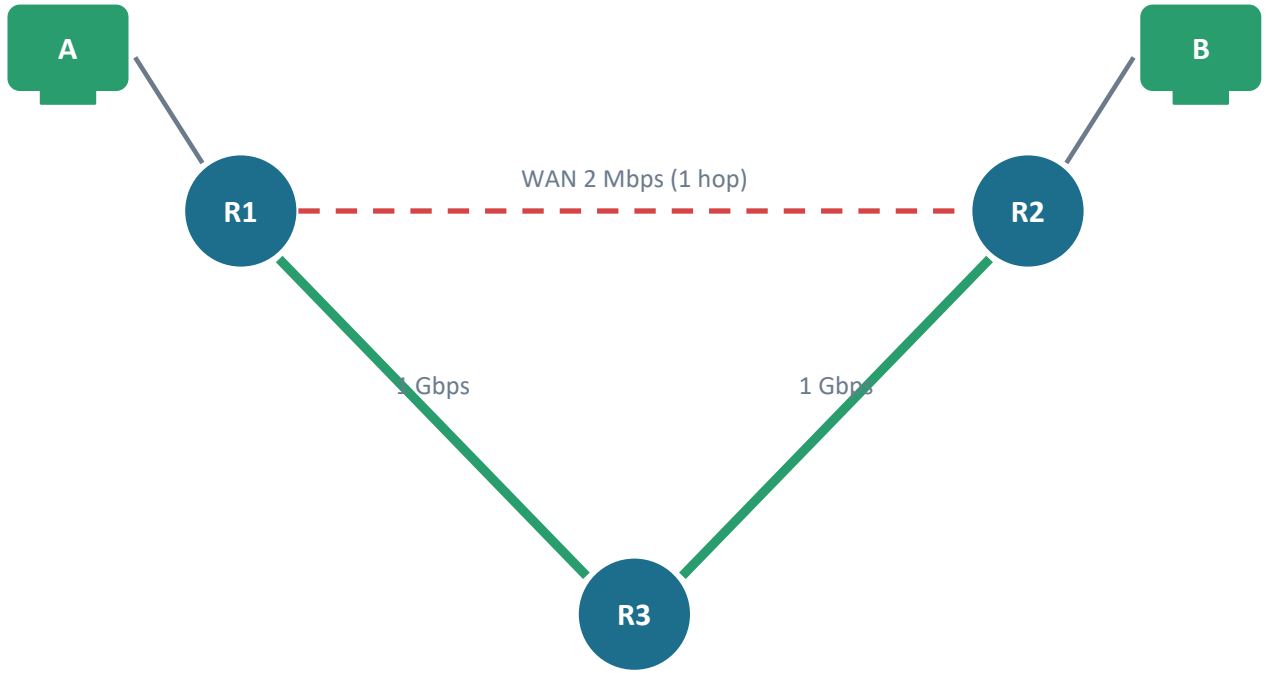


Net	Metric	Via
A	1	R1

R2 learns 'A at 1 hop via R1' from R1's periodic advertisement

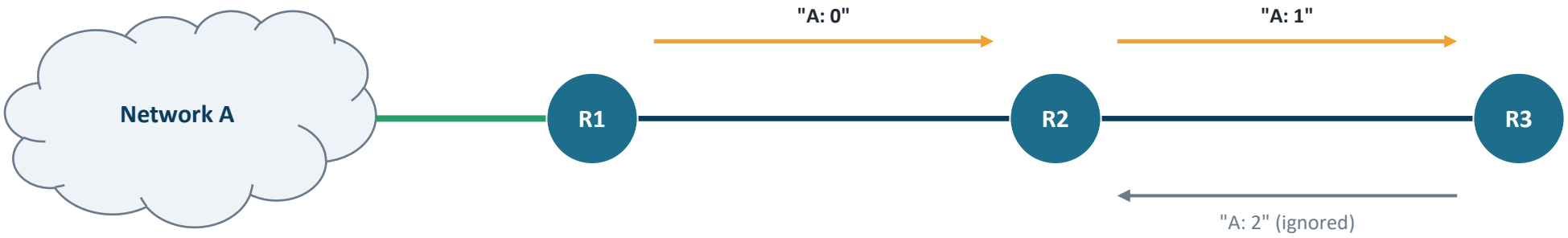
Routing metrics

- How the 'distance' between source and destination is measured
- **Hop count** (RIP), **bandwidth and delay** (EIGRP), administrative **cost** (OSPF, IS-IS), reliability, load
- The metric defines what 'best path' means to the protocol
- Hops ignore capacity: 1 hop at 2 Mbps may be worse than 2 hops at 1 Gbps



Path A → B	Hops (RIP)	Minimum bandwidth	Choice
R1 → R2	1	2 Mbps	RIP
R1 → R3 → R2	2	1 Gbps	EIGRP / OSPF

The slow convergence problem (count to infinity)



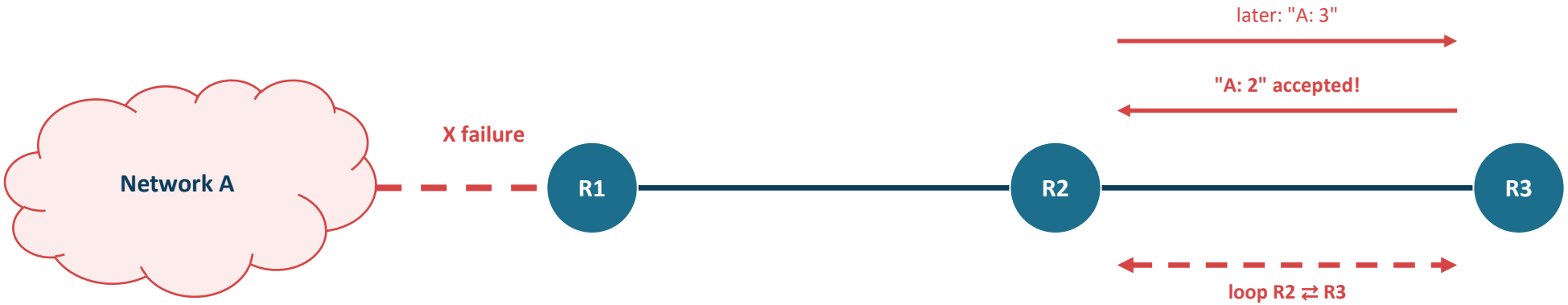
Net	Metr.	Via
A	0	direct

Net	Metr.	Via
A	1	R1

Net	Metr.	Via
A	2	R2

- Three routers (R1 to R3), each with a route to network A
- Periodic advertisements in every direction: R1 → R2 'A: 0', R2 → R3 'A: 1', and also R3 → R2 'A: 2'
- While everything works, R2 ignores R3's advertisement because it already has a better route (1 via R1)

Slow convergence: the failure creates a loop



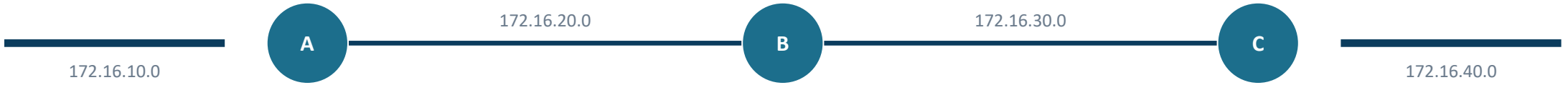
Net	Metr.	Via
A	16 (∞)	-

Net	Metr.	Via
A	3	R3

Net	Metr.	Via
A	2	R2

- The link to A fails; R1 marks A as **unreachable** (RIP: metric 16 = infinity)
- R2 loses the route via R1, but R3 still advertises 'A: 2' (via R2!); R2 accepts: 'A: 3 via R3' → **loop**
- Every cycle the metric climbs by 1 (3, 4, 5...) up to infinity: **count to infinity**; with advertisements every 30 s, convergence takes minutes

Fix 1: split horizon

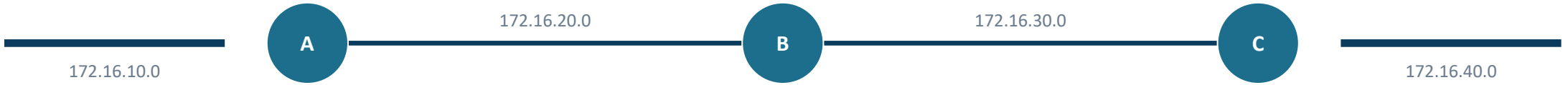


Advertisements from A	Advertisements from B	Advertisements from C
→ to the right: 10.0, 20.0	→ to the right: 10.0, 20.0, 30.0	→ to the right: (no neighbor)
← to the left: (no neighbor)	← to the left: 30.0, 40.0	← to the left: 30.0, 40.0

Rule: a route learned on an interface is never advertised back out that same interface.

- B learned 10.0 and 20.0 from A: does not advertise them back to A; learned 40.0 from C: does not advertise it to C
- Eliminates the loop between two direct neighbors; does not solve loops in rings of three or more routers
- Used by RIP and EIGRP; enabled by default in most implementations

Fix 2: split horizon with poison reverse



Advertisements from A

→ right: 10.0, 20.0, 30.0 (∞), 40.0 (∞)

← left: (no neighbor)

Advertisements from B

→ right: 10.0, 20.0, 30.0, 40.0 (∞)

← left: 10.0 (∞), 20.0 (∞), 30.0, 40.0

Advertisements from C

→ right: (no neighbor)

← left: 10.0 (∞), 20.0 (∞), 30.0, 40.0

Rule: the route learned on an interface is advertised back out of it with an infinite metric (∞ = 16 in RIP).

- Makes it explicit to the neighbor that it must **not** use this router for that destination
- Breaks loops faster than simple split horizon, at the cost of larger updates
- Default in RIP on many implementations; EIGRP uses simple split horizon and route poisoning when withdrawing a destination

Fix 3: hold-down timer

1. **Network goes down**
   The router receives (or generates) the 'destination unreachable' advertisement and starts the hold-down timer for that destination (Cisco RIP: 180 s).

2. **Waiting period**
   Ignores any advertisement claiming the destination is back with an equal or worse metric; keeps forwarding on the old route (marked possibly down).

3. **Exceptions**
   Immediately accepts an advertisement with a better metric or one coming from the neighbor that originally reported the route.

4. **Expiration**
   When the timer expires, advertisements are accepted normally again. The news of the failure has reached every router.

Goal

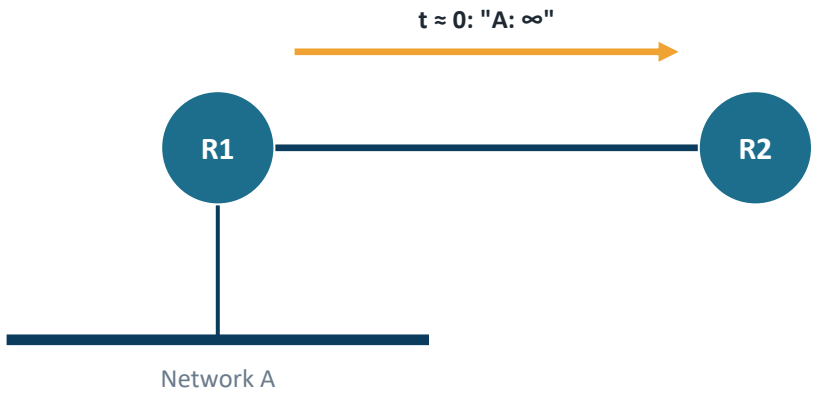
Wait long enough to make sure all routers are notified of the down route and not mistakenly accept an out-of-date message (still circulating) that claims the route exists.

Cost

Delays the convergence of routes that legitimately came back. Modern protocols (EIGRP/DUAL, OSPF) do without long timers.

Fix 4: triggered updates

- The router sends the update **immediately** on detecting a change in the routing table, without waiting for the period
- The triggered update contains only the **routes that changed**
- Speeds up convergence and shrinks the window in which stale information circulates
- Typical RIP combination: split horizon + poison reverse + hold-down + triggered updates
- EIGRP: **only** triggered, reliable updates (RTP); no periodic table advertisements



Without a triggered update, R2 would only learn of the failure in the next periodic advertisement (up to 30 s later).

Distance vector, example (1 of 4): initial state



Table of A

Net	Metric	Via
1	1	DC
2	1	DC

Table of B

Net	Metric	Via
2	1	DC
3	1	DC

Table of C

Net	Metric	Via
3	1	DC
4	1	DC

Table of D

Net	Metric	Via
4	1	DC
5	1	DC

Before the exchange of information, each router knows only its directly connected networks (DC).

Distance vector, example (2 of 4): first exchange



Table of A

Net	Metric	Via
1	1	DC
2	1	DC
3	**2**	**B**

Table of B

Net	Metric	Via
2	1	DC
3	1	DC
1	**2**	**A**
4	**2**	**C**

Table of C

Net	Metric	Via
3	1	DC
4	1	DC
2	**2**	**B**
5	**2**	**D**

Table of D

Net	Metric	Via
4	1	DC
5	1	DC
3	**2**	**C**

The first exchange of advertisements, with simple split horizon, adds to the tables the networks one neighbor away (highlighted).

Distance vector, example (3 of 4): second exchange



Table of A

Net	Metric	Via
1	1	DC
2	1	DC
3	2	B
4	**3**	**B**

Table of B

Net	Metric	Via
2	1	DC
3	1	DC
1	2	A
4	2	C
5	**3**	**C**

Table of C

Net	Metric	Via
3	1	DC
4	1	DC
2	2	B
5	2	D
1	**3**	**B**

Table of D

Net	Metric	Via
4	1	DC
5	1	DC
3	2	C
2	**3**	**C**

The second periodic update adds the networks two neighbors away. A and D still do not know the whole topology.

Distance vector, example (4 of 4): convergence



Table of A

Net	Metric	Via
1	1	DC
2	1	DC
3	2	B
4	3	B
5	**4**	**B**

Table of B

Net	Metric	Via
2	1	DC
3	1	DC
1	2	A
4	2	C
5	3	C

Table of C

Net	Metric	Via
3	1	DC
4	1	DC
2	2	B
5	2	D
1	3	B

Table of D

Net	Metric	Via
4	1	DC
5	1	DC
3	2	C
2	3	C
1	**4**	**C**

It took three iterations (three update periods) for this network to converge. Diameter of N hops ≈ N-1 cycles.

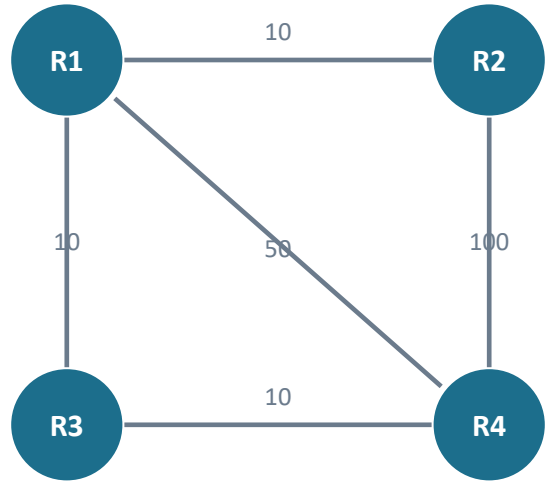
Link-state routing: principles (1 of 2)

1. **Meet the neighbors**
   Each router discovers the adjacent routers and learns their identifiers (Hello protocol).

2. **Describe the links**
   Builds a link-state advertisement (LSA) with the list of neighbors and the cost of each link.

3. **Flood**
   Transmits the LSA to all other routers (flooding); each one stores the most recent LSA from each originator in the link-state database (LSDB).

4. **Compute**
   With the complete topology map, each router runs SPF (Dijkstra) and computes the route to each destination with itself as the root.



LSA of R1: {R2: 10, R3: 10, R4: 50}

LSA of R4: {R1: 50, R2: 100, R3: 10}

Every router has the same LSDB and computes its own SPF tree

Link-state routing: principles (2 of 2)

What the link-state router does NOT do

- Does not send updates with a list of destinations and distances (the whole routing table)
- Does not depend on 'rumors': computes everything from primary data

What it does

- Actively tests the state of every neighbor (Hello)
- Propagates the message describing the state of its links (LSA): periodically (OSPF: 30 min) and immediately on every change

1 **On receiving an LSA**
Updates the internetwork map, marking the links as up or down (and stores the most recent version, by sequence number).

2 **When a link's state changes**
Recompiles the routes by running Dijkstra's algorithm (SPF) on the database; the resulting routing table is installed in the RIB/FIB.

Link state: advantages over distance vector

Independent computation

Each router computes the routes by itself, from the same original state data (identical LSDB). No accumulation of errors from neighbor to neighbor.

Fast convergence

A router recognizes a new LSA and forwards it before recomputing routes; the news spreads at flooding speed, not at the pace of update periods.

Better scaling

Each LSA carries only one router's direct connections. Areas (OSPF) and levels (IS-IS) limit the scope of flooding and of SPF.

Criterion	Distance vector (RIP)	Link state (OSPF, IS-IS)
What is advertised	Routing table (destination + metric) to neighbors	State of its own links (LSA) to everyone in the area
View of the network	Only the neighbors' 'rumor'	Complete map of the area
Convergence	Slow (30 s cycles, hold-down)	Fast (flooding + SPF; subsecond with BFD)

UPDATED

Link-state protocols in use

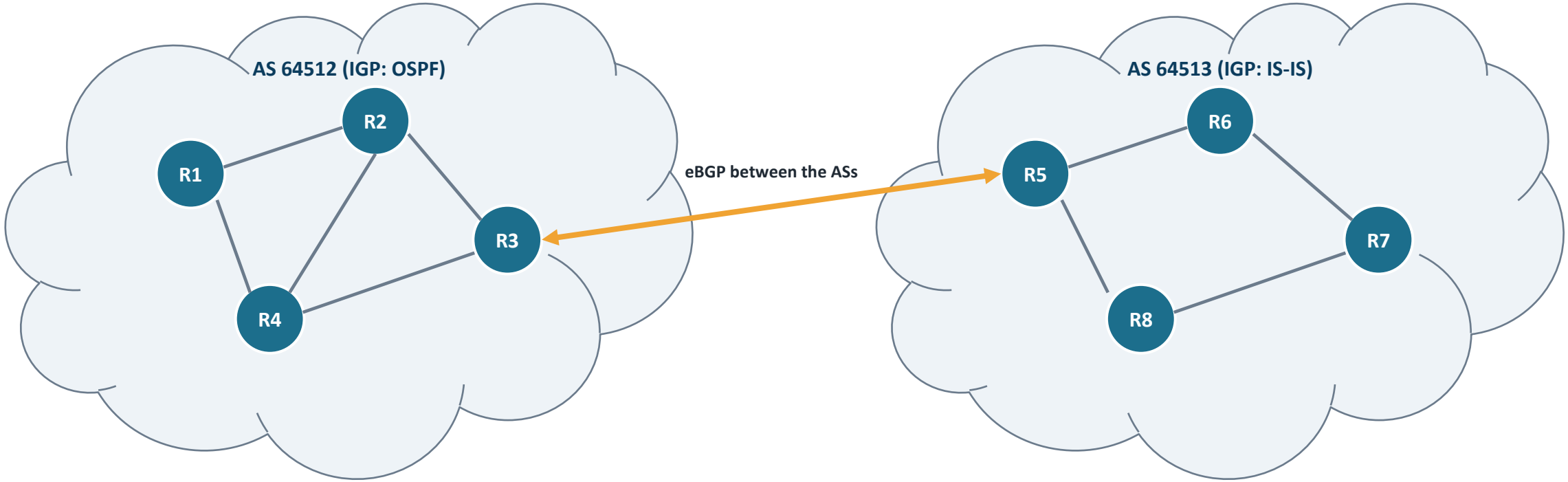
Protocol	Origin and standard	Status in 2026
OSPF	IETF; OSPFv2 RFC 2328 (1998, successor of RFC 1247 from 1991); OSPFv3 RFC 5340 (2008) for IPv6	Most used IGP in enterprise and campus networks; detailed in Module 2
IS-IS	ISO 10589 (designed at Digital for DECnet Phase V); integrated IS-IS for IP: RFC 1195; IPv6: RFC 5308	De facto standard in large providers and data centers; base of Segment Routing (SR-MPLS, SRv6)
NLSP	Novell Link State Protocol, for IPX routing	Extinct with NetWare/IPX (legacy)

OSPF and IS-IS implement the same algorithm (Dijkstra) with different encapsulations: OSPF runs over IP (protocol 89); IS-IS runs directly over the data link layer (it does not need IP to form adjacencies), which makes it attractive in provider cores and fabrics.

Also link state, outside IP routing

- TRILL and SPB (IEEE 802.1aq) used IS-IS for Layer 2 forwarding; today the dominant approach in data centers is BGP + EVPN/VXLAN
- SDN controllers and Wi-Fi fabric protocols reuse the idea of a synchronized topology database

Autonomous system (AS)



- **Routers under a common administration**, with a single routing policy toward the outside (RFC 1930)
- **Inside** the AS: interior protocol (IGP): OSPF, IS-IS, EIGRP, RIP; **between** ASs: exterior protocol (EGP): BGP-4
- AS number (ASN): 16 bits until 2007; **32 bits** since RFC 4893 / RFC 6793; private: 64512 to 65534 and 4,200,000,000 to 4,294,967,294 (RFC 6996)
- Assigned by the RIRs (ARIN, LACNIC...) and, in Brazil, by Registro.br / NIC.br

Module 1 summary

Internetworking

- L2 switch decides by MAC; router decides by IP; L3 switch does both in ASIC
- Routers bound broadcast domains and allow multiple paths
- Today only IPv4 and IPv6 are 'routable'; ARP, STP and LLDP stay on the link

Addressing and encapsulation

- MAC: flat, link scope; IP: hierarchical (prefix + host), global scope
- Each layer adds its header; EtherType identifies IPv4/IPv6/ARP
- Minimum IPv4 header of 20 bytes; Ethernet MTU of 1,500 bytes

Distance vector

- Advertises the table to neighbors; Bellman-Ford; converges in N-1 cycles
- Count to infinity mitigated by split horizon, poison reverse, hold-down and triggered updates
- RIP (legacy/small networks), EIGRP (advanced)

Link state and AS

- Floods LSAs; identical LSDB; Dijkstra; converges fast and scales with areas
- OSPF and IS-IS in use; NLSP extinct
- AS = common administration; IGP inside, BGP-4 between ASs; 32-bit ASN

MODULE 2

TCP/IP

Protocol stack, IPv4 addressing and subnetting, host behavior (route determination and ARP), routing across an internetwork, routing protocols (RIP, EIGRP, OSPF, BGP), access control, IPv6, multicast, MPLS and VRRP

Module 2 objectives (1 of 2)

1. **Characteristics of TCP/IP**
   Protocol stack, IPv4 datagram, protocol and port numbers, addressing, subnetting and VLSM, host-to-host communication, ARP and Proxy ARP.

2. **The IP routing process**
   Describe in detail, step by step, what happens to a packet between the source station and the destination station in a routed internetwork.

3. **RIP and EIGRP**
   Recall the fundamentals of RIP (v1, v2, RIPng) and of EIGRP (composite metric, DUAL, multipath) and the role of administrative distance.

4. **Redistribution and OSPF**
   Discuss route redistribution between routing processes and recall the principles of OSPF (areas, router types, Hello, DR, LSAs).

Module 2 objectives (2 of 2)

1. **BGP-4**
   Distinguish interior from exterior protocols and describe the fundamentals of BGP-4 (path vector, eBGP/iBGP, attributes).

2. **Access control**
   Recall how access control lists (ACLs) on routers filter IP traffic, and how they evolved into stateful firewalls.

3. **Operational topics**
   Broadcast handling (DHCP relay), secondary addresses, unnumbered interfaces, static and default routes.

4. **IPv6 and complementary technologies**
   Understand IPv6 (header, addressing, ND/SLAAC, routing, transition), IP multicast (IGMP/MLD, PIM), RSVP, MPLS and VRRP.

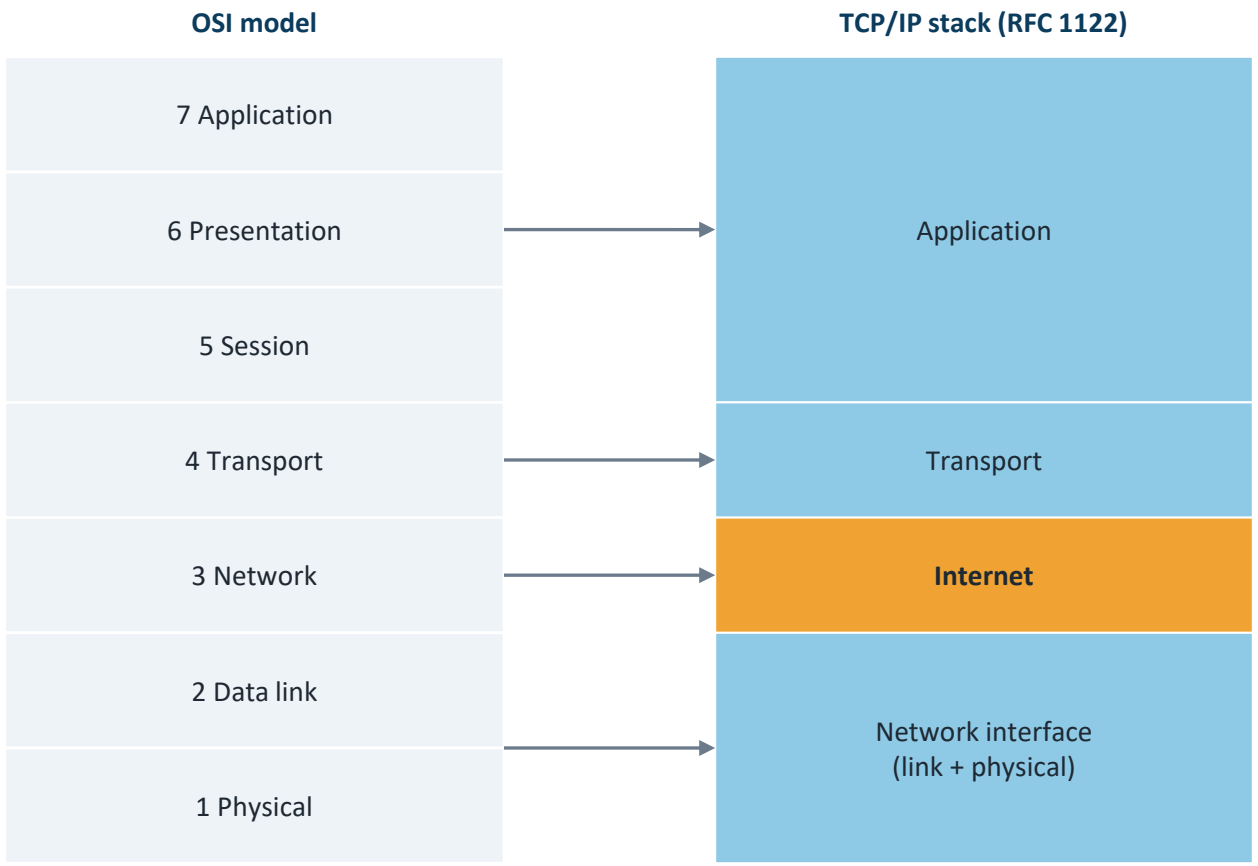
TCP/IP: origin and governance

- The most widely used protocol suite in the world: the foundation of the Internet and of enterprise networks
- Developed in a project funded by **ARPA/DARPA** (US); Cerf and Kahn, 1974; IP and TCP specified in 1981 (RFC 791; RFC 793, since replaced); ARPANET switches over on January 1, 1983
- The **RFCs** (Requests for Comments) define the processing of the protocols and are public; standardized by the **IETF**; numbers administered by **IANA**
- Includes not only Layers 3 and 4 but also applications (DNS, HTTP, SMTP, SSH, SNMP) and auxiliary protocols (ICMP, ARP, DHCP)
- TCP consolidated in 2022 in **RFC 9293**; IPv4 is still RFC 791; IPv6 is **RFC 8200** (2017)

Timeline

- **1969:** first ARPANET nodes
- **1974:** Cerf and Kahn, 'A Protocol for Packet Network Intercommunication'
- **1981:** RFC 791 (IP) and RFC 793 (TCP, since replaced)
- **1983:** ARPANET 'flag day';
  DNS (RFC 882/883, since replaced)
- **1985:** subnetting (RFC 950); 1986: IETF
- **1993:** CIDR (RFC 1519, since replaced);
  1998: IPv6 (RFC 2460, since replaced)
- **2011:** IANA runs out of free IPv4
- **2017:** IPv6 Internet Standard (RFC 8200)
- **2026:** > 50% of Google access over IPv6

The TCP/IP protocol stack and the OSI model

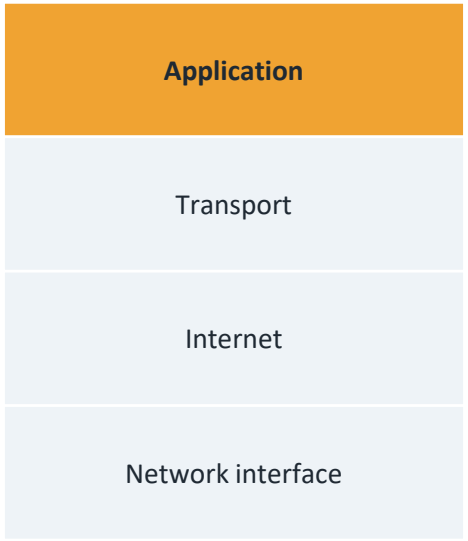


The Internet layer (IP) is the 'narrow waist of the hourglass': everything above runs over IP, and IP runs over any link.

Examples at each layer

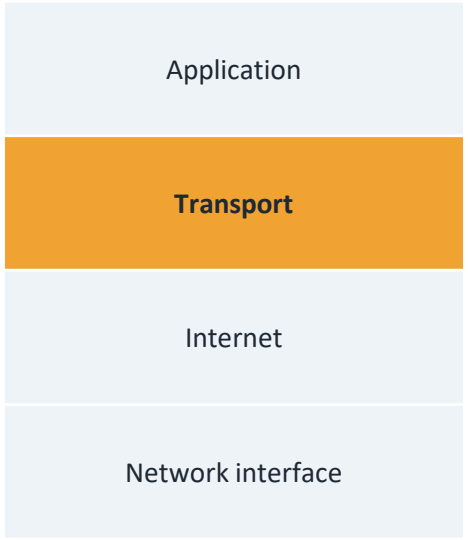
- **Application:** HTTP/HTTPS, DNS, SMTP, IMAP, SSH, SNMP, NTP, DHCP, TLS (security between application and transport)
- **Transport:** TCP, UDP, QUIC (over UDP), SCTP
- **Internet:** IPv4, IPv6, ICMP/ICMPv6, IGMP/MLD, IPsec; routing protocols (OSPF, BGP) run over IP or TCP
- **Network interface:** Ethernet, Wi-Fi, PPP/PPPoE, MPLS (Layer '2.5'), VXLAN (encapsulation over UDP)
- **Legacy:** Token Ring, FDDI, Frame Relay, ATM

Application layer



Function	Current protocols (port)	Legacy / replaced
Web and APIs	HTTPS: HTTP/1.1 and HTTP/2 over TLS/TCP 443; HTTP/3 over QUIC/UDP 443	Cleartext HTTP (80)
Names	DNS (UDP/TCP 53); DNS over TLS (853) and over HTTPS; DNSSEC	hosts file, static tables
Remote access	SSH (TCP 22); RDP (3389)	Telnet (23), rlogin (513)
Files	SFTP/SCP (22), HTTPS, SMB (445), NFSv4 (2049)	FTP (20/21), TFTP (69) still in network boot
E-mail	SMTP submission (587, TLS), IMAPS (993), server-to-server SMTP (25)	POP3 (110), SMTP without TLS
Management and time	SNMPv3 (UDP 161/162), NETCONF/RESTCONF, gNMI, syslog (514), NTP (123)	SNMPv1/v2c (cleartext communities)
Host configuration	DHCP (UDP 67/68), DHCPv6 (546/547), SLAAC (IPv6)	BOOTP, RARP

Transport layer



TCP: Transmission Control Protocol

- Connection oriented (three-way handshake), reliable and ordered byte stream
- Flow control (window) and congestion control
- Header of 20 bytes or more; RFC 9293 (2022)

UDP: User Datagram Protocol

- Connectionless, no delivery or ordering guarantee
- 8-byte header; minimal latency
- DNS, DHCP, SNMP, NTP, voice and video, QUIC; RFC 768

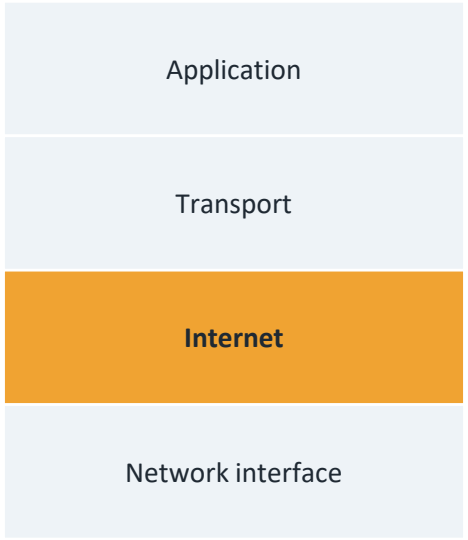
QUIC: the transport of HTTP/3

- Runs over UDP (port 443) with integrated TLS 1.3
- Multiple streams without head-of-line blocking; connection migration
- RFC 9000 (2021); a significant share of web traffic (HTTP/3)

SCTP and others

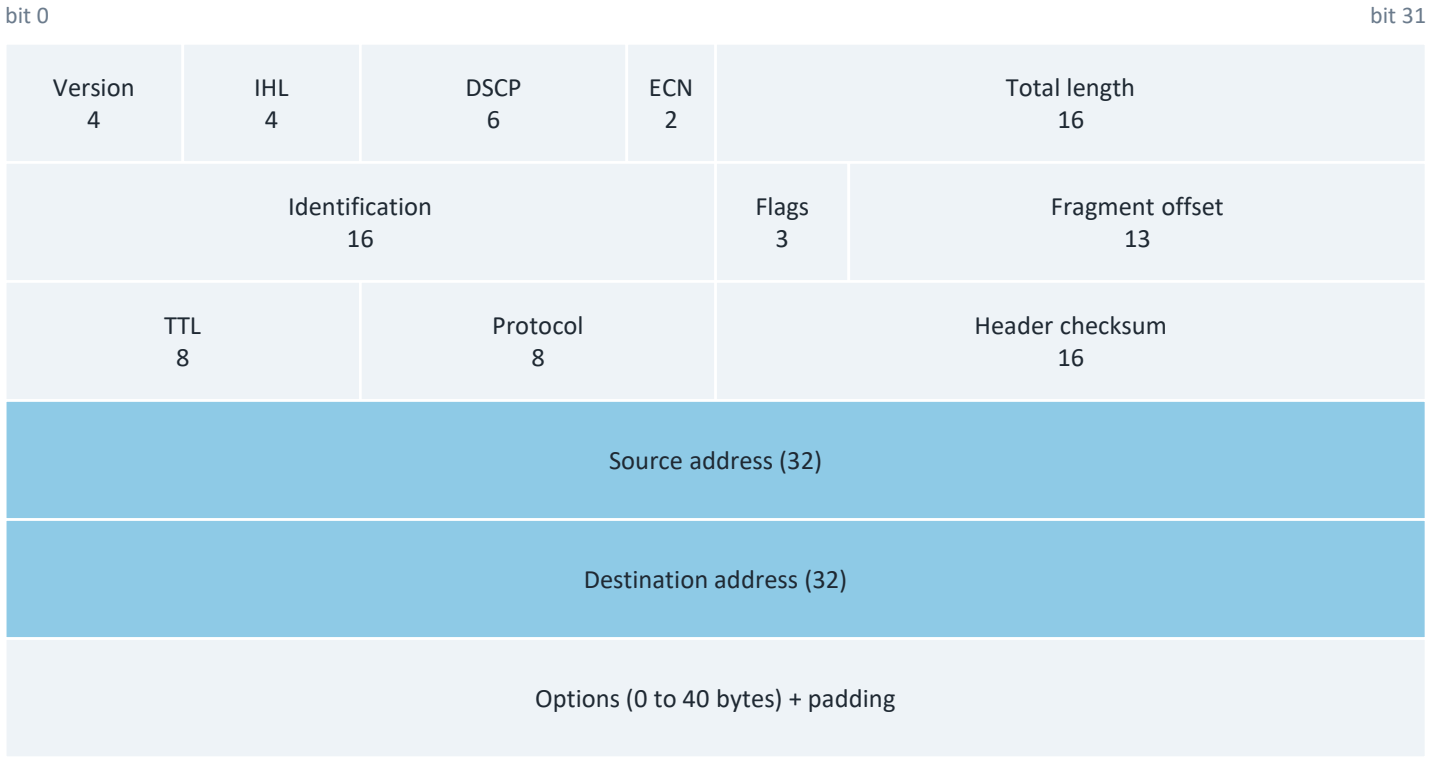
- SCTP (RFC 9260): messages, multihoming; telephony signaling (SIGTRAN), 4G/5G networks
- DCCP (RFC 4340): rarely used
- 16-bit ports identify the application in TCP, UDP, SCTP and QUIC

Internet layer



Protocol	Function	Reference
IPv4	32-bit addressing, forwarding, fragmentation; connectionless, best effort	RFC 791 (1981)
IPv6	128-bit addressing, simplified header, extension headers, no fragmentation in routers	RFC 8200 (2017)
ICMP / ICMPv6	Errors (destination unreachable, TTL expired), diagnostics (echo), MTU (Packet Too Big); ICMPv6 includes ND and MLD	RFC 792 / RFC 4443
ARP	IPv4 → MAC resolution on broadcast links (encapsulated directly in Ethernet, 0x0806)	RFC 826 (1982)
ND (IPv6)	Neighbor and router discovery, address resolution, SLAAC; replaces ARP and part of ICMP	RFC 4861, RFC 4862
IGMP / MLD	Hosts tell routers which multicast groups they want to receive	RFC 9776 / RFC 9777
IPsec (AH, ESP)	Authentication, integrity and confidentiality at the network layer; VPNs	RFC 4301 to 4303
RARP	MAC → IP resolution for diskless stations; replaced by BOOTP and DHCP (legacy)	RFC 903

The IPv4 datagram in detail



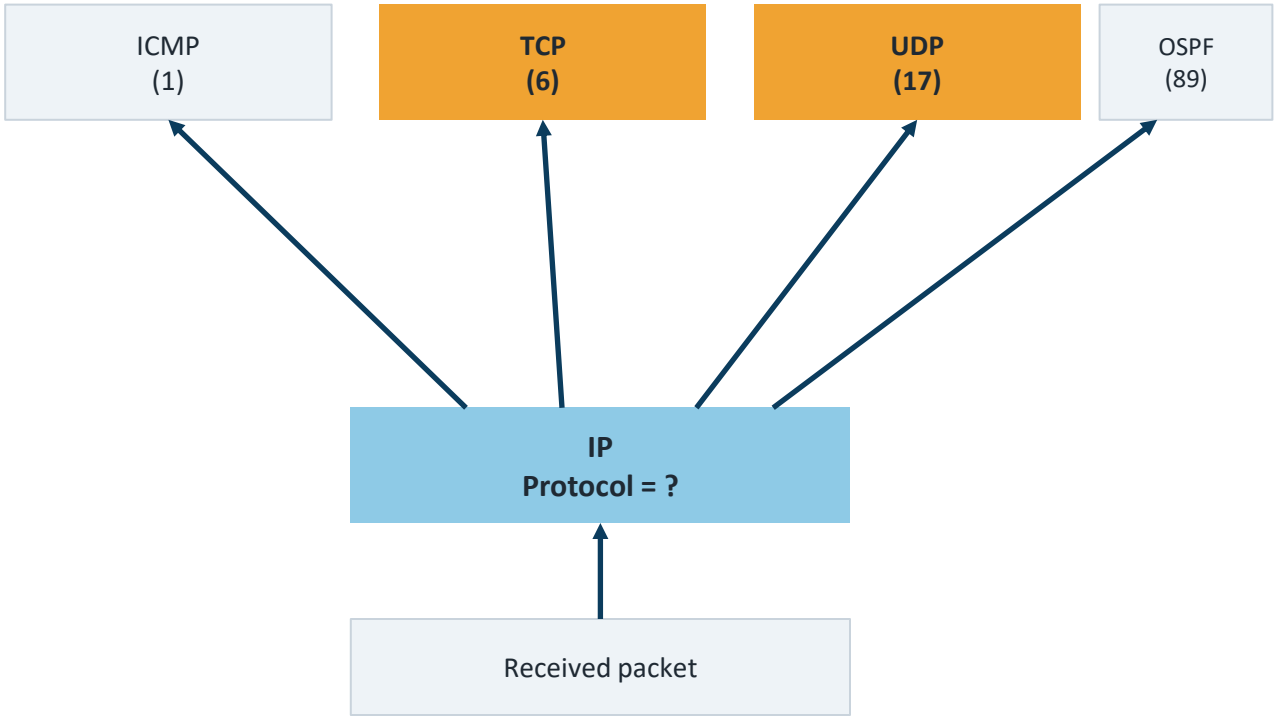
Field	Note
IHL	32-bit words; 5 = 20 bytes (no options)
DSCP / ECN	Former ToS; QoS (RFC 2474) and congestion (RFC 3168)
Flags	bit 1 DF (don't fragment), bit 2 MF (more fragments)
TTL	Remaining hops; 0 → drop + ICMP; traceroute
Protocol	6 TCP, 17 UDP, 1 ICMP, 89 OSPF...
Checksum	Header only; recomputed at every hop (TTL changes)
Options	Rare; many networks drop packets with options

Fragmentation: a packet larger than the MTU of the egress link is split by the router (IPv4) using Identification, MF and Offset; with DF set, it is dropped with ICMP 'Fragmentation Needed', the basis of Path MTU Discovery (RFC 1191). In IPv6 only the source fragments.

Protocol numbers: identifying the IP payload

No.	Protocol	Use
1	ICMP	Errors and diagnostics
2	IGMP	Multicast group membership
6	TCP	Reliable transport
17	UDP	Connectionless transport
41	IPv6	IPv6 encapsulated in IPv4 (6in4)
47	GRE	Generic tunnels
50 / 51	ESP / AH	IPsec
58	ICMPv6	ICMP for IPv6 (includes ND, MLD)
88	EIGRP	Routing (Cisco, RFC 7868)
89	OSPF	Link-state routing
103	PIM	Multicast routing
112	VRRP	Gateway redundancy
132	SCTP	Multistream transport

Demultiplexing at the Internet layer



Official registry: IANA, 'Assigned Internet Protocol Numbers' (replaces RFC 1700 from 1994).

Port numbers: identifying the application

Ranges (RFC 6335)

- **0 to 1023:** system ('well-known') ports
- **1024 to 49151:** user (registered) ports
- **49152 to 65535:** dynamic / ephemeral (client side)

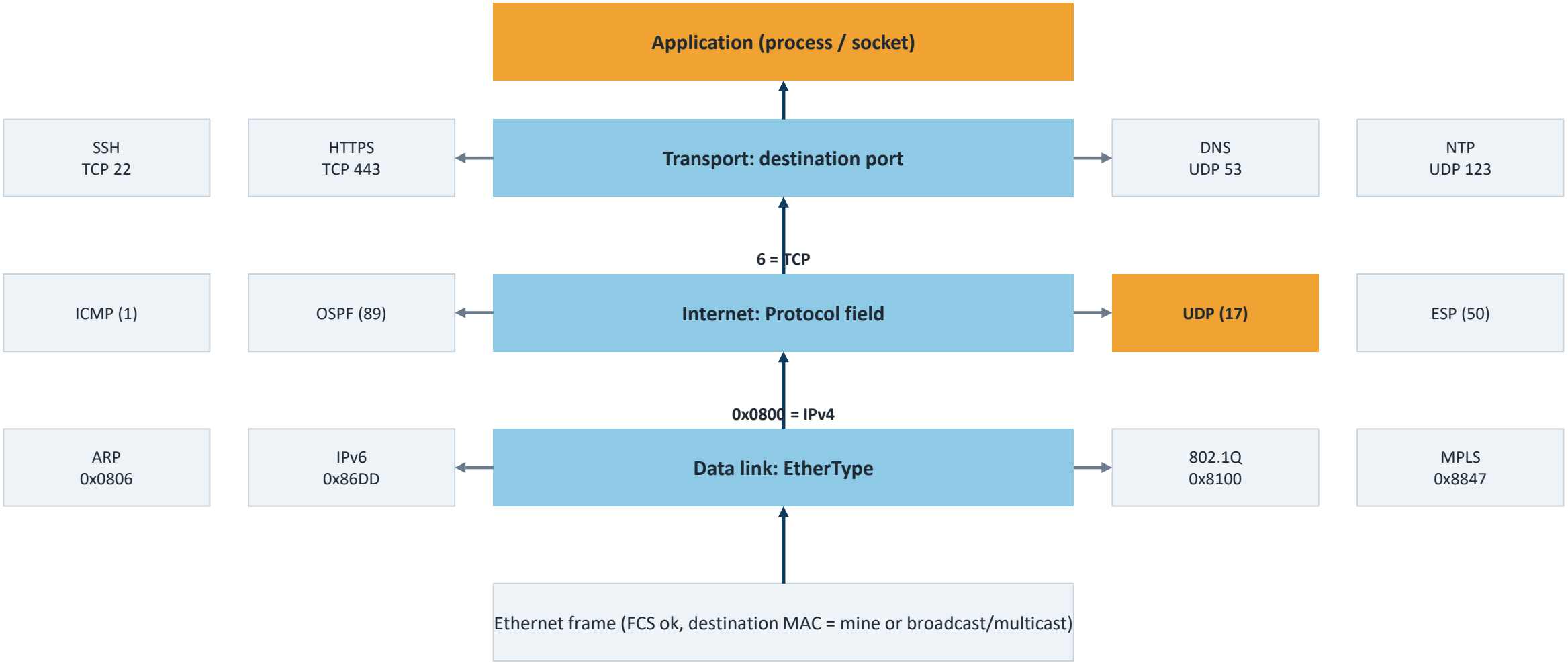
Socket = 5-tuple

protocol + source IP + source port + destination IP + destination port. It is what NAT, stateful firewalls and load balancers track.

Port	Transport	Service	Port	Transport	Service
22	TCP	SSH, SFTP, SCP	161 / 162	UDP	SNMP / traps
25	TCP	SMTP (server to server)	179	TCP	BGP
53	UDP, TCP	DNS	443	TCP, UDP	HTTPS; HTTP/3 (QUIC)
67 / 68	UDP	DHCP server / client	445	TCP	SMB
69	UDP	TFTP (network boot)	500 / 4500	UDP	IKE / IPsec NAT-T
80	TCP	HTTP (cleartext)	514	UDP	syslog
123	UDP	NTP	520 / 521	UDP	RIP / RIPng
143 / 993	TCP	IMAP / IMAPS	587	TCP	SMTP submission (TLS)
23	TCP	Telnet (legacy)	20 / 21	TCP	FTP (legacy)

Official registry: IANA Service Name and Transport Protocol Port Number Registry (successor of RFC 1060 / RFC 1700).

Receiving a frame: demultiplexing at three levels



Decimal-to-binary conversion

Position	8	7	6	5	4	3	2	1
Weight	128	64	32	16	8	4	2	1
172 =	**1**	0	**1**	0	**1**	**1**	0	0
255 =	1	1	1	1	1	1	1	1

Subtraction method

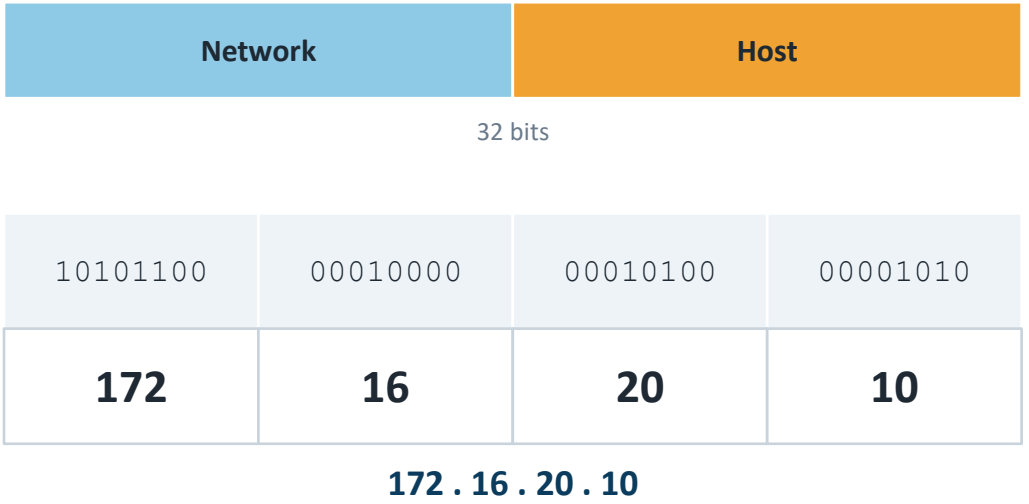
- 172 - 128 = 44 → bit 128 on
- 44 - 64 < 0 → bit 64 off
- 44 - 32 = 12 → bit 32 on
- 12 - 16 < 0 → bit 16 off
- 12 - 8 = 4 → bit 8 on
- 4 - 4 = 0 → bit 4 on; bits 2 and 1 off
- **Result: 172 = 10101100**

Mask (octet)	Binary	1 bits
0	00000000	0
128	10000000	1
192	11000000	2
224	11100000	3
240	11110000	4
248	11111000	5
252	11111100	6
254	11111110	7
255	11111111	8

Sum of all weights = 255. If a position is 0, nothing is added. The nine values in the table are the only ones that appear in contiguous masks.

The IPv4 address

- Every IPv4 interface is identified by a **32-bit** address
- Defined in 1981 (RFC 791) with two parts: **network identifier** and **host number**
- Network part assigned by the numbering hierarchy: **IANA → RIRs** (ARIN in the US; LACNIC with NIC.br in Brazil) → providers → organizations
- Host part assigned by the network administrator (or by DHCP)
- Written as **four decimal numbers separated by dots**; each number is one octet (0 to 255)



Where is the network/host boundary?

- 1981 to 1993: set by the **class** (first bits)
- Since 1993: set by the **mask** or the **prefix length** (CIDR), e.g., 172.16.20.10/24

LEGACY

Classful addressing (1981 to 1993)

Class	Leading bits	1st octet	Format (N = network, H = host)	Natural mask	Networks × hosts
A	`0xxxxxxx`	1 to 126	N.H.H.H (8 + 24 bits)	255.0.0.0 (/8)	126 × 16,777,214
B	`10xxxxxx`	128 to 191	N.N.H.H (16 + 16 bits)	255.255.0.0 (/16)	16,384 × 65,534
C	`110xxxxx`	192 to 223	N.N.N.H (24 + 8 bits)	255.255.255.0 (/24)	2,097,152 × 254
D	`1110xxxx`	224 to 239	Multicast (RFC 1112): 28 group bits	n/a	Multicast groups
E	`1111xxxx`	240 to 255	Reserved / experimental	n/a	Not routed on the Internet

First-octet rule: the value of the first octet reveals the class and therefore the 'natural' mask. 0.0.0.0/8 and 127.0.0.0/8 (loopback) are reserved.

Why it ended

- Waste: a class B (65,534 hosts) for a company with 300 machines
- Explosion of the Internet routing table with thousands of class Cs

What replaced it

- CIDR (RFC 1519, 1993; current RFC 4632) and VLSM: the mask sets the boundary
- Private addresses (RFC 1918) + NAT; then IPv6

UPDATED

CIDR: classless routing and prefix notation

Prefix notation

- **172.16.20.10/24** = mask 255.255.255.0 (24 one bits)
- **/n**: the n leftmost bits identify the network; the remaining 32 - n, the host
- Usable hosts = 2^(32-n) - 2 (except /31 and /32)

Aggregation (supernetting)

- 192.0.2.0/24 + 192.0.3.0/24 → advertised as **192.0.2.0/23**
- A provider holding 198.51.100.0/22 advertises a single prefix for 1,022 customer hosts
- Global IPv4 table ≈ 1 million prefixes (2026); unmanageable without CIDR

Prefix	Mask	Host bits	Addresses	Usable hosts	Typical use
/8	255.0.0.0	24	16,777,216	16,777,214	Former class A; 10.0.0.0/8 private
/16	255.255.0.0	16	65,536	65,534	Former class B; large campus
/24	255.255.255.0	8	256	254	User subnet / VLAN
/26	255.255.255.192	6	64	62	Small subnet
/30	255.255.255.252	2	4	2	Point-to-point link
/31	255.255.255.254	1	2	2 (RFC 3021)	Point-to-point link with no waste
/32	255.255.255.255	0	1	1	Router loopback; host route

Special-purpose IPv4 addresses

Block	Meaning	Reference
0.0.0.0/8 (0.0.0.0)	'This host on this network'; source only (e.g., DHCP client without an address)	RFC 1122
127.0.0.0/8	Loopback: packets never leave the host (127.0.0.1; IPv6 ::1)	RFC 1122
10/8, 172.16/12, 192.168/16	Private addresses: not routed on the Internet; used with NAT	RFC 1918
100.64.0.0/10	Shared address space for provider CGNAT (between the customer and the carrier's NAT)	RFC 6598
169.254.0.0/16	Link-local (autoconfiguration without DHCP, 'APIPA'); not routed	RFC 3927
192.0.2/24, 198.51.100/24, 203.0.113/24	Documentation and examples (used in this course)	RFC 5737
224.0.0.0/4	Multicast (former class D); 224.0.0.0/24 is link local	RFC 5771
240.0.0.0/4	Reserved (former class E); not usable on the public Internet	RFC 1112
255.255.255.255	Limited broadcast: destination only; does not cross routers	RFC 919
x.y.z.255 (/24)	Directed broadcast: all host bits set to 1; routers do not forward it by default	RFC 2644

Consolidated registry: RFC 6890 and the IANA IPv4 Special-Purpose Address Registry. Exhaustion: IANA in 2011, LACNIC in 2020 (final phase); hence NAT, CGNAT and IPv6.

Subnets: a third level of hierarchy

Before subnetting (1981)



With subnetting (RFC 950, 1985): bits borrowed from the host part



Example: 172.16.0.0/16 split with 8 subnet bits → 256 /24 subnets (172.16.0.0/24 to 172.16.255.0/24), 254 hosts each

- A subnet is a division of the space reserved for hosts, made by the organization itself; **one subnet per link** (VLAN, segment, WAN link)
- To the outside, the network is still **a single prefix** (aggregation); only the internal routers know the subnets
- The subnet field can have **any size**; it is specified by a **32-bit mask** (or by the prefix length)
- With CIDR/VLSM the 'network versus subnet' distinction lost its technical meaning: everything is a prefix; the term subnet remains in everyday use

The subnet mask and the comparison under mask

- Mask: **32 bits**, with 1s in the network and subnet positions and 0s in the host positions; always **contiguous** (e.g., 255.255.255.0 = /24)
- **Comparison under mask:** bitwise AND between address and mask; the host bits are zeroed and the subnet identifier remains
- The subnet field can have any size; **/n** says how many bits are 1
- Used by the host (local or remote destination?) and by the router (which route matches the destination? **longest prefix**)

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

AND truth table: only 1 AND 1 gives 1

Practical rules

- Mask 255 in an octet: copy the address octet
- Mask 0: zero the octet
- Intermediate mask (128 to 254): convert to binary and apply the AND bit by bit
- Prefix /n ⇔ mask with n leading 1 bits

Comparison under mask: example

172.16.128.2	10101100	00010000	10000000	00000010	address
AND					
255.255.255.0	11111111	11111111	11111111	00000000	mask /24
result	**10101100**	**00010000**	**10000000**	**00000000**	**= 172.16.128.0**
	172	**16**	**128**	**0**	

Classic reading (1999)

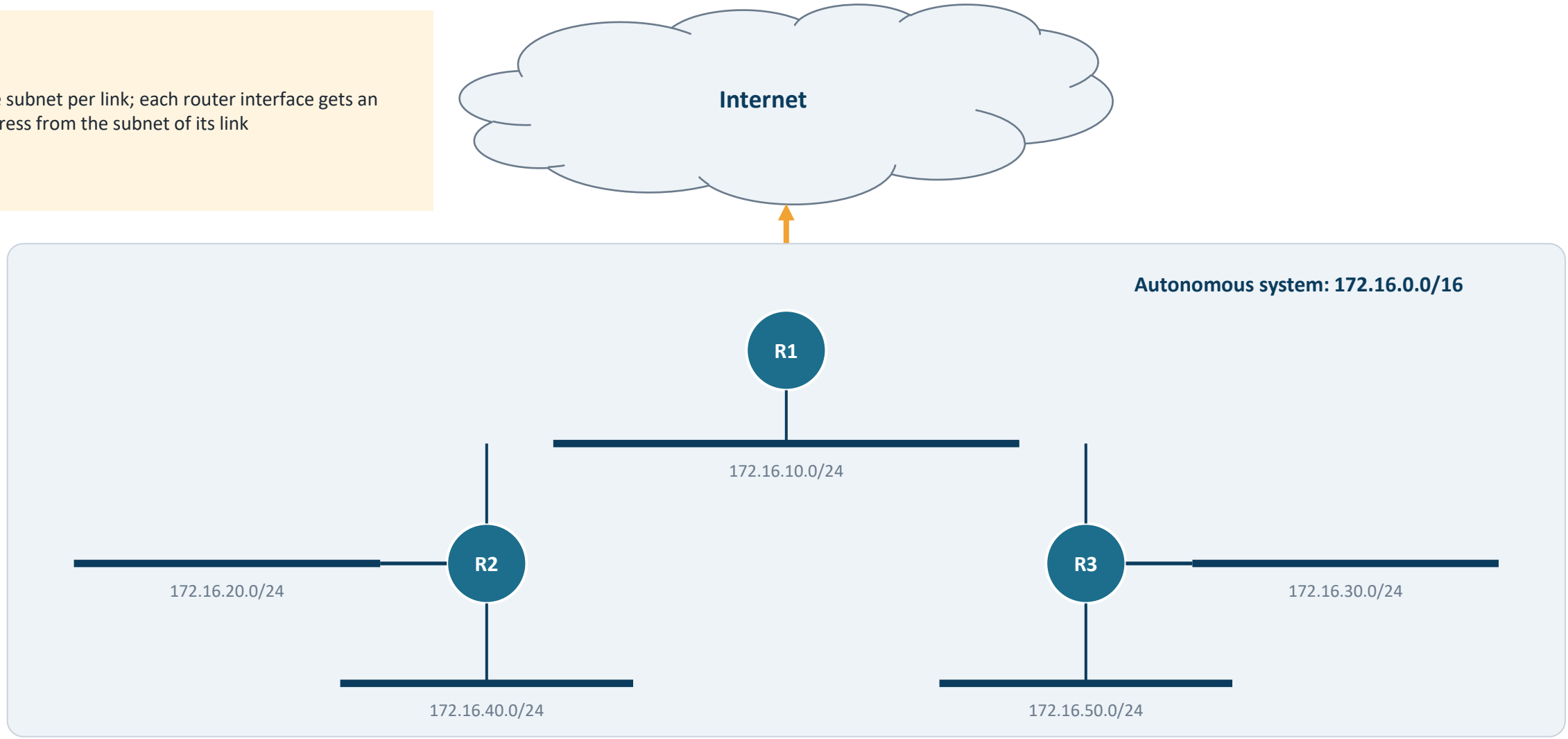
- Network (class B): 172.16.0.0
- Subnet: 172.16.128.0 (8 borrowed bits)
- Host: .2

CIDR reading (2026)

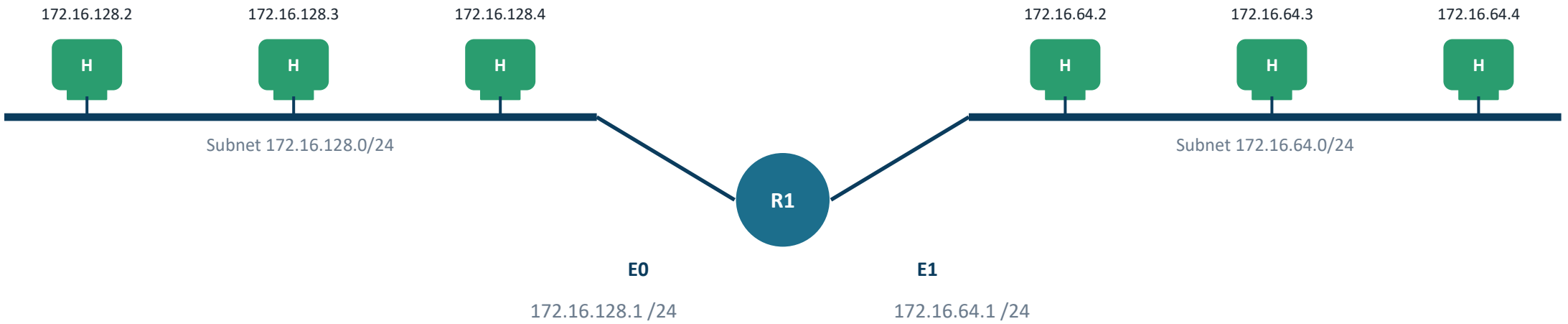
- The host belongs to prefix **172.16.128.0/24**
- Host range: 172.16.128.1 to 172.16.128.254
- Directed broadcast: 172.16.128.255

Addressing with subnets in an internetwork

One subnet per link; each router interface gets an address from the subnet of its link



Host addressing with subnets



Breakdown of 172.16.128.2 /24

172	16	128	2
Network (16 bits, historical class B)		Subnet (8 bits)	Host (8 bits)

Every host needs three parameters: address, mask (or prefix) and default gateway (the router's address on its subnet). Today DHCP delivers all three; in IPv6, RA/SLAAC delivers prefix and router and DHCPv6 complements.

Special addresses in a subnetted environment

In every subnet

- **All host bits = 1:** directed broadcast of the subnet (172.16.128.255 for 172.16.128.0/24)
- **All host bits = 0:** subnet identifier (172.16.128.0)
- Neither can be assigned to a host: usable = 2^h - 2

Example: 172.16.128.0/24

- Subnet: 172.16.128.0
- Hosts: 172.16.128.1 to 172.16.128.254 (254)
- Directed broadcast: 172.16.128.255
- Gateway (convention): .1 or .254

Old rule (RFC 950, 1985)	Current rule	Reference
Do not use the first subnet ('subnet zero') nor the last ('all-ones')	Both usable; default on all modern equipment	RFC 1878 (1995)
Point-to-point links with /30 (4 addresses, 2 usable)	/31 allowed: 2 addresses, both usable, no broadcast	RFC 3021 (2000)
Router loopback in an ordinary subnet	/32 per loopback; host route	Current practice
Directed broadcast forwarded by routers	Disabled by default ('smurf' attacks)	RFC 2644 (1999)

Masks off the octet boundary (VLSM)

Example: 203.0.113.77 /29 (mask 255.255.255.248)

203.0.113.77	11001011	00000000	01110001	01001|101	address
AND					
255.255.255.248	11111111	11111111	11111111	11111|000	mask /29
subnet	**11001011**	**00000000**	**01110001**	**01001|000**	**= 203.0.113.72**

subnet | host

Result

- Subnet: **203.0.113.72/29**
- Hosts: .73 to .78 (6)
- Broadcast: .79

The 'step' shortcut

- 256 - 248 = 8 → subnets every 8: .0, .8, .16 ... .72, .80
- 77 lies between 72 and 80 → subnet .72

Other masks

- /25 step 128; /26 step 64; /27 step 32; /28 step 16; /30 step 4
- In the 3rd octet (/17 to /23) the step counts in multiples of 256 addresses

VLSM in practice: addressing plan for a /24

Need	Hosts	Prefix	Subnet	Host range	Broadcast
Users	100	/25 (126)	172.16.20.0/25	.1 to .126	.127
Servers	50	/26 (62)	172.16.20.128/26	.129 to .190	.191
Management	25	/27 (30)	172.16.20.192/27	.193 to .222	.223
Printers	10	/28 (14)	172.16.20.224/28	.225 to .238	.239
WAN link 1	2	/30 (2)	172.16.20.240/30	.241 to .242	.243
WAN link 2	2	/30 (2)	172.16.20.244/30	.245 to .246	.247
WAN link 3	2	/30 (2)	172.16.20.248/30	.249 to .250	.251
Spare	-	/30	172.16.20.252/30	.253 to .254	.255

VLSM rules: allocate from largest to smallest; each subnet starts at a multiple of its size; no overlap; the IGP must be classless (RIPv2, EIGRP, OSPF, IS-IS).

Modern variant: point-to-point links with /31 (RFC 3021) free up 8 addresses; router loopbacks use /32 in a separate block (e.g., 172.16.255.0/24).

MODULE 2 | PART 2

Internet Protocol behavior

How a host decides where to send a packet: route determination (local or remote), next-hop lookup and address resolution with ARP

IP host communication: always two steps

1

Route determination

Is the destination local (same prefix as mine) or remote? If remote, what is the next hop (gateway) for it? Decided from the host's address, mask and route table.

2

Address resolution

What is the data-link (MAC) address of the local destination, or of the next hop? ARP in IPv4; Neighbor Discovery (ND) in IPv6. Result kept in a cache.

Always, on every host

IP hosts perform these steps regardless of configuration or topology. Understanding them is the basis of connectivity troubleshooting.

Verification tools

- Linux: ip route get <destination>; ip neigh
- Windows: route print; arp -a
- macOS: netstat -rn; arp -an
- Routers: show ip route; show ip arp

Route determination: local or remote?

- First step: is the destination IP **local** or **remote**?
- **Local:** host inside the same network or subnet address space (prefix) as the source
- **Remote:** host outside the source's prefix; reachable only through a router
- The source uses **its IP, its mask and the destination IP** in the decision (comparison under mask of both addresses)
- Modern implementation: lookup in the host's route table with **longest prefix match** (equivalent)

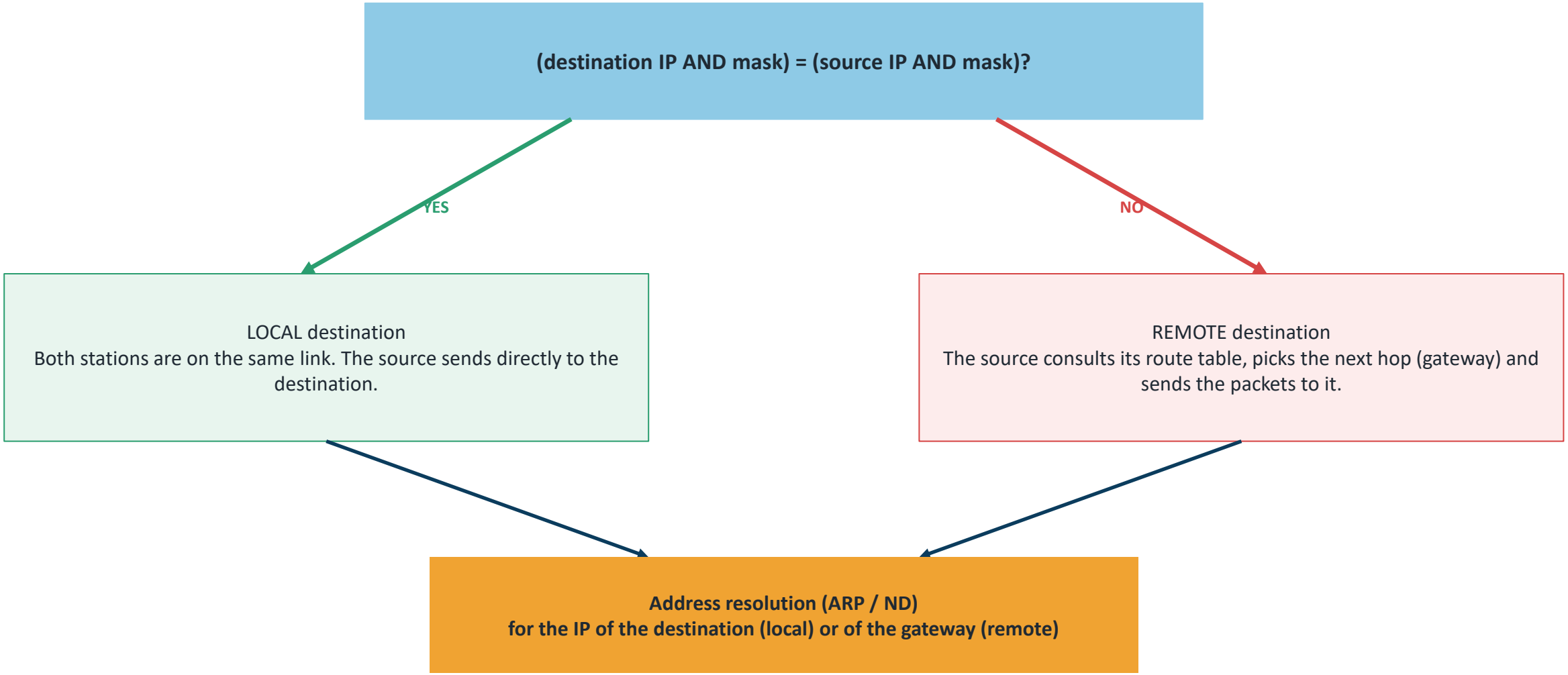
Inputs of the decision

- Source IP: 172.16.128.7
- Mask: 255.255.255.0 (/24)
- Destination IP: 172.16.128.2 (or 172.16.64.2)

Possible outcomes

- **Local** → resolve the destination's MAC (ARP)
- **Remote** → look up the next hop (gateway) and resolve its MAC
- No route → 'network unreachable' error on the host itself

Route determination logic



Local or remote: example with AND

Destination: 172.16.128.2

172.16.128.2 (destination)	10101100	00010000	10000000	00000010	
AND 255.255.255.0	11111111	11111111	11111111	00000000	
result	**10101100**	**00010000**	**10000000**	**00000000**	**172.16.128.0**
172.16.128.7 (source)	10101100	00010000	10000000	00000111	
AND 255.255.255.0	11111111	11111111	11111111	00000000	
result	**10101100**	**00010000**	**10000000**	**00000000**	**172.16.128.0**

Equal results → LOCAL destination (same link)

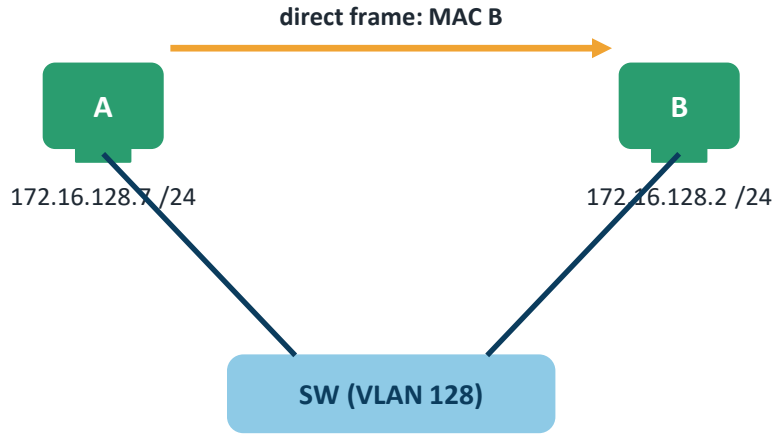
What if the destination were 172.16.64.2? AND → 172.16.64.0 ≠ 172.16.128.0 → REMOTE

Check it on the operating system

- Linux: ip route get 172.16.128.2 → 'dev eth0' (local); ip route get 172.16.64.2 → 'via 172.16.128.1' (remote, through the gateway)

Local destinations

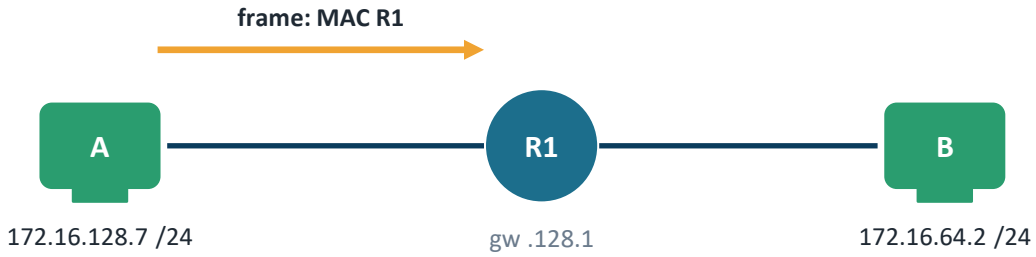
- If the destination subnet is the same as the source's, communication happens **without a router**
- Frames are transmitted directly on the **attached link** (L2 switches along the way are transparent; 'same link' = same VLAN)
- The source moves immediately to **address resolution** (ARP) to obtain the destination's MAC
- Common troubleshooting mistake: a wrong mask makes the host believe a remote destination is local (and the ARP is never answered)



Subnet 172.16.128.0/24: no router involved

Remote destinations

- Remote destination: an **intermediate step** is required before address resolution
- **Next-hop gateway lookup**: which router to use to reach the remote network?
- The source then resolves the MAC **of the gateway**, not of the final destination
- The IP packet is still addressed to the final destination; only the Ethernet frame is addressed to the gateway



IP packet: source 172.16.128.7 → destination 172.16.64.2
Ethernet frame: MAC A → MAC R1

Different subnets (128.0/24 and 64.0/24): a router is mandatory

Next-hop gateway lookup

- The source consults its **route table** to find the next hop for the remote network
- Sources of a host's table: the **connected-subnet route**, the **default gateway** (DHCP or RA), static routes, and on servers/routers, dynamic protocols (RIP, OSPF, BGP)
- Selection by **longest prefix match**: the most specific route wins; /0 (default) matches everything and is used last
- Result: the IP of a router **on the same link** as the source

Destination	Next hop	Interface	Route source
172.16.128.0/24	connected	eth0	interface address
172.16.64.0/24	172.16.128.1	eth0	static
10.0.0.0/8	172.16.128.2	eth0	static (VPN)
0.0.0.0/0	172.16.128.1	eth0	DHCP (default gateway)

Route table of a host (ip route / route print)

Lookup examples

- 172.16.128.9 → matches the connected /24 → local
- 172.16.64.2 → matches the static /24 → via 172.16.128.1
- 10.1.2.3 → matches the /8 → via 172.16.128.2
- 8.8.8.8 → only the /0 → via 172.16.128.1

IP communication: what the host knows at this point

Already determined (network layer)

- Whether the destination is local or remote
- If remote, the IP address of the next hop (router)
- The IP packet can be built: source, destination, TTL, protocol...

Still missing (data link layer)

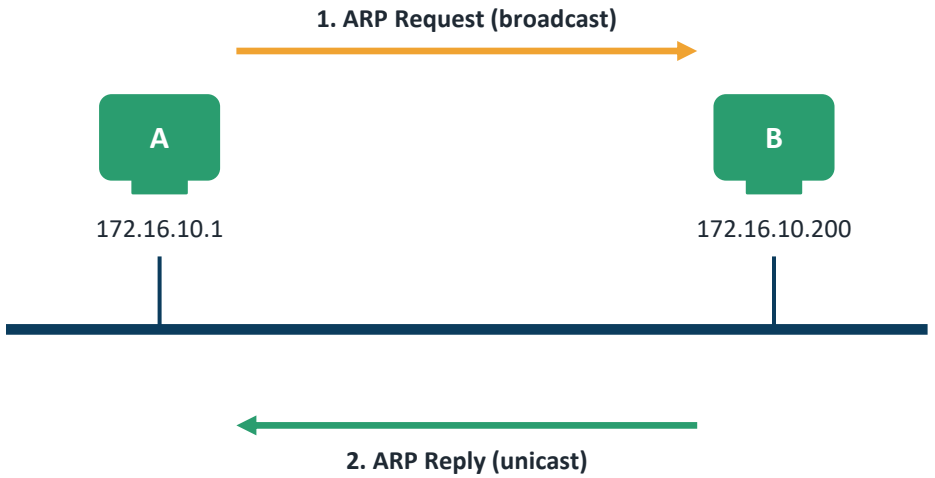
- The MAC address of the destination (if local) or of the next hop (if remote)
- Without it, the Ethernet frame cannot be built
- Solution: ARP (IPv4) or ND (IPv6), with a cache



The frame is ready except for the destination MAC: ARP resolves that

Address resolution: ARP (RFC 826)

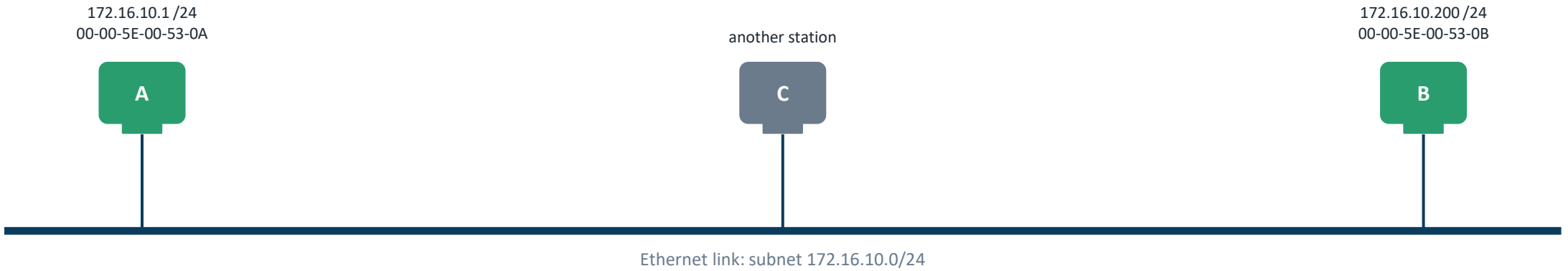
- Direct transmission or via the gateway, communication on the link is always **MAC to MAC**
- The source must **resolve the MAC** of the destination or of the next hop before transmitting
- **ARP** (RFC 826): a broadcast question 'who has IP X?'; the owner replies in unicast with its MAC
- Every host keeps an **ARP cache** with the IP → MAC mappings (entries expire: Linux revalidates after ≈ 30 s; Cisco IOS 4 h)
- Encapsulated directly in Ethernet (**0x0806**), never crosses routers; in IPv6 it is replaced by **Neighbor Discovery** (ND)



IP	MAC	Age
172.16.10.200	00-00-5E-00-53-0B	0 s

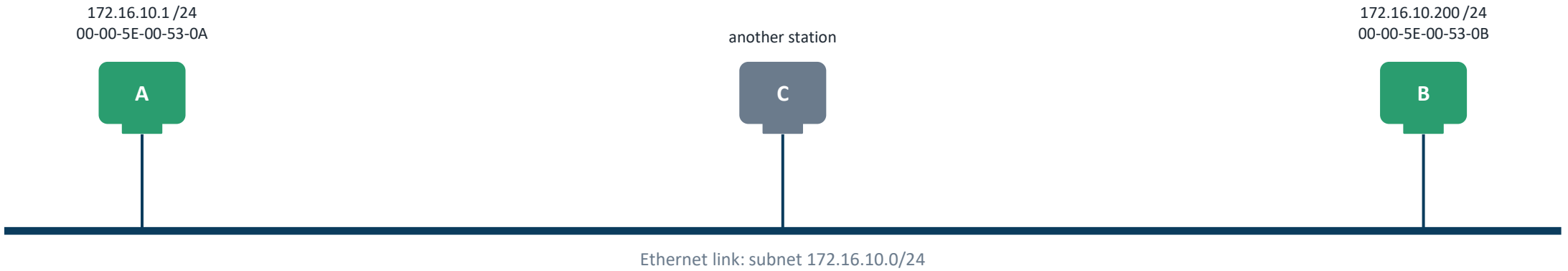
ARP cache of A after the exchange (arp -a / ip neigh)

ARP, local host example (1 of 6): the scenario



- Host A has yet to communicate with host B
- A wants to open an **SSH** session (TCP 22) to B
- A's TCP/IP stack applies the **route determination logic**: is B local or remote?

ARP, local host example (2 of 6): B is local; empty cache



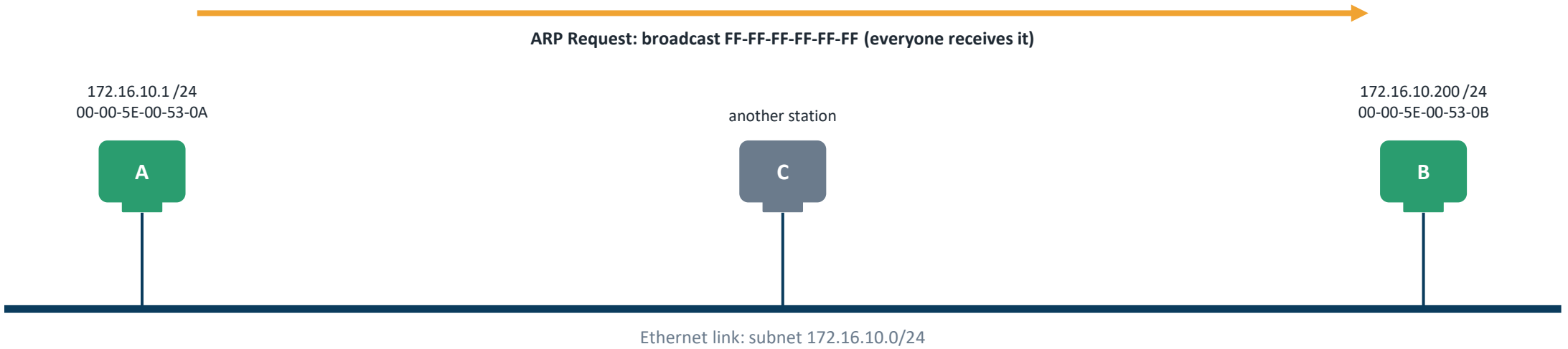
Route determination

- 172.16.10.200 AND /24 = 172.16.10.0
- 172.16.10.1 AND /24 = 172.16.10.0
- **Equal → B is local**

ARP cache of A

- Looks up 172.16.10.200 → **not found**
- A must send an ARP Request
- The SSH packet waits in the queue until the reply

ARP, local host (3 of 6): ARP Request in broadcast



- A sends an **ARP Request** in a **broadcast** Ethernet frame (FF-FF-FF-FF-FF-FF, EtherType 0x0806)
- Every station on the link receives the frame and examines the request (including C)
- Question: 'who has 172.16.10.200? reply to 172.16.10.1 (00-00-5E-00-53-0A)'

The ARP Request packet

ARP packet (RFC 826), operation = 1 (Request)

Hardware type = 1 (Ethernet)		Protocol type = 0x0800 (IPv4)
HLEN = 6	PLEN = 4	**Operation = 1 (Request)**
Sender MAC (A): 00-00-5E-00-53-0A		
Sender IP (A): 172.16.10.1		
Target MAC: 00-00-00-00-00-00 (unknown)		
Target IP (B): 172.16.10.200		

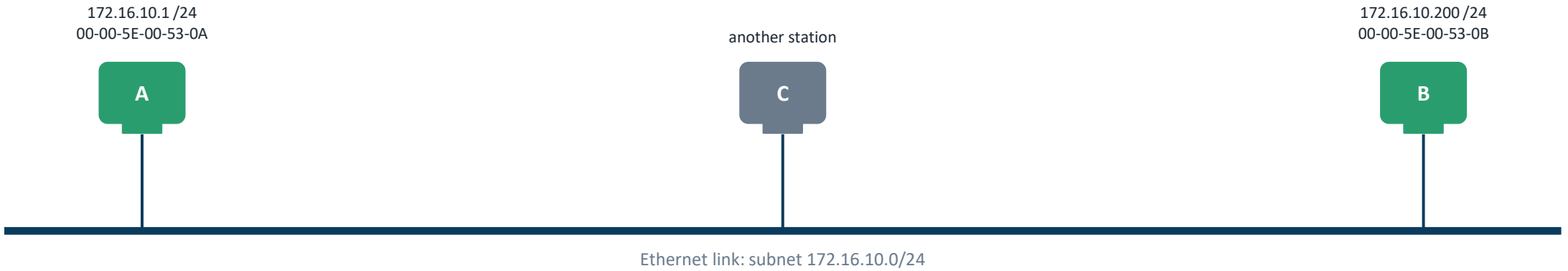
Ethernet frame that carries it

Destination FF-FF-FF-FF-FF-FF	Source 00-00-5E-00-53-0A	Type 0x0806	ARP (28 B) + padding (18 B)	FCS

Notes

- The target MAC is zeroed: it is exactly what we want to discover
- The hardware and protocol fields allow ARP for other pairs (e.g., Token Ring/IPX in 1999); today only Ethernet/Wi-Fi + IPv4
- Whoever receives the request also learns the sender's IP/MAC pair (many stacks cache it)
- Wireshark capture: filter 'arp'

ARP, local host example (4 of 6): B processes the request

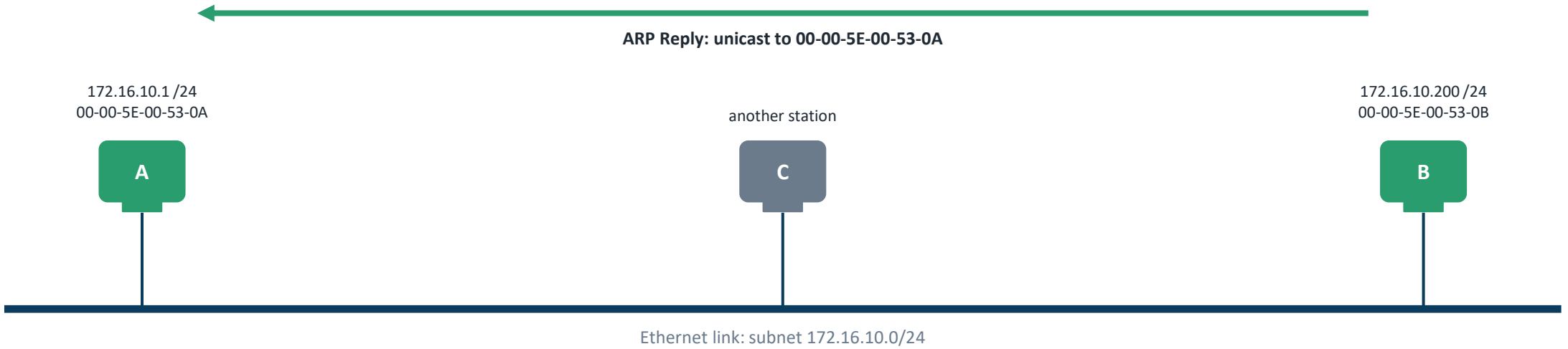


1. **Delivery to ARP**
   B's MAC layer forwards the ARP Request to the ARP software in B's IP stack (EtherType 0x0806).

2. **Recognition**
   B's ARP recognizes the target IP, 172.16.10.200, as its own; the other stations (C) discard the request.

3. **Reply**
   B's ARP process formulates an ARP Reply and hands it to the MAC layer for unicast transmission to A.

ARP, local host example (5 of 6): ARP Reply in unicast



- B's ARP sends the **ARP Reply** to A, providing B's MAC (00-00-5E-00-53-0B)
- The reply is **unicast**: a frame from MAC B to MAC A; the other stations do not see it
- B normally stores A's IP/MAC pair in its own cache (learned from the request)

The ARP Reply packet

ARP packet (RFC 826), operation = 2 (Reply)

Hardware type = 1 (Ethernet)		Protocol type = 0x0800 (IPv4)
HLEN = 6	PLEN = 4	**Operation = 2 (Reply)**
Sender MAC (B): 00-00-5E-00-53-0B		
Sender IP (B): 172.16.10.200		
Target MAC (A): 00-00-5E-00-53-0A		
Target IP (A): 172.16.10.1		

Ethernet frame that carries it

Destination 00-00-5E-00-53-0A	**Source** **00-00-5E-00-53-0B**	Type 0x0806	ARP (28 B) + padding	FCS

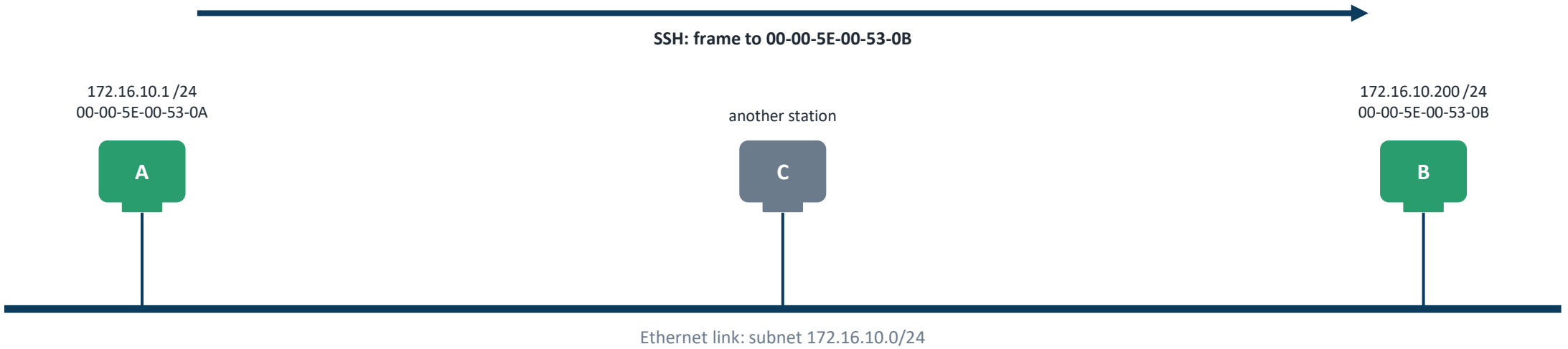
IP	MAC	Type
172.16.10.200	`00-00-5E-00-53-0B`	dynamic

ARP cache of A after receiving the reply

Security

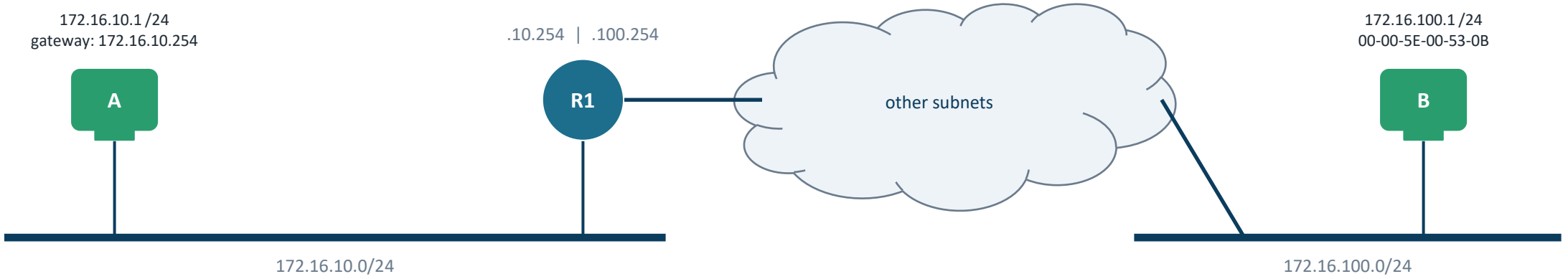
- ARP authenticates nothing: any station can answer for any IP (ARP spoofing / poisoning)
- Defenses: Dynamic ARP Inspection + DHCP snooping on the switches, static entries on critical hosts

ARP, local host example (6 of 6): communication proceeds



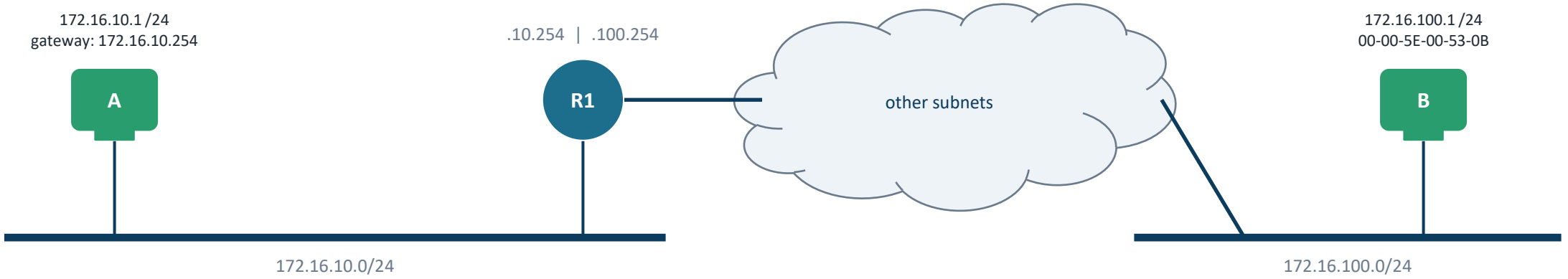
- A installs the entry **172.16.10.200 → 00-00-5E-00-53-0B** in its ARP cache
- A's stack now has destination IP and MAC: the queued SSH segment is transmitted
- Subsequent packets use the cache (no new ARP) until the entry expires or is invalidated

ARP, remote host example (1 of 4): the scenario



- Host A has yet to communicate with B and wants to open an **SSH** session
- A has a subnet mask (/24) and a **default gateway**: R1, 172.16.10.254
- The default gateway gives A a next hop for **every destination outside subnet 172.16.10.0/24**

ARP, remote host example (2 of 4): B is remote



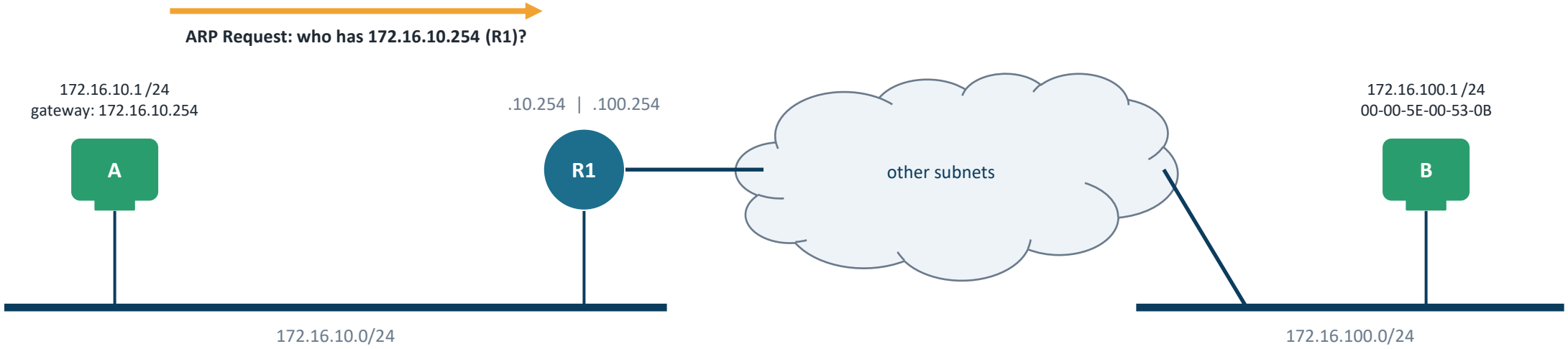
Route determination

- 172.16.100.1 AND /24 = 172.16.100.0
- 172.16.10.1 AND /24 = 172.16.10.0
- **Different → B is remote**

Next step

- Next-hop lookup in A's route table
- Goal: a router on A's link that reaches 172.16.100.0/24

ARP, remote host (3 of 4): default route and ARP for the gateway



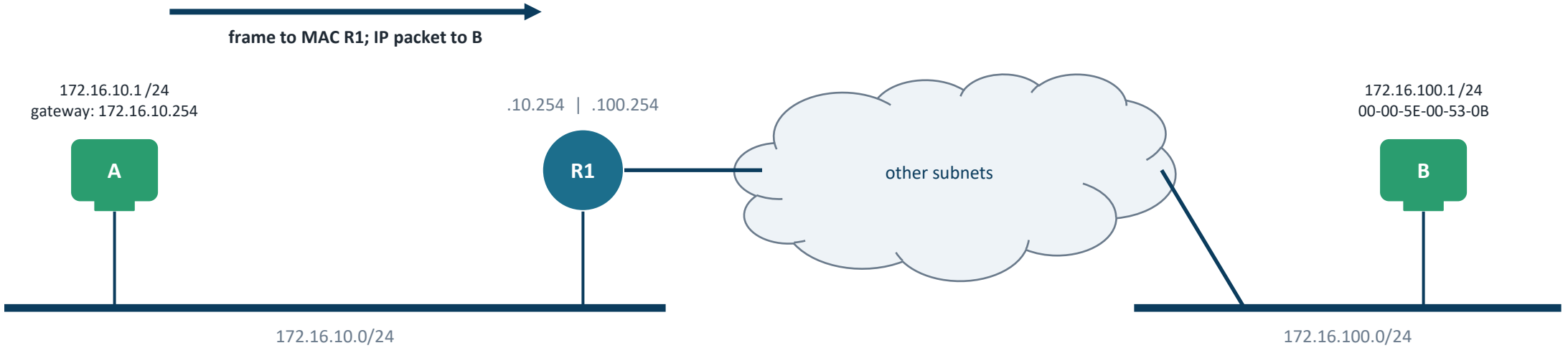
Destination	Next hop	Source
172.16.10.0/24	connected	interface
0.0.0.0/0	**172.16.10.254**	default gateway

Route table of host A

Key point

- A ARPs for **R1 (172.16.10.254)**, NOT for B
- The ARP broadcast does not cross the router: an ARP for B would go unanswered

ARP, remote host example (4 of 4): communication via R1



Destination MAC: R1 (00-00-5E-00-53-01)	Source MAC: A	0x0800	Source IP 172.16.10.1 → destination IP 172.16.100.1 (B) | TCP 22	FCS

- At the data link layer, A talks to **R1's interface**; at the network layer, to **B**
- R1 receives the frame, discards the Ethernet header and routes by destination IP; on B's link, R1 does its own ARP for B
- Resulting caches: A knows R1; R1 knows A and B; B knows R1 (and never A's MAC)

LEGACY

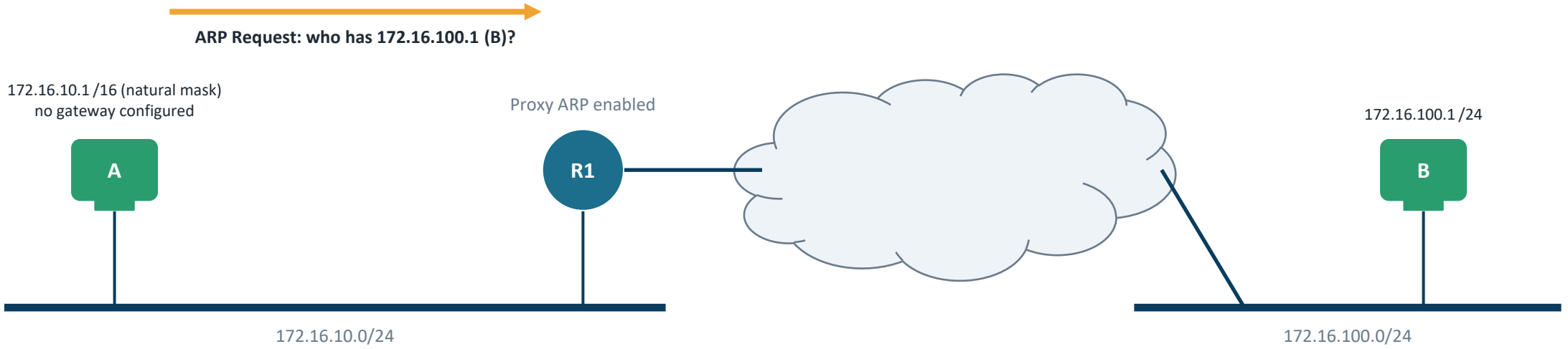
Proxy ARP

- An extension of ARP that routers can run: **answer on behalf of hosts on other subnets** (RFC 1027)
- Original goal: minimize the routing knowledge required of hosts in a subnetted network
- With Proxy ARP on the interface, the hosts on that subnet communicate with the whole network **without a static route or a configured gateway**
- The hosts use the **natural mask** (e.g., /16) even though the network is divided into /24 subnets
- In 2026: unnecessary with DHCP/RA; kept for special cases (VPN, migrations, legacy hosts); **turn it off if you do not need it**

Where it still shows up

- VPN concentrators answering for remote clients with LAN IPs
- Migration from a flat (bridged) network to a routed one without reconfiguring every host
- 'Local proxy ARP' to force traffic between hosts on the same VLAN through the router (private VLANs)
- Risk: ARP caches full of entries pointing at the router; traffic 'mysteriously' routed

Proxy ARP, example (1 of 3): host with a 'flat' mask



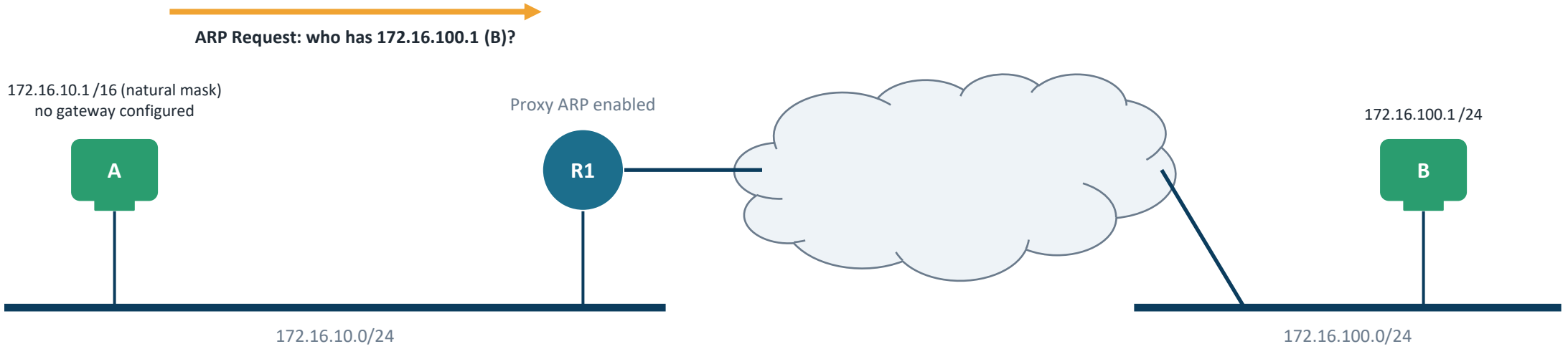
Route determination on A (mask /16)

- 172.16.100.1 AND /16 = 172.16.0.0
- 172.16.10.1 AND /16 = 172.16.0.0
- **Equal → A believes B is local** and ARPs for B

Without Proxy ARP

- The ARP broadcast dies on A's link; B never receives it
- A has no gateway: communication fails ('host unreachable')

Proxy ARP, example (2 of 3): R1 steps in



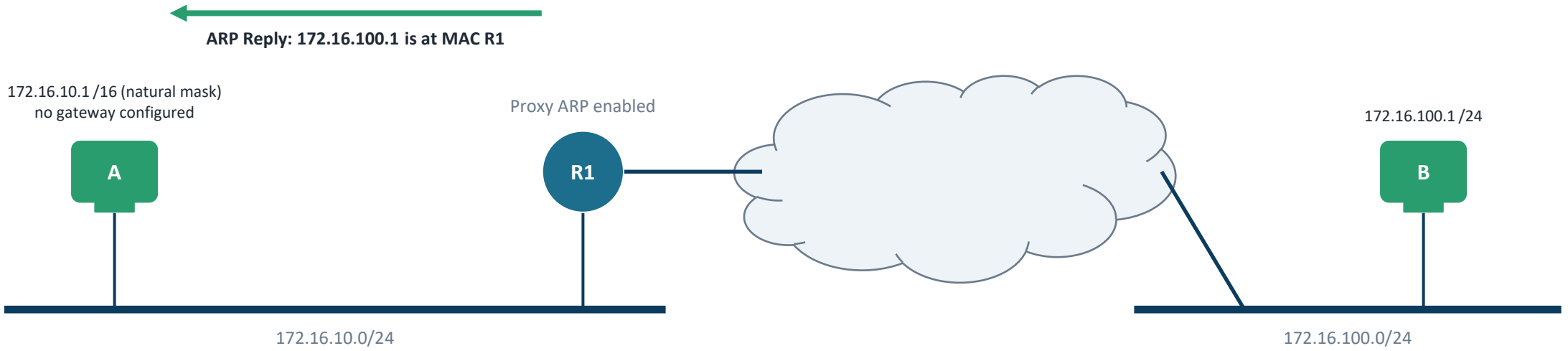
Destination	Next hop / interface
172.16.10.0/24	connected, E0 (A's side)
172.16.100.0/24	**connected, E1 (B's side)**
172.16.11.0/24	via 172.16.10.230

Route table of R1

The proxy's decision

- Is there a route to the target IP? **Yes**
- Does the route leave through another interface? **Yes** → R1 answers on behalf of B

Proxy ARP, example (3 of 3): R1 answers with its own MAC



IP	MAC
172.16.100.1 (B)	**00-00-5E-00-53-01 (R1!)**
172.16.100.9	00-00-5E-00-53-01
172.16.200.3	00-00-5E-00-53-01

ARP cache of A: several remote IPs → the same MAC (R1)

- R1 answers with the **MAC of its interface** instead of B's
- A records the entry and starts sending frames to R1, which routes them
- Troubleshooting: many IPs with the same MAC in the cache = Proxy ARP

UPDATED

ARP in 2026: gratuitous ARP, cache, security and IPv6

Gratuitous ARP

- A host announces its own IP/MAC pair (sender = target)
- Uses: detect a duplicate IP; update caches after a MAC change; VRRP/HSRP failover; VM migration

Cache and timers

- Linux: entry 'reachable' for ≈ 30 s (15 to 45 s), then revalidated (ip neigh)
- Cisco IOS: 4 h by default (arp timeout); Windows: ≈ 15 to 45 s
- Large tables in data centers: thousands of VMs per VLAN

Security

- ARP does not authenticate: cache poisoning allows interception (man in the middle)
- Defenses: Dynamic ARP Inspection + DHCP snooping; static entries; segmentation

IPv6: Neighbor Discovery instead of ARP

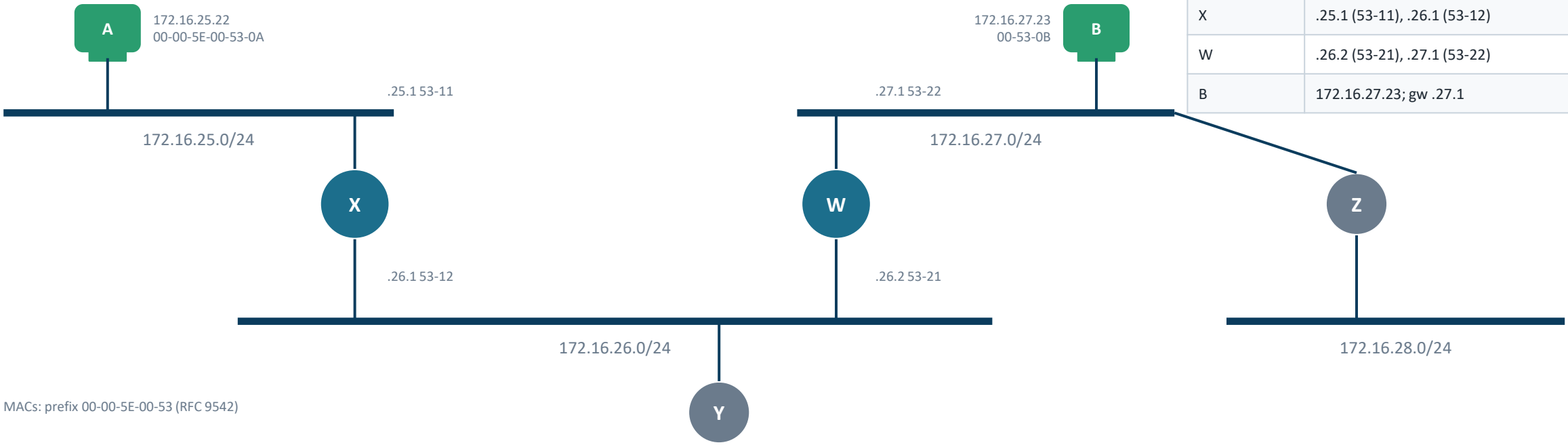
- ICMPv6 Neighbor Solicitation / Advertisement (RFC 4861)
- Sent to a 'solicited-node' multicast (ff02::1:ffxx:xxxx), not to broadcast: only hosts with that suffix process it
- Per-neighbor states: reachable, stale, delay, probe; also Router Discovery and redirects

MODULE 2 | PART 3

IP in a routed internetwork: a complete example

Four links, four routers (W, X, Y, Z) and two users (A and B): what happens to the frames and the packets at every hop

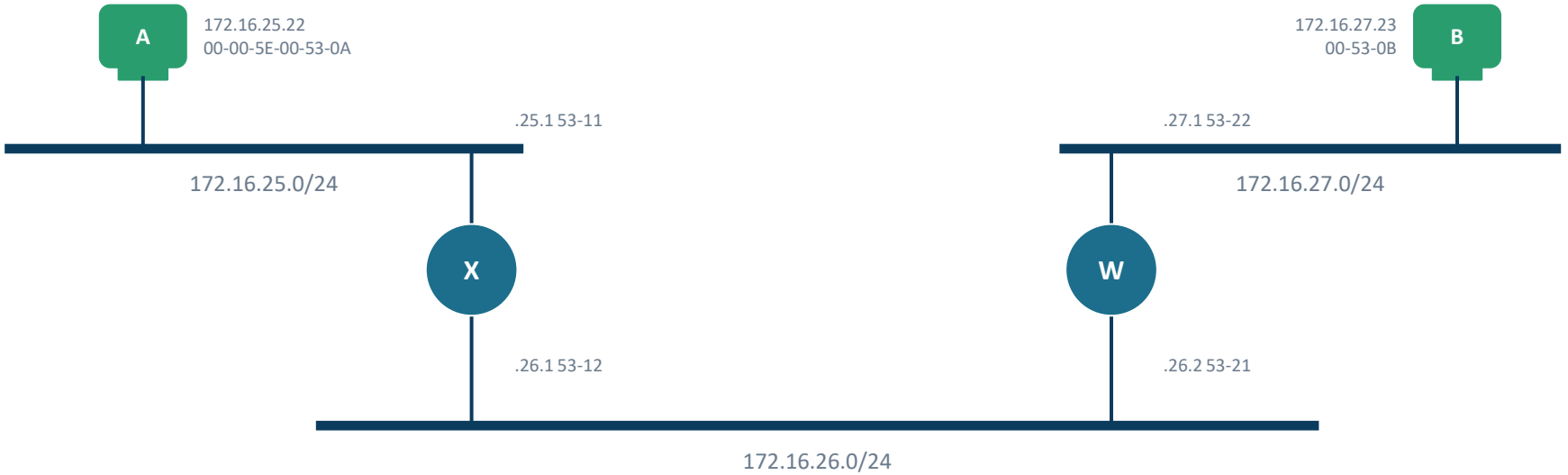
IP in a routed internetwork: the topology



Element	Addresses
A	172.16.25.22; gw .25.1
X	.25.1 (53-11), .26.1 (53-12)
W	.26.2 (53-21), .27.1 (53-22)
B	172.16.27.23; gw .27.1

- A starts the communication by applying **route determination**: is B local or remote?
- B is remote: A looks up the next hop and finds the **default route** (X); resolves X's MAC with ARP
- Two routing hops (X and W) to B's LAN

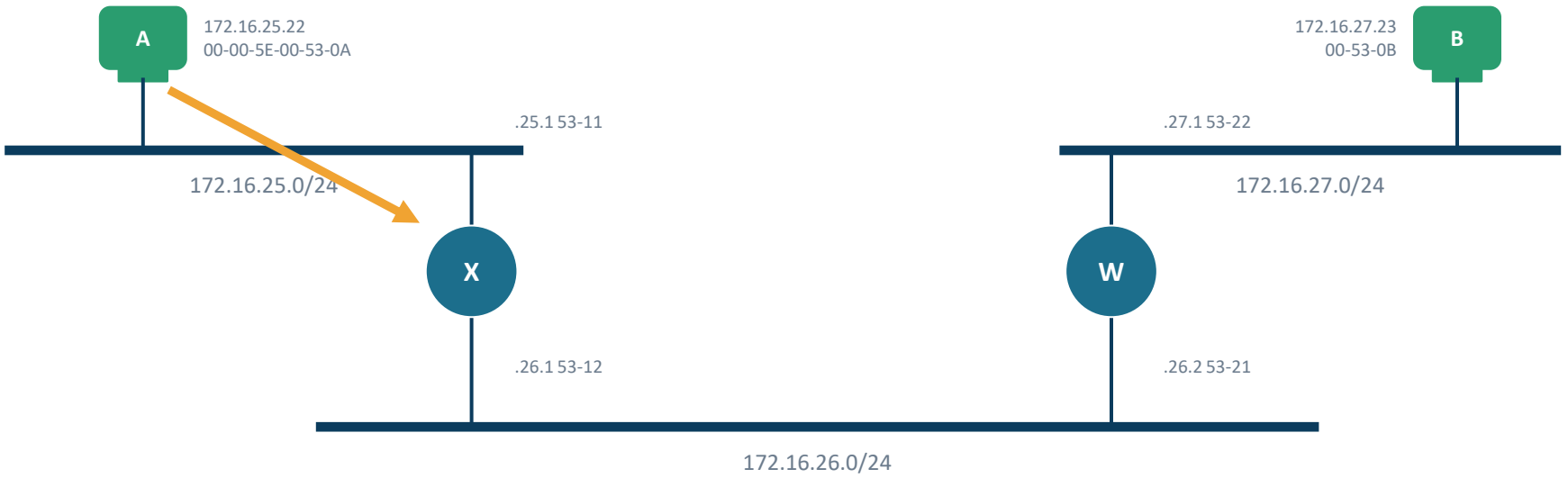
The data transfer process: the six steps



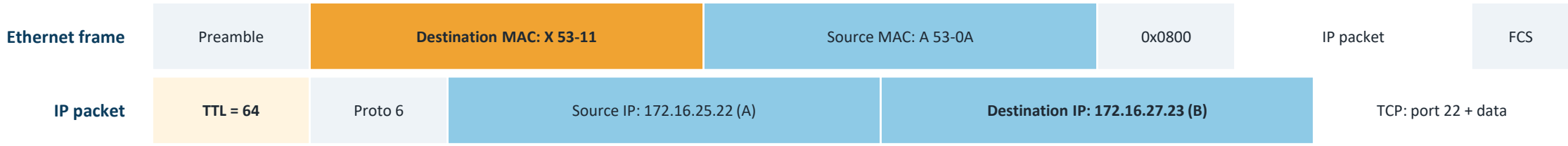
1. **A sends the initial packet**
   Frame to X's MAC; IP packet to B.
2. **X receives and processes**
   Strips the Ethernet header; consults the route table.
3. **X forwards to the next hop**
   New frame to W's MAC; IP packet intact (TTL-1).
4. **W receives the packet**
   Strips the Ethernet header; destination connected.
5. **W delivers to B**
   ARP for B if needed; new frame to B's MAC.
6. **B receives the packet**
   Own MAC and IP: delivery to transport (port).

Invariants: source IP (172.16.25.22) and destination IP (172.16.27.23) never change (no NAT). Variables: source and destination MACs on each link; TTL decremented at X and at W.

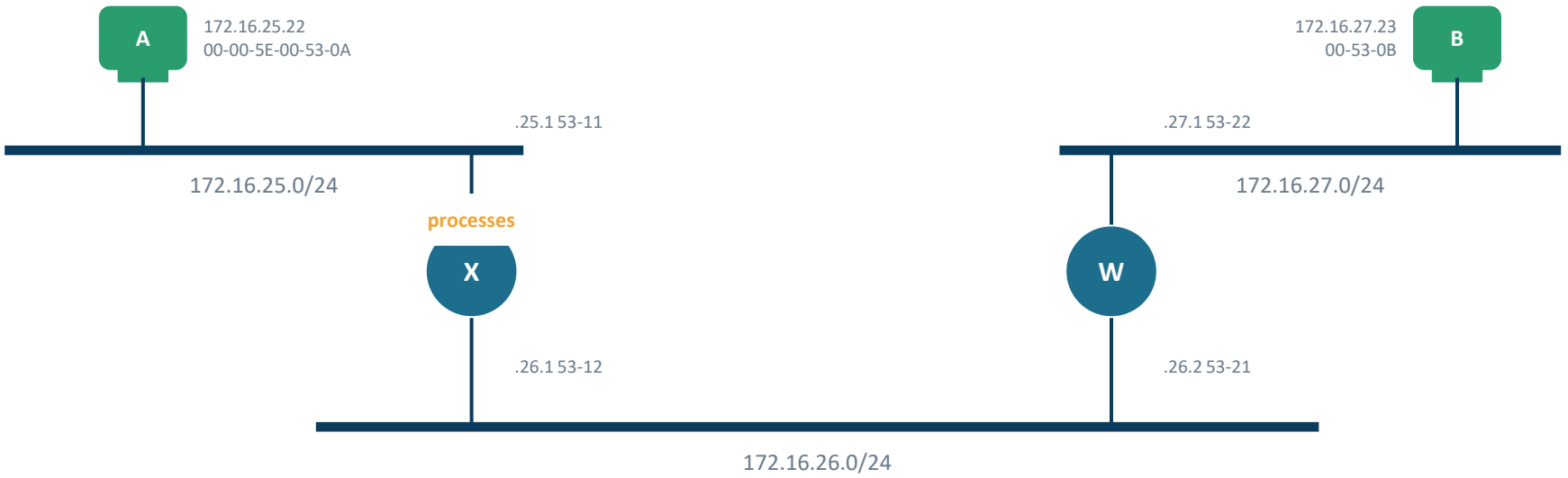
Step 1: A sends the initial packet



- A formats an IP packet for B
- To reach B, A sends to X: default route + ARP for X
- Frame: A → X; packet: A → B; TTL = 64



Step 2: X receives and processes the packet

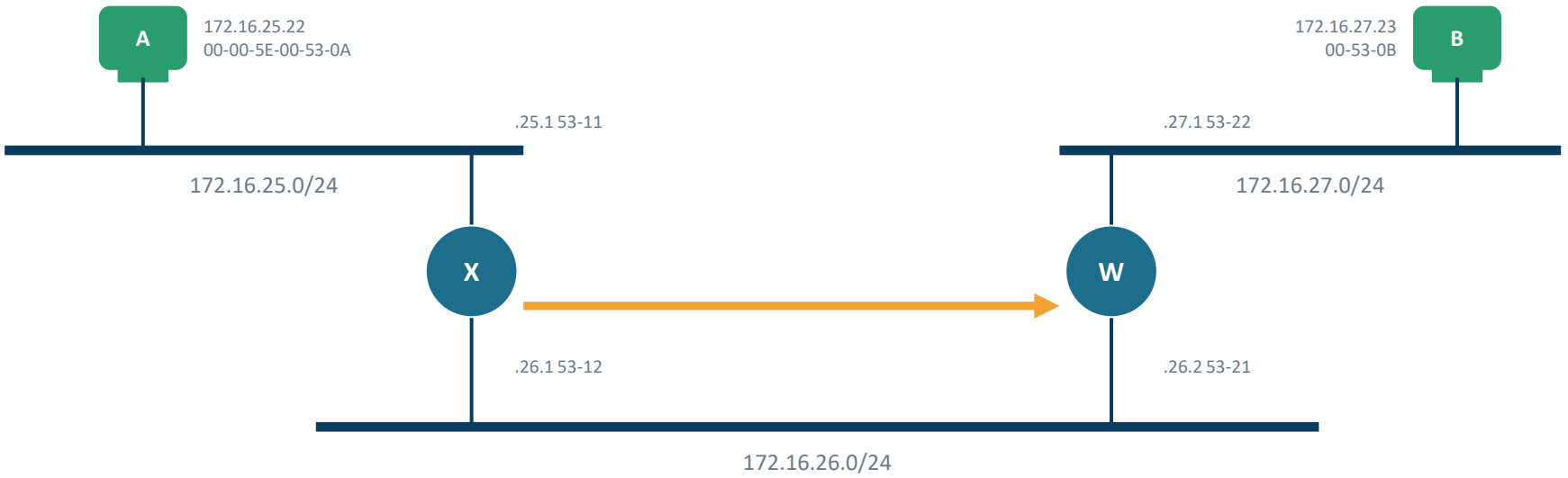


Destination	Next hop
172.16.25.0/24	connected
172.16.26.0/24	connected
172.16.27.0/24	**172.16.26.2 (W)**
172.16.28.0/24	172.16.26.3 (Y)

Route table of X (OSPF or static)

1. **Accepts the frame**
   Destination MAC = its own; discards the Ethernet header.
2. **Validates the packet**
   Checksum ok, TTL > 1, destination is not X → routing.
3. **Looks up the route**
   172.16.27.0/24 via 172.16.26.2 (W): binary AND with the mask.

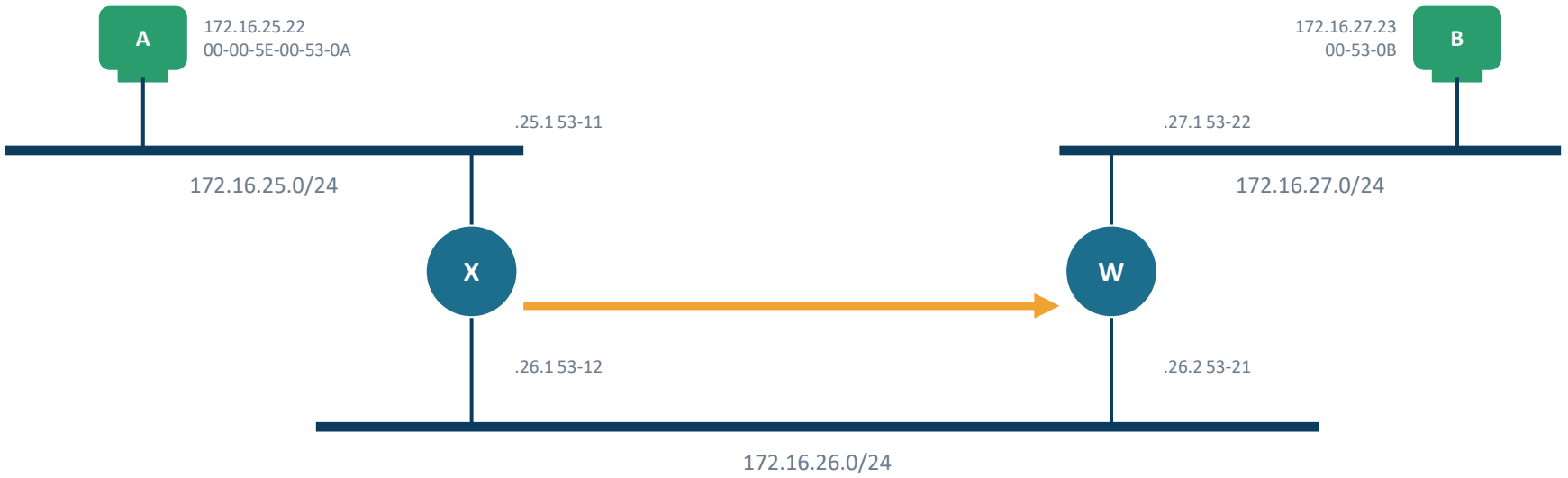
Step 3: X forwards the packet to the next hop (W)



- X knows the next hop for 172.16.27.0 is W (routing protocol or static)
- X ARPs for 172.16.26.2 on LAN 26 (if needed)
- New frame X → W; same packet A → B; **TTL 63**

Ethernet frame	Preamble	**Destination MAC: W 53-21**	Source MAC: X 53-12	0x0800	IP packet	FCS
IP packet	**TTL = 63**	Proto 6	Source IP: 172.16.25.22 (A)	**Destination IP: 172.16.27.23 (B)**	TCP: port 22 + data	

Step 4: W receives the packet



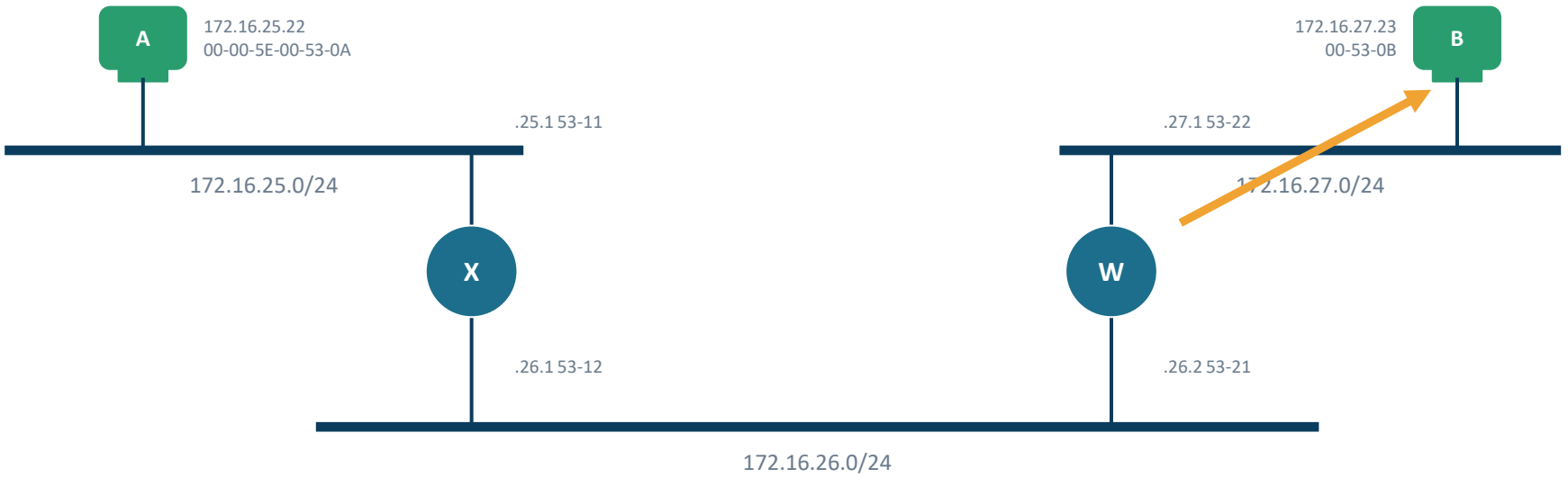
Destination	Next hop
172.16.26.0/24	connected
172.16.27.0/24	**connected (E1)**
172.16.25.0/24	172.16.26.1 (X)
172.16.28.0/24	172.16.27.9 (Z)

Route table of W

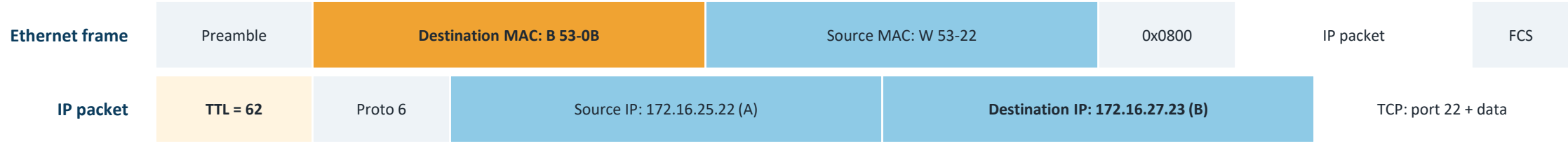
1 **Accepts the frame**
Destination MAC = W's interface on LAN 26; discards the Ethernet header.

2 **Routes**
Destination 172.16.27.23 matches 172.16.27.0/24, directly connected: the next hop is B itself.

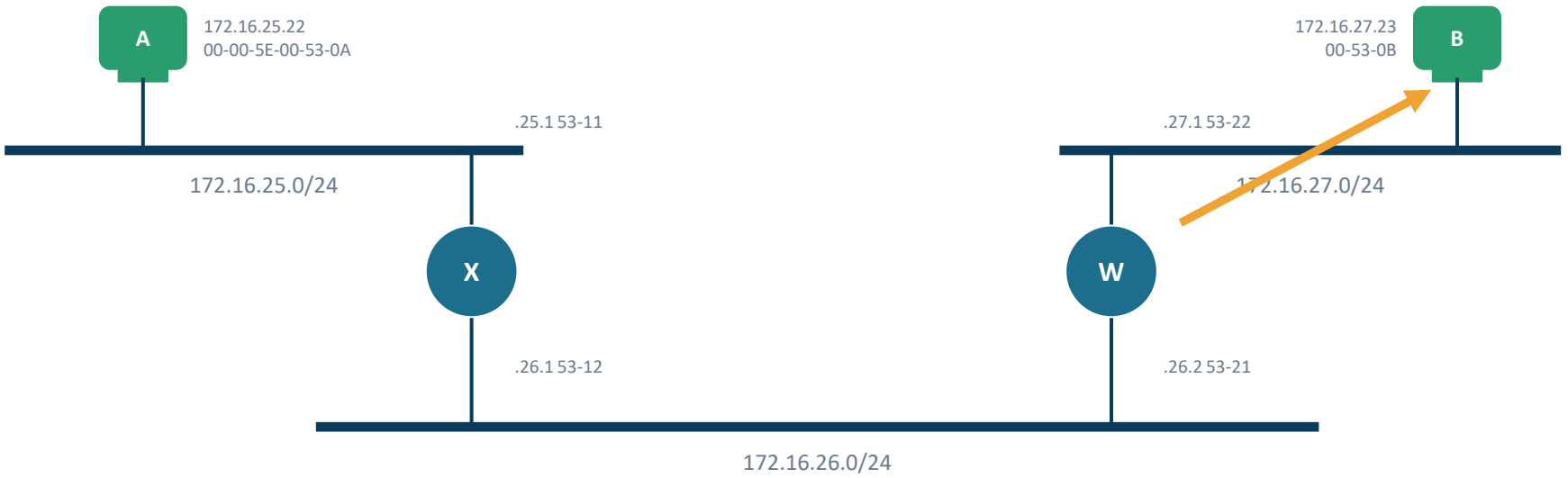
Step 5: W delivers the packet to B



- Destination on a connected network: W needs B's MAC
- ARP Request from W on LAN 27 for 172.16.27.23 (or cache)
- Frame W → B; packet A → B; **TTL 62**



Step 6: B receives the packet



Tally

- 3 links, 2 routers, 3 different frames
- 1 IP packet (addresses intact), TTL 64 → 62
- Up to 6 ARP resolutions, all cached

1 **Accepts the frame**
Destination MAC = B; copies the packet to the buffer.

2 **Network layer**
Destination IP = its own; Protocol 6 → delivers to TCP.

3 **Transport and application**
Port 22 → SSH server. The reply retraces the path in reverse (B → W → X → A).

UPDATED

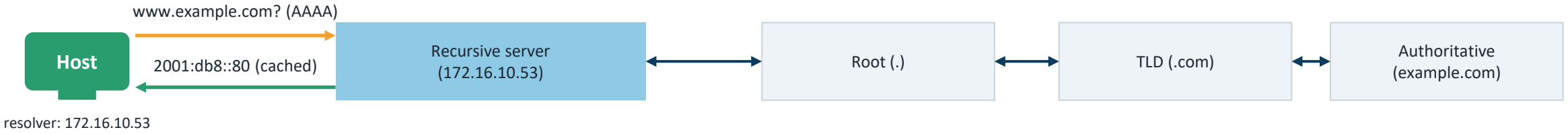
Host names: from the hosts file to DNS

1999: local tables

- Router: ip host name address
- Hosts: hosts file
- Local scope; no scale

2026: DNS (RFC 1034/1035 and successors)

- Hierarchical namespace: root → TLD (.com, .br) → domain → host
- Host resolver (DHCP/RA) → recursive server → authoritative servers
- Records: **A** (IPv4), **AAAA** (IPv6), **PTR** (reverse), CNAME, MX, SRV, TXT
- UDP/TCP 53; DNS over TLS (853) and HTTPS; DNSSEC



Typical precedence on the host: local cache → hosts file → DNS. The hosts file is still handy for testing (and causes trouble when forgotten).

Proxy ARP configuration: vendor defaults

Platform	Default	How to enable / disable	Note
Cisco IOS / IOS-XE	**Enabled**	interface X / [no] ip proxy-arp	Also 'ip local-proxy-arp' for hosts on the same VLAN
Linux	Disabled	sysctl -w net.ipv4.conf.eth0.proxy_arp=1	Used on VPN concentrators and virtualization
Juniper Junos	Disabled	set interfaces ge-0/0/0 unit 0 proxy-arp restricted | unrestricted	'restricted' answers only for hosts on other subnets
Cabletron (1999)	Disabled	IP General Config → Port Level Setup	Legacy

```
! Cisco IOS: disable Proxy ARP on the interface
configure terminal
 interface GigabitEthernet0/0
  no ip proxy-arp
! verify:
show ip interface GigabitEthernet0/0 | include Proxy
```

```
# Linux: enable Proxy ARP on eth0 (temporary)
sysctl -w net.ipv4.conf.eth0.proxy_arp=1
# persistent: /etc/sysctl.d/99-proxyarp.conf
net.ipv4.conf.eth0.proxy_arp = 1
# check neighbors:
ip neigh show dev eth0
```

Recommendation: Proxy ARP off by default; enable it only with a documented need (VPN, migration) and per interface.

MODULE 2 | PART 4

IP routing protocols

RIP, administrative distance, EIGRP, multipath routing and redistribution; OSPF in detail; OSPFv3 and IS-IS; EGP and BGP-4

RIP: Routing Information Protocol

- One of the **historically** most used interior protocols; a direct implementation of distance vector
- Advertises the **whole table every 30 s**: v1 in broadcast, v2 in multicast 224.0.0.9 (UDP 520)
- Metric: **hop count**; 16 = infinity → maximum diameter of 15 hops
- Timers (Cisco): update 30 s, invalid 180 s, hold-down 180 s, flush 240 s
- Loops mitigated by split horizon, poison reverse, hold-down and triggered updates
- 2026: suitable only for small networks; great for teaching

Version	RFC	Characteristics
RIPv1 (1988)	1058	Classful: no mask; broadcast; no authentication
RIPv2 (1998)	2453	Classless (VLSM/CIDR); multicast 224.0.0.9; authentication; next hop; tag
RIPng (1997)	2080	IPv6; UDP 521; ff02::9; authentication via IPsec

Table update criteria

- Unknown destination → install
- Metric lower than the current one → replace
- Same outgoing neighbor with a different metric → accept (even if worse)
- Route not advertised for 180 s → invalid; removed at 240 s

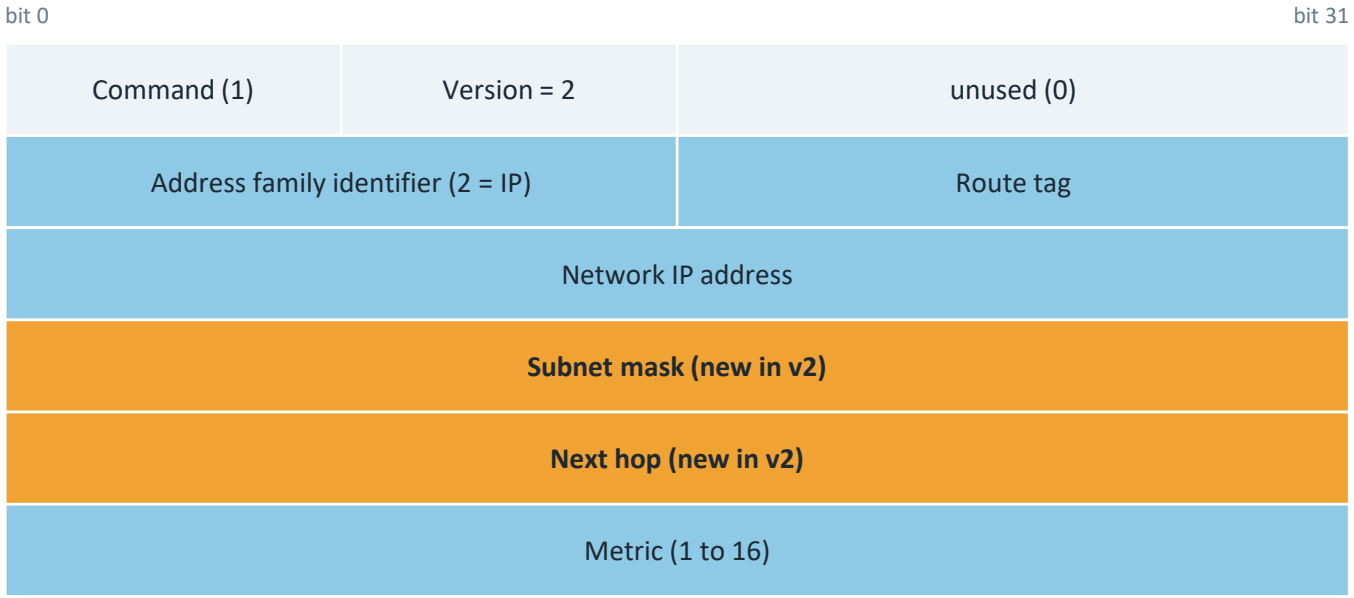
LEGACY

RIP: origin and variants

Protocol	Family	Relation to RIP	Status
XNS RIP	Xerox Network Systems (1970s)	Origin of the algorithm and the format	Extinct
routed (BSD UNIX 4.2, 1983)	TCP/IP	The implementation that popularized RIP on IP before RFC 1058	Evolved into RIPv2 / RIPng
IPX RIP	Novell NetWare	Very similar to Xerox RIP; metric in ticks and hops	Extinct with IPX
RTMP	AppleTalk	Modified version of RIP	Extinct (Apple moved to IP in 2009)
RTP	Banyan VINES	Slight variation of Xerox RIP	Extinct

Lesson: a routing algorithm (Bellman-Ford) is independent of the network protocol. The same holds today: OSPF and IS-IS implement Dijkstra for IPv4 and IPv6; BGP carries IPv4, IPv6, VPNs and EVPN with the same path-vector mechanism.

RIPv2: message format and what is new



4-byte header + up to 25 entries of 20 bytes (504 bytes per message)

RIPng (RFC 2080) is RIPv2 for IPv6: 128-bit prefixes with length, UDP 521, group ff02::9, no authentication of its own (uses IPsec). Same limitations of 15 hops and 30 s.

What version 2 added (RFC 2453)

- **Subnet mask** per route: VLSM and CIDR
- **Next hop:** avoids an extra hop when several routers share the link
- **Route tag:** marks external/redistributed routes
- **Authentication:** simple password; MD5 (RFC 2082); SHA (RFC 4822)
- **Multicast 224.0.0.9** instead of broadcast: hosts without RIP are not interrupted
- Keeps: UDP 520, 30 s, 15 hops, Bellman-Ford

Administrative distance: choosing among route sources

Route source	AD (Cisco)
Connected interface	0
Static route	1
EIGRP summary	5
External BGP (eBGP)	20
Internal EIGRP	90
IGRP	100 (legacy)
OSPF	110
IS-IS	115
RIP	120
EGP	140 (legacy)
External EIGRP	170
Internal BGP (iBGP)	200
Unknown / unusable	255

Selection rule

- 1st: longest prefix (always wins, regardless of source)
- 2nd: among equal routes, the **lowest AD** enters the table
- 3rd: among routes from the same source, the **lowest metric**; tie → ECMP

Practical use: floating static

- ip route 0.0.0.0 0.0.0.0 203.0.113.1 250
- Only installed if the BGP-learned default route (AD 20) disappears: automatic backup
- Junos uses 'preference' with its own scale; the concept is the same

UPDATED

From IGRP to EIGRP

IGRP (Cisco, 1980s): legacy

- Created to overcome RIP's 15-hop limit (IGRP: up to 255)
- Distance vector with a **composite metric**: delay, bandwidth, reliability and load, combined with adjustable weights
- Multipath (up to 4 paths)
- Classful (no mask in updates); advertisements every 90 s
- Proprietary; removed from IOS in 2005

EIGRP (Cisco, 1992; RFC 7868, 2016)

- **Advanced** distance vector: the **DUAL** algorithm guarantees loop-free routes and subsecond convergence, with no hold-down
- IGRP's composite metric × 256; K1 and K3 (bandwidth and delay) by default
- Only **triggered, reliable** updates (RTP); adjacencies by Hello (IP protocol 88, multicast 224.0.0.10)
- Classless (VLSM/CIDR), summarization on any interface, IPv6 (ff02::a), unequal-cost multipath (variance)
- AD 90 (internal) and 170 (external); 'open' through RFC 7868, but still essentially Cisco

The EIGRP composite metric

Full formula (K1 to K5 configurable)

$$\text{metric} = 256 \times \left[K1 \times B + \frac{K2 \times B}{256 - L} + K3 \times D \right] \times \left[\frac{K5}{R + K4} \right] \quad \text{(the last factor only applies if } K5 \neq 0\text{)}$$

With the default weights (K1 = K3 = 1, K2 = K4 = K5 = 0):

$$\text{metric} = 256 \times \left(\frac{10{,}000{,}000}{\text{minimum_bandwidth[kbps]}} + \Sigma\, \text{delay}[\mu s] / 10 \right)$$

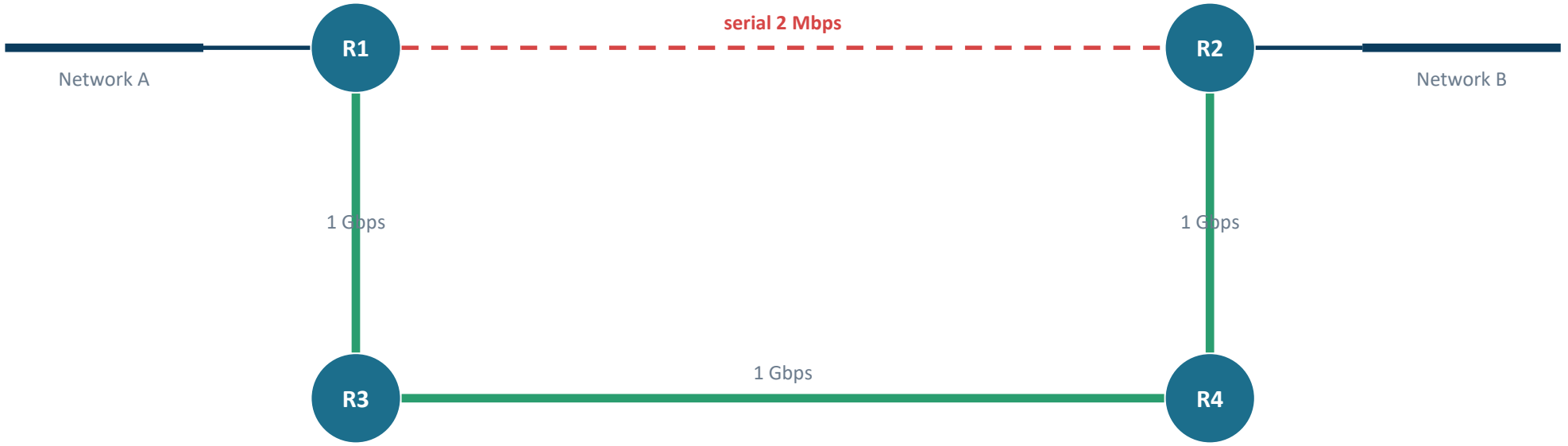
Example: three 1 Gbps links

- Minimum bandwidth 1,000,000 kbps → 10^7 / 10^6 = 10
- Delay: 3 × 10 µs = 30 µs → 30 / 10 = 3
- **metric = 256 × (10 + 3) = 3,328**

Example: one 2 Mbps serial link

- Bandwidth 2,000 kbps → 10^7 / 2,000 = 5,000
- Default serial delay: 20,000 µs → 2,000
- **metric = 256 × (5,000 + 2,000) = 1,792,000**

Example: the same topology with RIP and with EIGRP



Protocol	Path chosen from A to B	Metric	Alternative
RIP	R1 → R2 (direct, 2 Mbps)	1 hop	R1 → R3 → R4 → R2 = 3 hops (worse for RIP)
EIGRP	R1 → R3 → R4 → R2 (1 Gbps)	3,328	R1 → R2 = 1,792,000 (much worse)

OSPF?

Cost = reference bandwidth / bandwidth; also picks the 1 Gbps path

Loop detection: from IGRP's hold-down to EIGRP's DUAL

IGRP (legacy)

- Split horizon and triggered updates, without poison reverse
- Early versions: hold-down
- Later versions: 'route poisoning': a path with an increasing metric is treated as unusable until confirmed by the following update

EIGRP: the DUAL algorithm

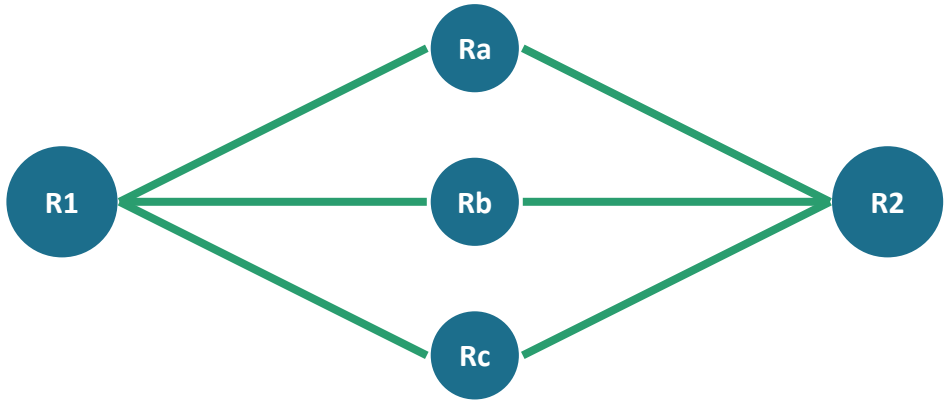
- **FD** (feasible distance): the lowest known metric to the destination
- **RD** (reported distance): the metric advertised by each neighbor
- **Feasibility condition:** RD < FD ⇒ the neighbor is a **feasible successor** (a guaranteed loop-free path)
- Successor fails and a feasible successor exists → local switch, no recomputation and no hold-down

When there is no feasible successor

- The route goes **active**: queries to the neighbors, which reply with their best metric
- Converges when every reply arrives; 'stuck in active' (SIA) if a neighbor is slow: 3-minute limit, mitigated by summarization and stub

Multipath routing (ECMP and unequal cost)

- Routers keep **several paths** per destination, not only the best one
- Two advantages: **immediate switchover** to the alternate path on failure and **load sharing** across equal-cost paths
- **ECMP** in every modern IGP and in BGP: 4 to 64 paths; distribution by **hash of the 5-tuple** (a flow is never reordered), RFC 2991/2992
- **EIGRP variance N:** also uses paths up to N times worse, if they are feasible successors (unequal cost)
- Data center leaf-spine fabrics: dozens of ECMP paths over BGP



Three equal-cost paths: ECMP distributes the flows by hash

Flow (5-tuple)	Hash	Path
10.1.1.5:40001 → 10.2.2.9:443	… 0	via Ra
10.1.1.5:40002 → 10.2.2.9:443	… 1	via Rb
10.1.1.7:51000 → 10.2.2.9:22	… 2	via Rc

Each flow always follows the same path: no TCP segment reordering.

Route redistribution (1 of 3): two routing domains

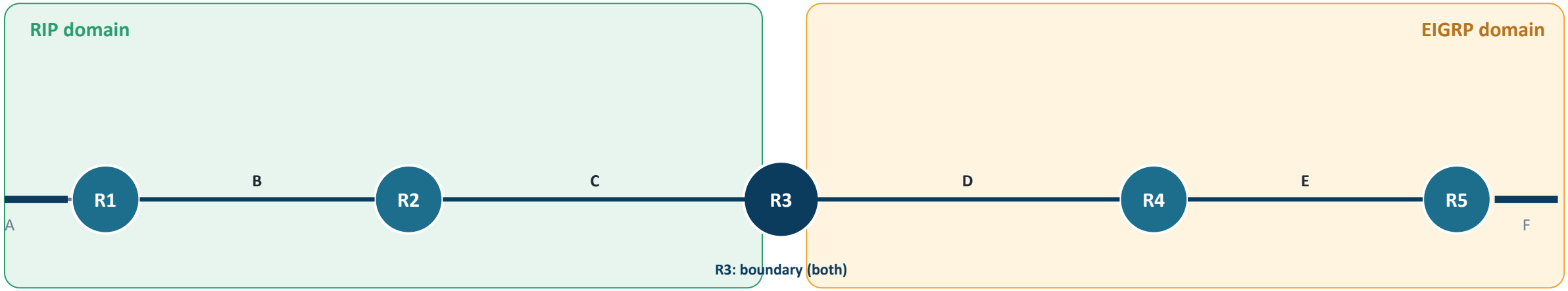


Table of R1 (RIP)

Net	Source	Metric
A	C	0
B	C	0
C	R	1
D, E, F	unknown	-

Table of R3 (RIP + EIGRP)

Net	Source	Metric
C	C	0
D	C	0
B	R	1
A	R	2
E	D	3,072
F	D	3,328

Table of R5 (EIGRP)

Net	Source	Metric
E	C	0
F	C	0
D	D	3,072
A, B, C	unknown	-

Without redistribution, only R3 (which runs both protocols) knows every network. The others reach only their own domain.

Route redistribution (2 of 3): EIGRP → RIP

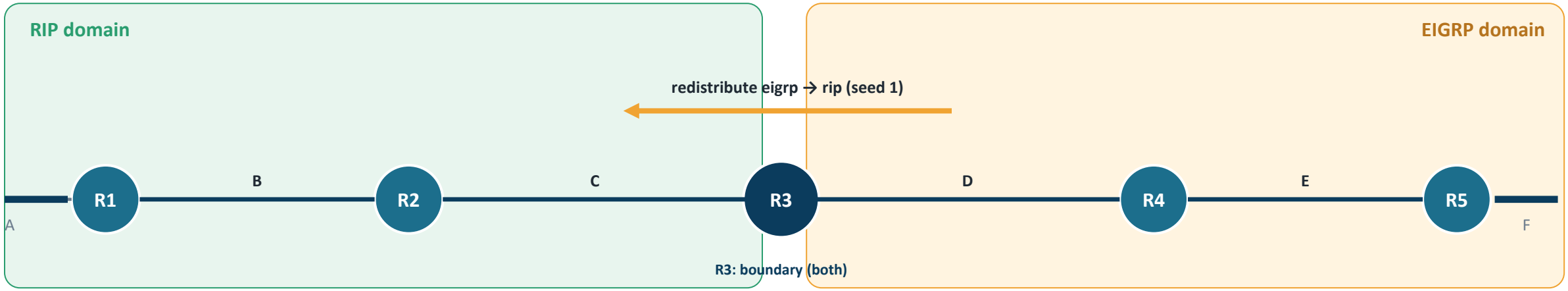


Table of R1 (RIP)

Net	Source	Metric
A	C	0
B	C	0
C	R	1
D	R (redistr.)	3
E	R (redistr.)	3
F	R (redistr.)	3

Table of R3

Net	Source	Metric
C	C	0
D	C	0
B	R	1
A	R	2
E	D	3,072
F	D	3,328

Table of R5 (EIGRP)

Net	Source	Metric
E	C	0
F	C	0
D	D	3,072
A, B, C	still unknown	-

```
router rip
 redistribute eigrp 100 metric 1
```

Seed metric is mandatory: the EIGRP routes enter RIP with 1 hop (R2 sees 2, R1 sees 3). Traffic from A to F leaves R1 toward R3.

Route redistribution (3 of 3): both directions and the risks

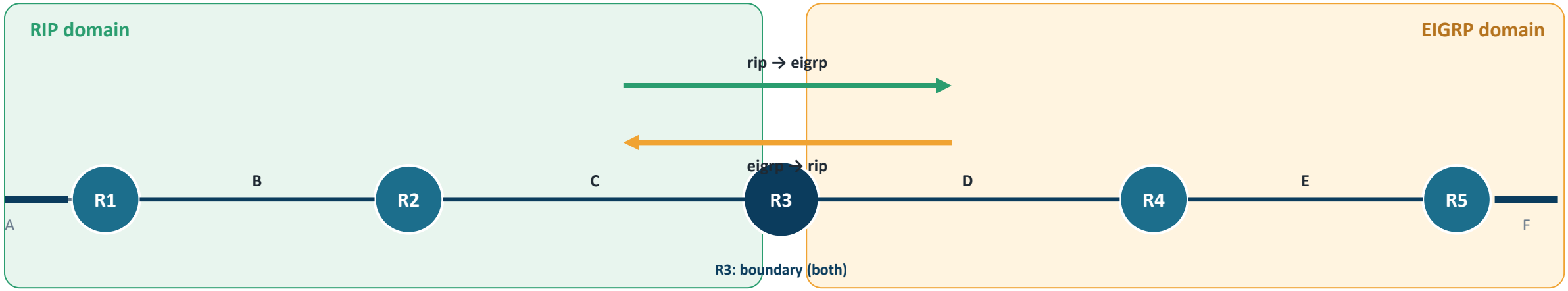


Table of R1 (RIP)

Net	Source	Metric
A	C	0
B	C	0
C	R	1
D	R	3
E	R	3
F	R	3

Table of R5 (EIGRP)

Net	Source	Metric
E	C	0
F	C	0
D	D	3,072
C	D EX (AD 170)	2,816,000
B	D EX	2,816,000
A	D EX	2,816,000

```
router eigrp 100
 redistribute rip metric 10000 100 255 1 1500 route-map RIP-to-EIGRP
```

Risks and defenses

- **Feedback:** a route returns to its domain of origin with a better metric → loop or suboptimal path
- **Defenses:** route tags + route-maps; prefix filters; AD (external 170 > RIP 120); one-way redistribution + default route

OSPF: Open Shortest Path First

- **Open:** public IETF specification (OSPF working group, late 1980s)
- **Shortest Path First:** Dijkstra's algorithm over a link-state database
- Created because RIP could not serve large, heterogeneous internetworks: 15 hops, slow convergence, no VLSM
- Versions: OSPFv1 (RFC 1131, 1989); OSPFv2 (RFC 1247, 1991 → **RFC 2328**, 1998) for IPv4; **OSPFv3** (RFC 5340, 2008) for IPv6 and, with address families (RFC 5838), also IPv4
- Runs directly over IP (protocol 89); multicast 224.0.0.5 (all OSPF routers) and 224.0.0.6 (DR/BDR); ff02::5 and ff02::6 in v3

Interface cost

- cost = reference bandwidth / interface bandwidth
- Default: 10^8 (100 Mbps) → 1 Gbps and 100 Gbps would both cost 1!
- Set 'auto-cost reference-bandwidth' (e.g., 100,000 Mbps) on every router

Why OSPF in 2026

- Multivendor, mature, scalable with areas
- Subsecond convergence with BFD and tuned timers
- Alternative in providers and data centers: IS-IS

OSPF: protocol overview (1 of 2)

1

Topology database (LSDB)

Every router maintains a database describing the topology of the area: every router, link and cost, in the form of LSAs.

2

The same algorithm everywhere

Every router runs exactly the same SPF (Dijkstra), in parallel, over the same LSDB; each computes the shortest-path tree with itself as the root.

3

Routes per type of service (legacy)

RFC 1247 allowed separate routes per ToS; RFC 2328 removed the feature. Today differentiation is done by QoS and traffic engineering, not by the IGP.

OSPF packet types (IP protocol 89)

- **1 Hello:** neighbor discovery and maintenance
- **2 Database Description:** LSDB summary
- **3 Link State Request:** asks for missing LSAs
- **4 Link State Update:** carries LSAs (flooding)
- **5 Link State Ack:** acknowledges receipt
- Reliable exchange: every LSA is acknowledged

OSPF: protocol overview (2 of 2)

Areas

- Sets of networks grouped together; an area's topology is hidden from the others
- Less routing traffic, smaller LSDB, faster SPF
- Area 0 (backbone) interconnects all the others

Flexible subnets (VLSM / CIDR)

- Every advertised route carries its mask
- Summarization configurable at area (ABR) and AS (ASBR) boundaries

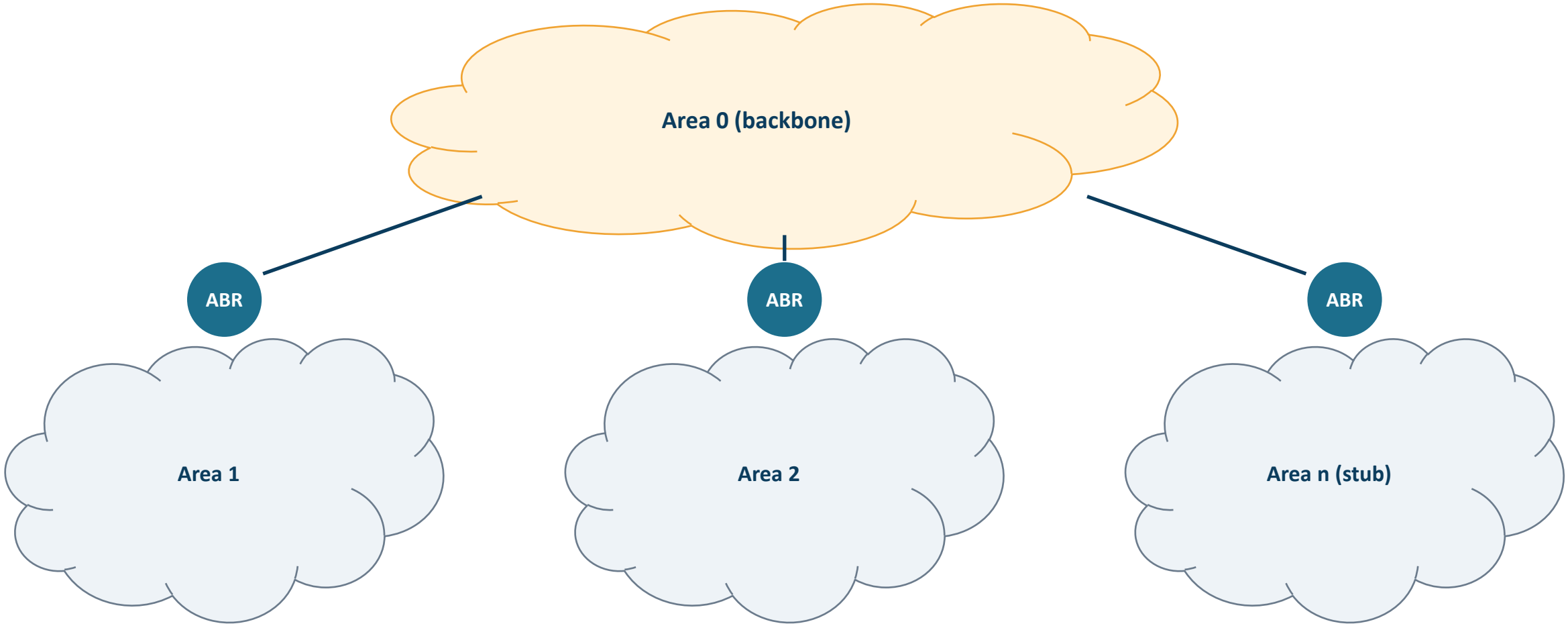
Authentication

- OSPFv2: null, simple password or cryptographic (MD5, RFC 2328; HMAC-SHA, RFC 5709)
- OSPFv3: IPsec (RFC 4552) or Authentication Trailer (RFC 7166)

External routes

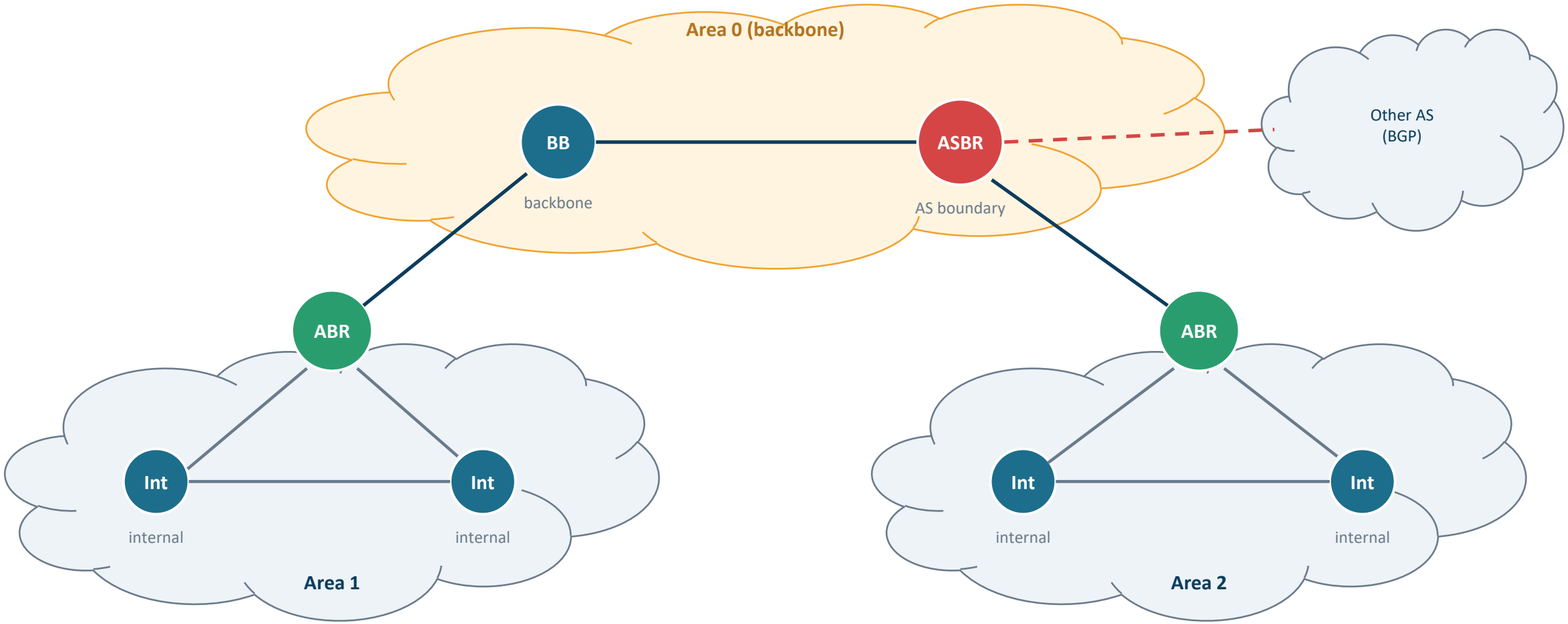
- Routes from another AS or another protocol (redistribution) cross the AS transparently
- Marked as external (E1/E2), with a route tag

OSPF topology: backbone and areas



- Every area connects to **area 0** through area border routers (ABRs); inter-area traffic always crosses the backbone
- Types: normal, **stub** (no externals), **totally stubby**, **NSSA** (a stub that injects externals, type 7 LSA)
- An area that cannot reach the backbone: virtual link (temporary fix)

OSPF router classification



Internal	Backbone	ABR (Area Border Router)	ASBR (AS Boundary Router)
All interfaces in a single area	At least one interface in area 0	Interfaces in more than one area; one LSDB per area; summarizes	Injects external routes (redistribution, BGP)

Network types supported by OSPF

Point-to-point

- Two routers; direct adjacency, no DR/BDR
- Hello 10 s, dead 40 s; multicast 224.0.0.5
- Also recommended on router-to-router Ethernet links (ip ospf network point-to-point)

Broadcast multi-access (Ethernet)

- Automatic neighbor discovery by multicast
- Elects DR and BDR; Hello 10 s, dead 40 s
- LANs, VLANs, switches between routers

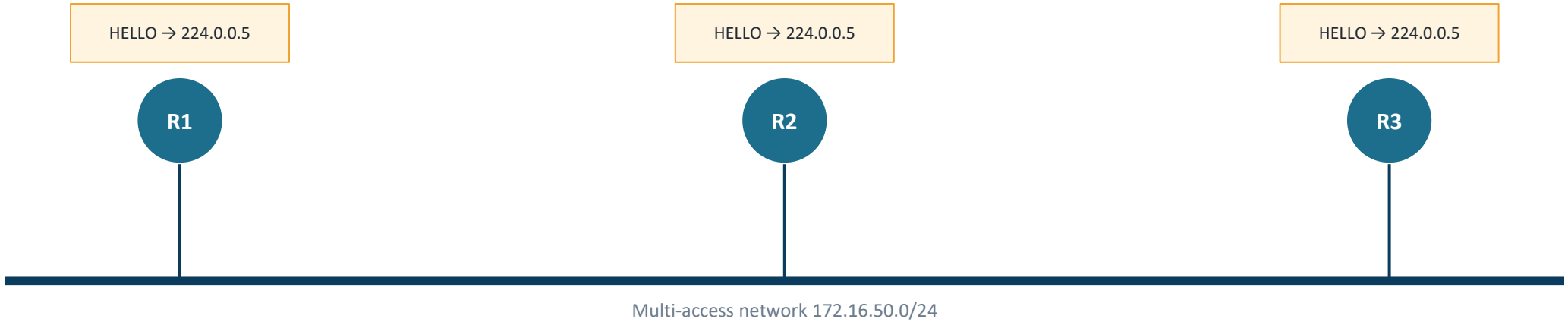
Non-broadcast multi-access (NBMA)

- Frame Relay, ATM, X.25 in 1999 (legacy)
- Neighbors configured manually; DR elected; Hello 30 s, dead 120 s
- Today: very rare (some satellite networks and multipoint tunnels)

Point-to-multipoint

- Treats a cloud as a collection of point-to-point links; no DR
- Hello 30 s by default; useful on hub-and-spoke tunnels (DMVPN)
- 'non-broadcast' variant with manual neighbors

The Hello protocol



Hello field	Must match?
Area ID	Yes
Hello / dead intervals	Yes
Authentication (type and key)	Yes
Network mask (broadcast)	Yes
Stub / NSSA flags	Yes
Priority, router ID, DR/BDR	No (used in the election)

Timers

- Broadcast and p2p: Hello 10 s, dead 40 s
- NBMA and point-to-multipoint: Hello 30 s, dead 120 s
- Fast detection: BFD (RFC 5880) in tens of ms, without touching the timers
- On multi-access, the Hello also elects DR and BDR

Neighbors and adjacencies

1. **Down → Init**
   I received a Hello from the neighbor, but it has not listed my router ID yet.

2. **2-Way**
   Bidirectional communication confirmed. On multi-access, DROthers stop here with each other; the DR/BDR election happens at this point.

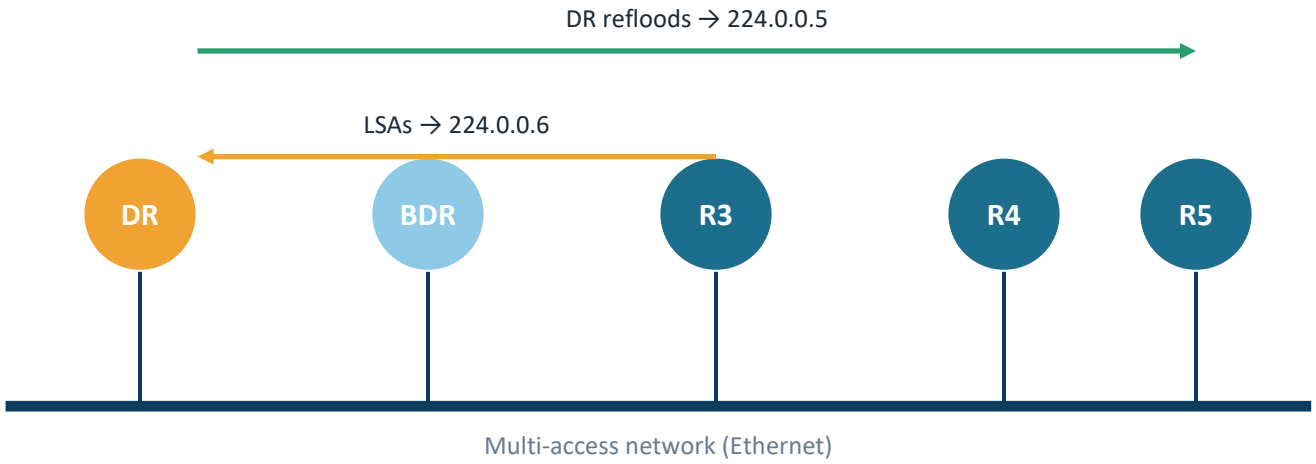
3. **ExStart → Exchange**
   Master/slave negotiation and exchange of Database Description (LSDB summary); MTU check.

4. **Loading → Full**
   Link State Requests for the missing LSAs; when done, the LSDBs are synchronized: full adjacency.

Who becomes adjacent to whom

- **Point-to-point and virtual links:** both routers, always
- **Multi-access:** everyone with the DR and with the BDR; DROthers stay in 2-Way with each other
- Adjacencies control flooding: LSAs flow only over them
- 'show ip ospf neighbor': FULL is the goal; 2-WAY/DROTHER is normal

Designated router (DR) and backup (BDR)



Routers (n)	Without DR: n(n-1)/2	With DR and BDR: 2(n-2)+1
5	10	7
10	45	17
20	190	37

Rules

- Election: highest interface **priority** (0 = never a DR); tie → highest **router ID**
- **Not preemptive:** a new router does not displace the current DR
- The DR generates the **network LSA (type 2)** of the multi-access network
- The BDR takes over if the DR fails; a new election only for the BDR
- Tip: pin the DR with a high priority on the most capable router; avoid DRs on access routers

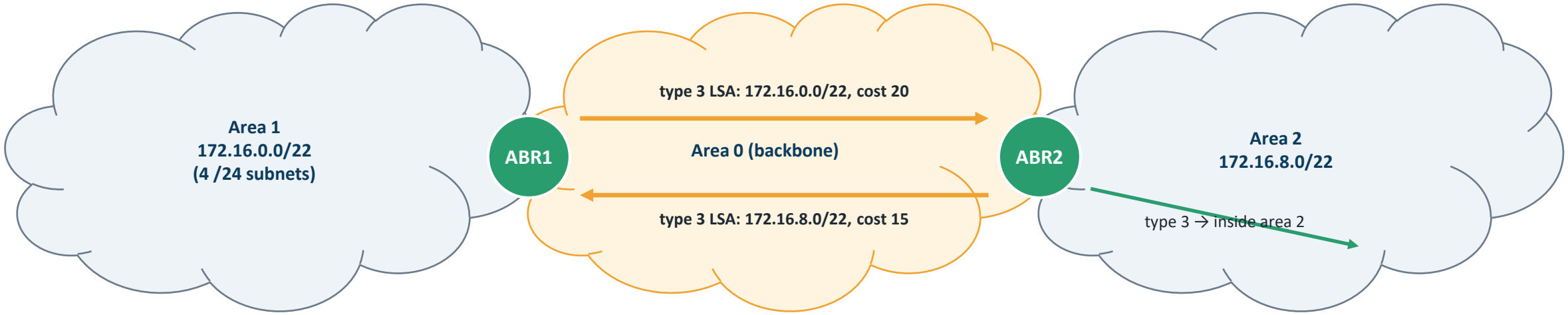
Link-state advertisements (LSAs) and the LSDB

Type	Name	Generated by	Scope	Contents
1	Router	Every router	Area	The router's links and costs
2	Network	DR	Area	Routers on the multi-access network
3	Summary (network)	ABR	Area	Prefixes of other areas
4	Summary (ASBR)	ABR	Area	How to reach the ASBR
5	AS External	ASBR	AS (except stub)	External prefixes (E1/E2)
7	NSSA External	ASBR in NSSA	NSSA area	Externals; the ABR converts them to type 5
8 / 9	Link / Intra-Area-Prefix	OSPFv3	Link / area	Link-local addresses and prefixes (v3)

Life cycle of an LSA

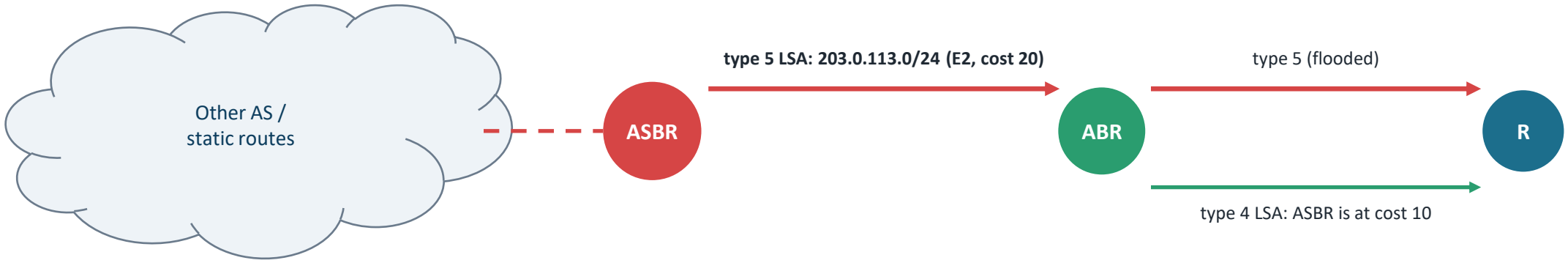
- **Flooding** throughout the area over adjacencies; every LSA acknowledged (Ack)
- **Sequence number:** the highest is the most recent
- **LS Age:** re-sent every 30 min (LSRefreshTime); discarded at 60 min (MaxAge)
- Identical LSDB on every router in the area → each runs **SPF** with itself as the root → intra-area routing table
- 'show ip ospf database' lists the LSAs by type

Inter-area routing



- Each **ABR summarizes** its areas in type 3 LSAs (prefixes and costs only) and sends them over the backbone to the other ABRs
- With that, every ABR computes paths to destinations outside its areas and **advertises them inside its areas**
- Inter-area computation = distance vector over the backbone → every area must touch area 0
- **area range** (manual summarization) reduces type 3 LSAs; **stub / totally stubby** areas receive only a default route from the ABR

AS external routes



Type	Metric	Use
E2 (default)	External cost only (the same across the AS); internal cost only breaks ties	'Far away' routes, when the internal cost is irrelevant
E1	External cost + internal cost to the ASBR	Several ASBRs advertising the same prefix: optimal choice

Rules

- The ASBR generates **type 5**; flooded throughout the AS, except stub areas
- The ABR generates **type 4** to say where the ASBR is
- In an NSSA: **type 7**, translated to 5 by the ABR
- Redistribution into OSPF: 'redistribute bgp 65000 subnets metric-type 1'

UPDATED

OSPFv3 and IS-IS: the link-state IGPs in 2026

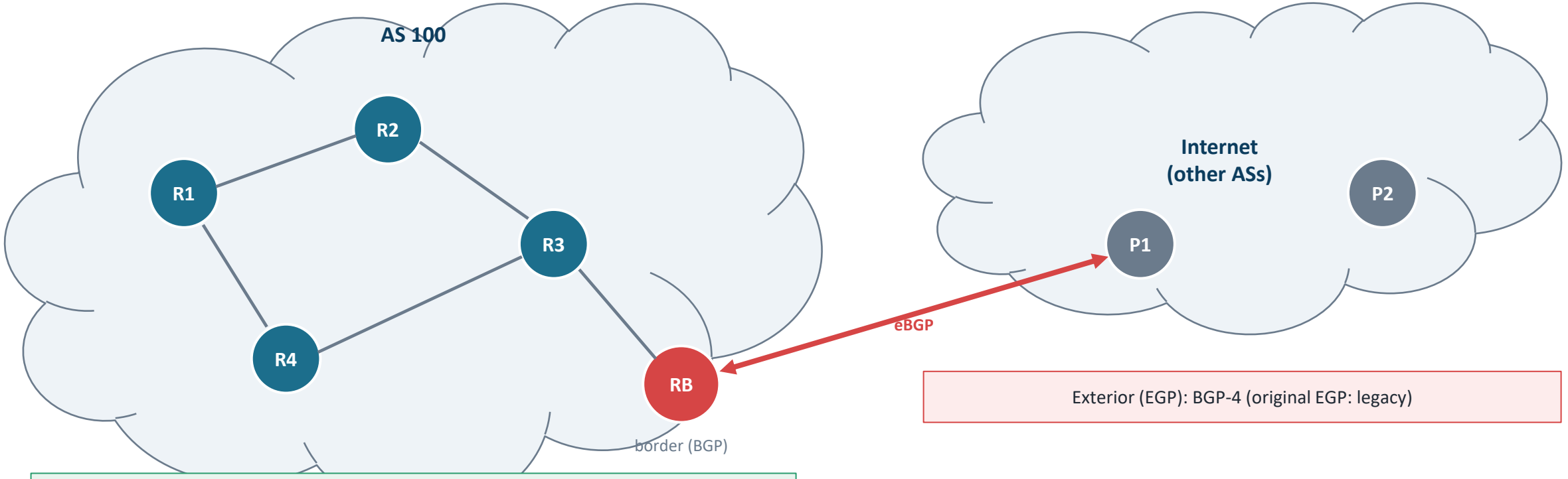
OSPFv3 (RFC 5340, 2008)

- Runs **per link**, with a link-local next hop (fe80::/10); transport ff02::5 and ff02::6
- Type 1 and 2 LSAs carry topology only; prefixes go in type 8 (link) and type 9 (intra-area prefix) LSAs
- Several instances per link (Instance ID)
- Authentication through IPsec (RFC 4552) or the Authentication Trailer (RFC 7166)
- **Address families (RFC 5838):** IPv4 and IPv6 in a single protocol, separate instances per AF

IS-IS (ISO 10589; RFC 1195; RFC 5308)

- Same algorithm (Dijkstra), encapsulated **directly at Layer 2**: it does not need IP to form adjacencies
- Two levels: **L1** (intra-area) and **L2** (backbone); L1/L2 routers form the boundary
- Identification by NSAP/NET; extensible through **TLVs**: IPv4, IPv6, multi-topology, TE, Segment Routing
- Preferred by large providers and in fabrics; base of **SR-MPLS and SRv6**
- Same concepts of LSP (the LSA equivalent), flooding, DIS (the DR equivalent, preemptive)

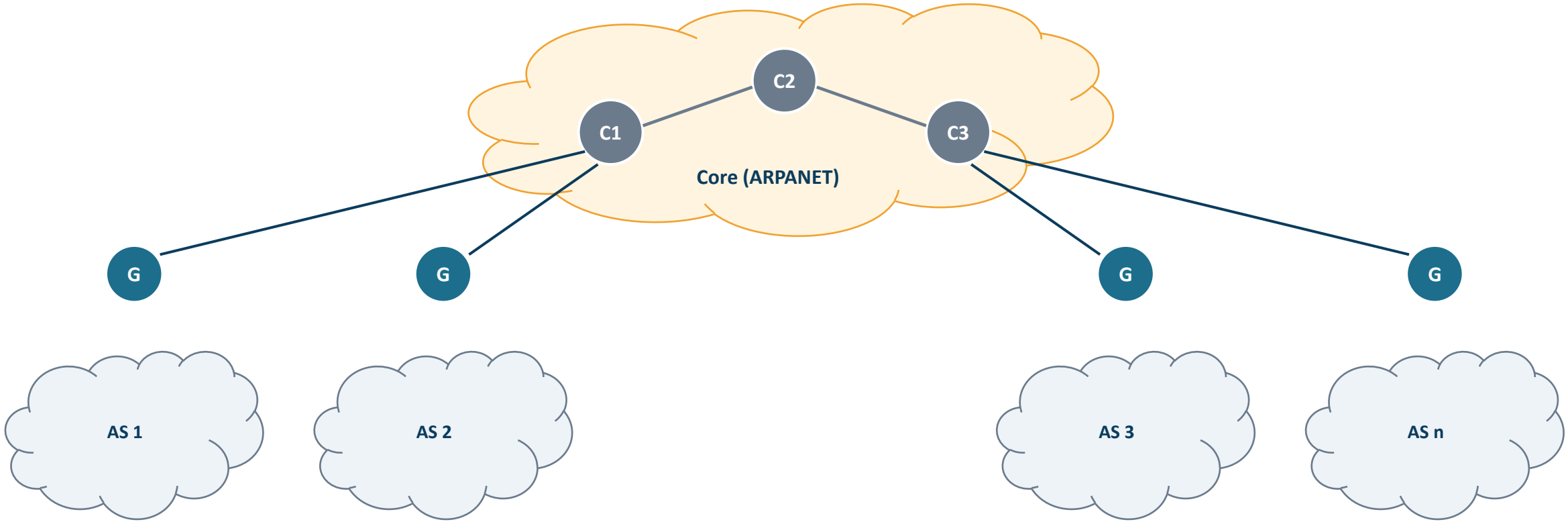
Interior and exterior routing protocols



	Interior (IGP)	Exterior (BGP)
Goal	Best path by metric inside the AS	Reachability between ASs according to business policies
Scale	Hundreds to thousands of routers	The whole Internet: ≈ 1 million IPv4 prefixes, > 200,000 IPv6

LEGACY

Exterior Gateway Protocol (EGP): the predecessor

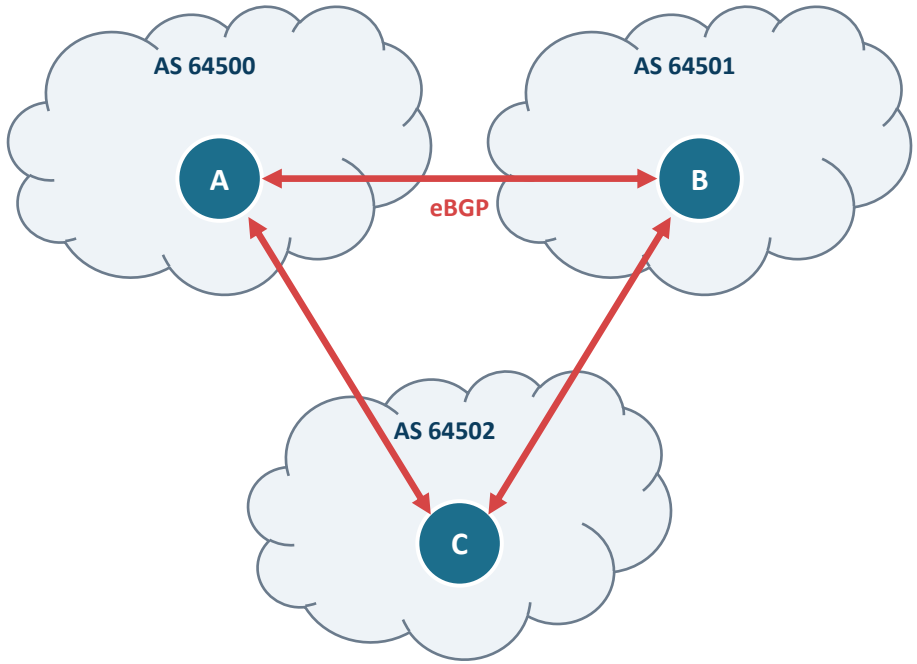


- Designed for a **very specific topology**: a core system (ARPANET) with the ASs hanging off it, each AS through **a single router**
- No multiple paths from an AS to the core; **no loop detection** (the topology had to be a tree); reachability only, no metric
- Unworkable once the Internet lost its single core (NSFNET and commercial providers); replaced by BGP by the mid-1990s

UPDATED

BGP-4: the routing protocol of the Internet

- Created to overcome EGP's limitations as the Internet grew; **BGP-4** since 1994 (RFC 1654 → RFC 1771 → **RFC 4271**, 2006)
- **Detects loops:** the AS_PATH lists the ASs traversed; a route that already contains my AS is rejected
- Policy-oriented **path vector**; explicitly configured **TCP 179** sessions (no automatic discovery)
- **eBGP** between different ASs (AD 20); **iBGP** inside the AS (AD 200), in full mesh or with **route reflectors** (RFC 4456)
- **MP-BGP** (RFC 4760): IPv6, VPNv4/VPNv6 (MPLS L3VPN), EVPN; **32-bit ASNs** (RFC 6793)
- Also inside data centers: BGP as the IGP of leaf-spine fabrics (RFC 7938)



A advertises 203.0.113.0/24 with AS_PATH 64500; B re-advertises it to C with AS_PATH 64501 64500; C would never accept back a route with 64502 in the path.

Global table 2026: ≈ 1 million IPv4 prefixes; > 200,000 IPv6 (bgp.potaroo.net).

BGP-4: attributes and best-path selection

Attribute	Category	Purpose
ORIGIN	Mandatory	How the route entered BGP (IGP, EGP, incomplete)
AS_PATH	Mandatory	List of ASs; loop detection; policy by prepending
NEXT_HOP	Mandatory	Next-hop IP (eBGP: the peer; iBGP: unchanged)
LOCAL_PREF	Discretionary	Exit preference inside the AS (highest wins)
MED	Optional non-trans.	Suggests an entry point to the neighbor (lowest wins)
COMMUNITY	Optional transitive	Tags for policy (RFC 1997, 8092)
AGGREGATOR / ATOMIC_AGG.	Optional / discr.	Route aggregation information

Selection order (Cisco)

- 1. Highest **weight** (local, Cisco) 2. Highest **LOCAL_PREF**
- 3. Locally originated route 4. Shortest **AS_PATH**
- 5. Lowest ORIGIN (IGP < EGP < incomplete) 6. Lowest **MED**
- 7. eBGP over iBGP 8. Lowest IGP metric to the NEXT_HOP
- 9. Oldest route 10. Lowest router ID / peer address

Routing security

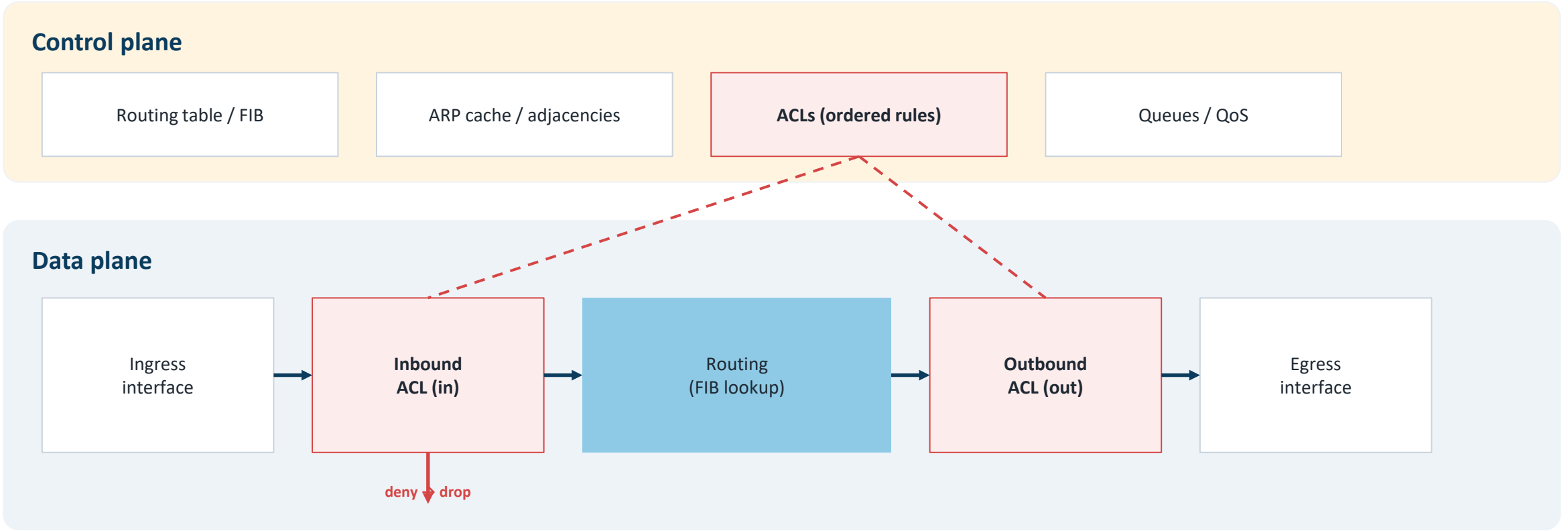
Prefix and AS_PATH filters, per-peer prefix limits, RPKI/ROV (RFC 6811), MANRS.

MODULE 2 | PART 5

Controlling IP access

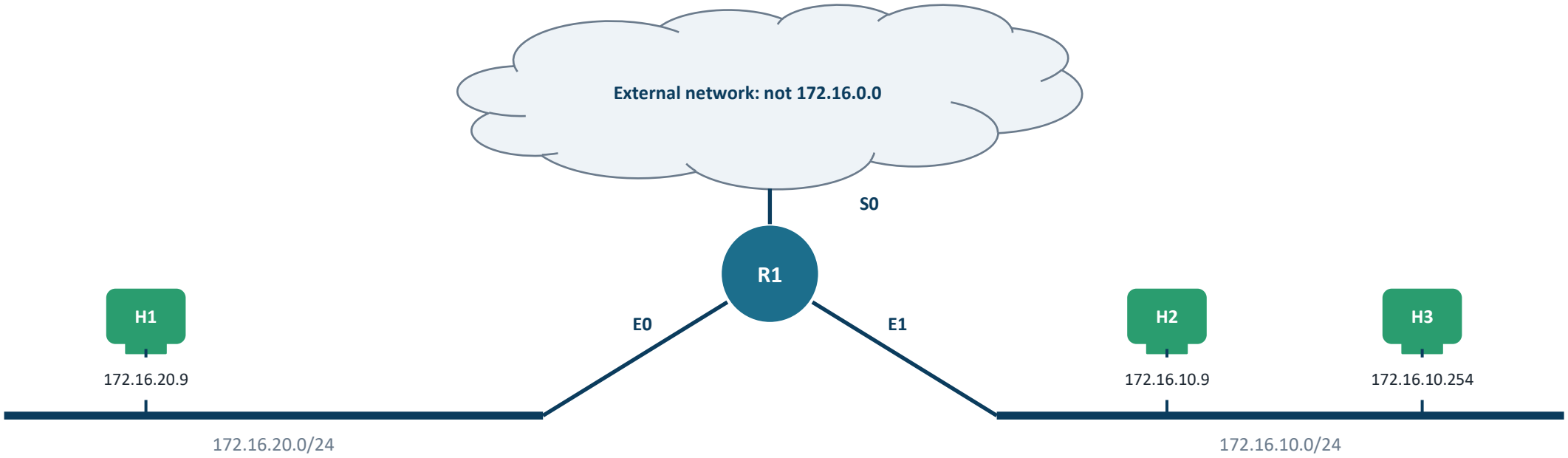
Access control lists (ACLs): how routers filter traffic, classic standard-ACL examples and the evolution toward stateful firewalls

ACLs in the router's processing flow



- Rules evaluated **in order**; the first match decides; **implicit deny** at the end
- Applied per **interface and direction** (in: before routing; out: after); a denied packet is dropped (optionally with ICMP)
- Uses beyond filtering: selecting traffic for NAT, QoS, VPN, route-maps, management access

Standard ACL example (1 of 3): permit only the internal network



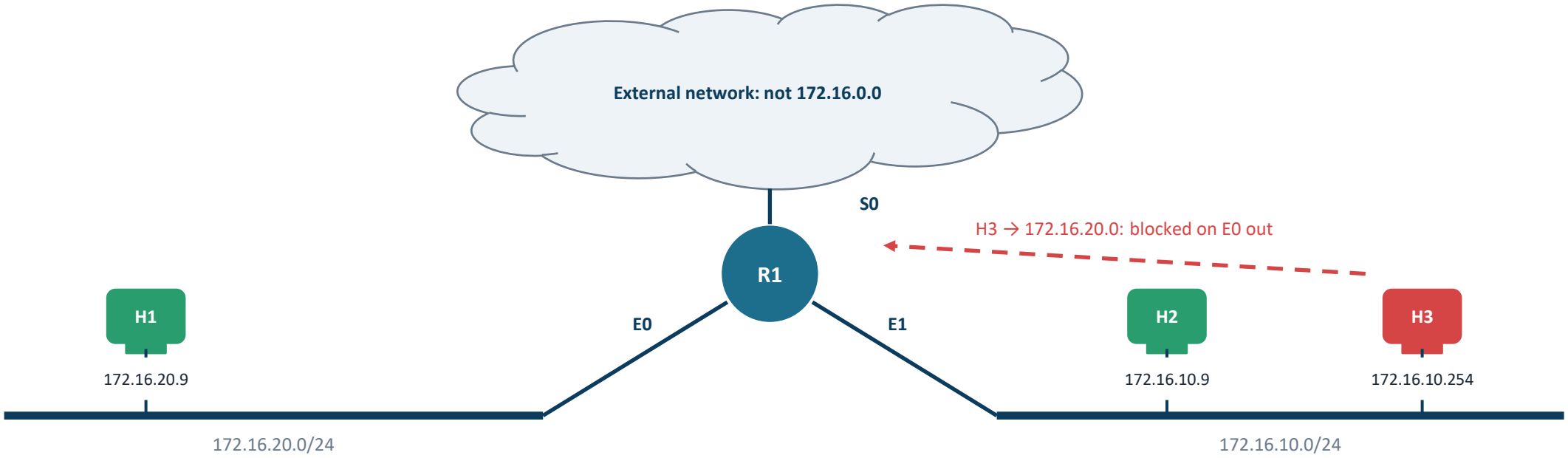

```
access-list 1 permit 172.16.0.0 0.0.255.255
! (implicit: access-list 1 deny 0.0.0.0 255.255.255.255)

interface Ethernet0
 ip access-group 1 out
interface Ethernet1
 ip access-group 1 out
```

Reading

- Wildcard 0.0.255.255 = 'the first two octets must be 172.16'
- Traffic from outside (via S0) to the LANs: denied outbound on E0/E1
- Traffic between the LANs: permitted (source 172.16.x.x)

Standard ACL example (2 of 3): deny a specific host

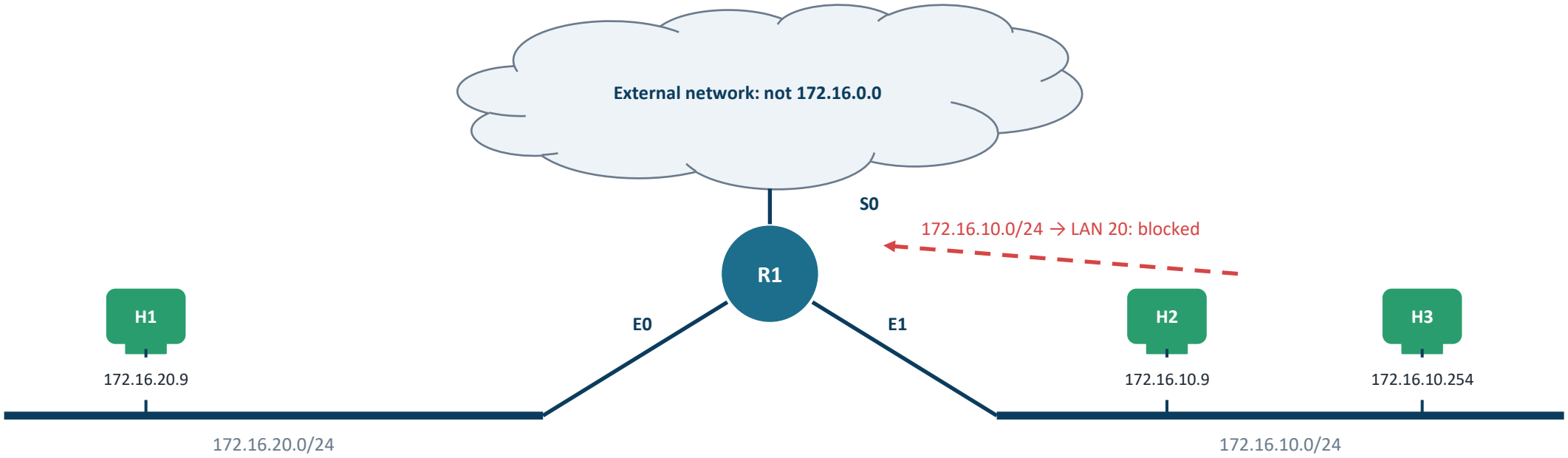



```
access-list 1 deny   172.16.10.254 0.0.0.0
access-list 1 permit 0.0.0.0 255.255.255.255
! modern equivalent:
!  access-list 1 deny host 172.16.10.254
!  access-list 1 permit any
interface Ethernet0
 ip access-group 1 out
```

Reading

- Wildcard 0.0.0.0 = match all 32 bits (a single host)
- Order matters: a 'permit any' before the 'deny' would void the rule
- H3 still reaches the Internet and its own LAN

Standard ACL example (3 of 3): deny a subnet




```
access-list 1 deny   172.16.10.0 0.0.0.255
access-list 1 permit 0.0.0.0 255.255.255.255

interface Ethernet0
 ip access-group 1 out

show access-lists 1   ! per-rule counters
```

Reading

- Wildcard 0.0.0.255 = 'the first three octets must be 172.16.10'
- One direction only: LAN 20 still reaches LAN 10 (the reply to traffic initiated by LAN 20 passes)
- To filter by protocol and port: extended ACL

UPDATED

ACLs in 2026: extended, named, IPv6 and stateful firewalls

Extended ACL

- Protocol, source, destination, ports, flags (established)
- Place close to the **source**; standard ACLs close to the **destination**
- Named: editing by sequence number; remarks

```
ip access-list extended LAN10-to-WEB
 remark permit HTTPS from LAN 10 to the server
 permit tcp 172.16.10.0 0.0.0.255 host 172.16.20.80 eq 443
 permit icmp 172.16.10.0 0.0.0.255 172.16.20.0 0.0.0.255 echo
 deny   ip any any log
interface GigabitEthernet0/1
 ip access-group LAN10-to-WEB in
```

IPv6

- ipv6 access-list NAME / permit tcp 2001:db8:10::/64 host 2001:db8:20::80 eq 443
- Never block ICMPv6 indiscriminately: ND and PMTUD depend on it (RFC 4890)

Beyond ACLs

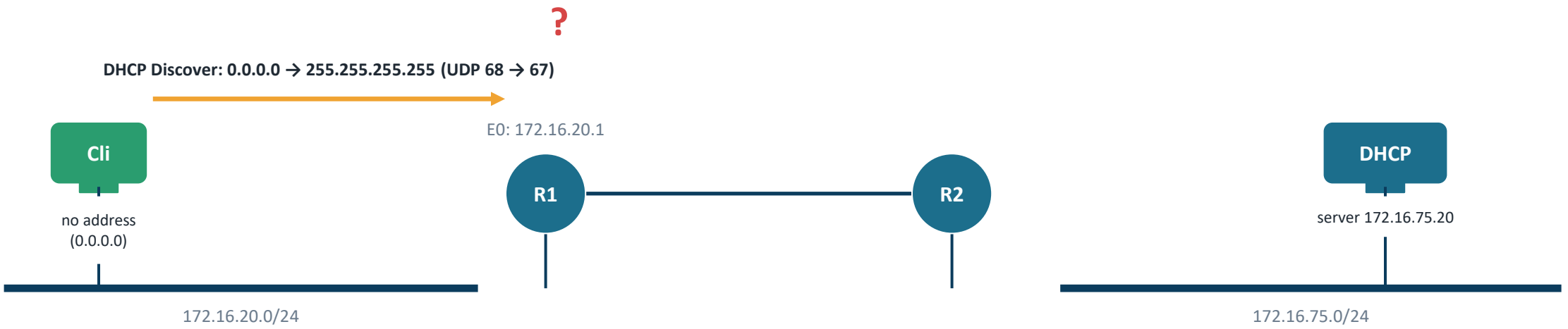
- ACLs are **stateless**: they do not know whether a packet is a reply to a legitimate connection
- Stateful firewall / NGFW: tracks the 5-tuple and TCP state; application inspection
- Zero trust: microsegmentation by identity; the principle (ordered rules, implicit deny) continues

MODULE 2 | PART 6

Additional IP routing topics

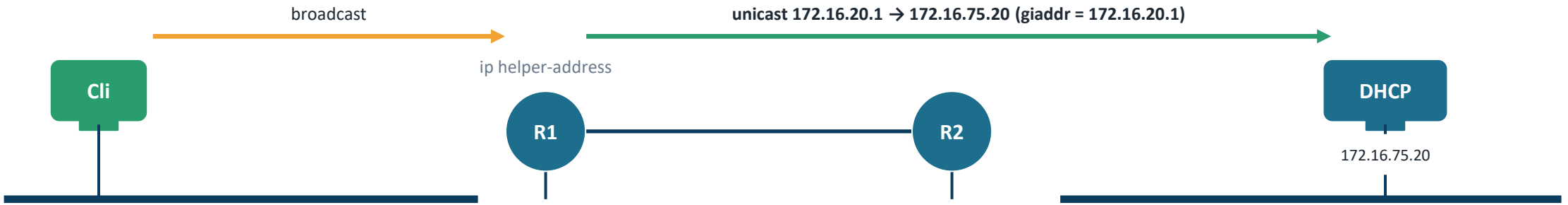
Broadcast handling and DHCP relay, directed broadcast, secondary addresses and unnumbered interfaces

IP broadcast handling: the DHCP problem



- What does the router do with the DHCP client's **broadcast** (formerly BootP)? By default, it **drops** it: routers do not forward 255.255.255.255
- Without a relay, the client never gets an address (unless there is a server per subnet)
- Solution: **DHCP relay** (RFC 1542 / RFC 2131): the router forwards the request **as unicast** to the server, filling **giaddr** with its address on the client's subnet
- Option 82 (RFC 3046): circuit/port identification for policies on the server

DHCP relay and directed broadcast: configuration

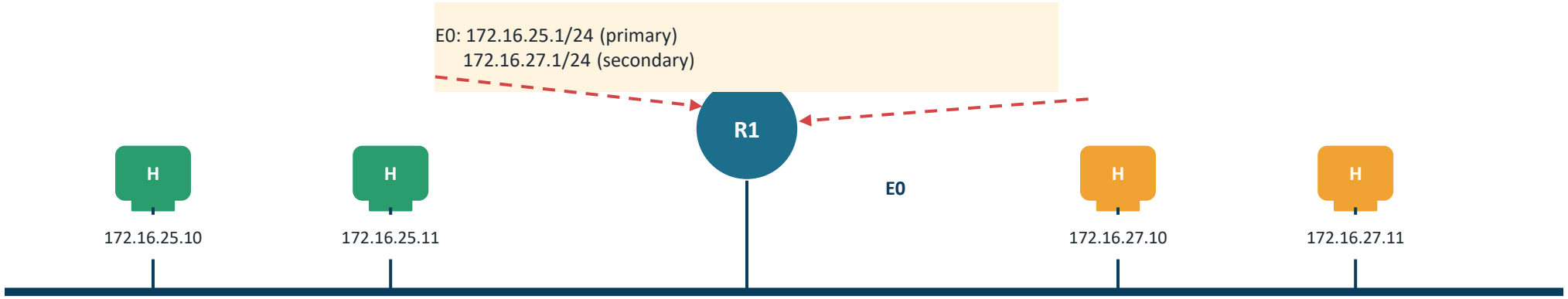



```
! Cisco IOS: relay on the clients' interface
interface GigabitEthernet0/0
 ip address 172.16.20.1 255.255.255.0
 ip helper-address 172.16.75.20
! list of forwarded UDP ports (default: 37 49 53 67 68 69 137 138)
ip forward-protocol udp 67
no ip forward-protocol udp 137
```

Platform	DHCP relay
Linux	isc-dhcp-relay / dhcrelay -i eth0 172.16.75.20
Juniper Junos	forwarding-options dhcp-relay server-group ...
L3 switch (several)	ip dhcp relay address 172.16.75.20 (per VLAN/SVI)

Directed broadcast: RFC 2644 (1999) requires routers NOT to forward it by default (smurf attack); 'no ip directed-broadcast' has been the IOS default since 12.0.

Secondary addresses



A single physical link (VLAN 25)

Traffic 172.16.25.x ↔ 172.16.27.x goes through the router, even on the same wire

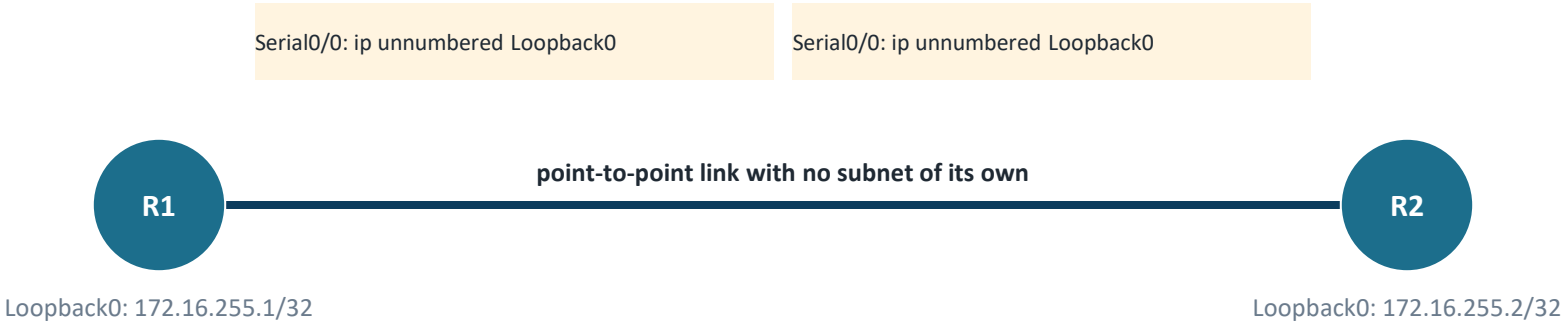
```
interface GigabitEthernet0/0
 ip address 172.16.25.1 255.255.255.0
 ip address 172.16.27.1 255.255.255.0 secondary
```

Legacy: the original configured 'interface tokenring 0' with 134.141.25.1 and 134.141.27.1 secondary.

When to use it (and when not)

- Hosts exhausted on the subnet with no immediate renumbering; bridged → routed migration
- Caveats: IGP adjacencies only over the primary; DHCP relay uses the primary as giaddr
- Better in 2026: a new VLAN + new subnet (or IPv6, where several addresses per interface are the rule)

Unnumbered interfaces (IP unnumbered)




```
interface Loopback0
 ip address 172.16.255.1 255.255.255.255
interface Serial0/0
 ip unnumbered Loopback0
```

```
! BGP unnumbered (FRRouting / Cumulus): no IPv4 on the link
router bgp 65001
 neighbor swp1 interface remote-as external
! session over IPv6 link-local; IPv4 via RFC 8950
```

Rules and limitations

- The lending interface must have an IP (use a loopback); you cannot ping the unnumbered interface nor point a static route at its 'IP'; saves a /30 per link (today: /31 or unnumbered)

MODULE 2 | PART 7

IPv6: from 'IPng' to the dominant protocol

Motivation, differences from IPv4, header and extension headers, addressing, neighbor discovery and autoconfiguration, routing, flows, security and transition

UPDATED

Why IPv6: from 'IPng' (1995) to RFC 8200 (2017)

Year	Milestone
1992 to 1994	IETF studies successors to IPv4 (IPng); CIDR and private addresses buy time
1995	RFC 1752 selects the IPng; RFC 1883 (since replaced) publishes IPv6 (version 6; version 5 had been used by ST)
1998	RFC 2460 (Draft Standard, since replaced); 6bone test network
2008 to 2012	Google, Facebook and others turn on IPv6; World IPv6 Launch (2012)
2011	IANA hands out the last IPv4 /8 blocks (February 3)
2017	RFC 8200: IPv6 becomes an Internet Standard (STD 86)
2020	LACNIC exhausts its final IPv4 phase (August)
2026	> 50% of accesses to Google over IPv6 (Mar 28); Brazil ≈ 59% IPv6-capable (APNIC, Oct 5)

The problem

- 32 bits = 4.29 billion addresses, many wasted by the classes
- NAT and CGNAT break the end-to-end model and raise operating costs
- The global routing table keeps growing

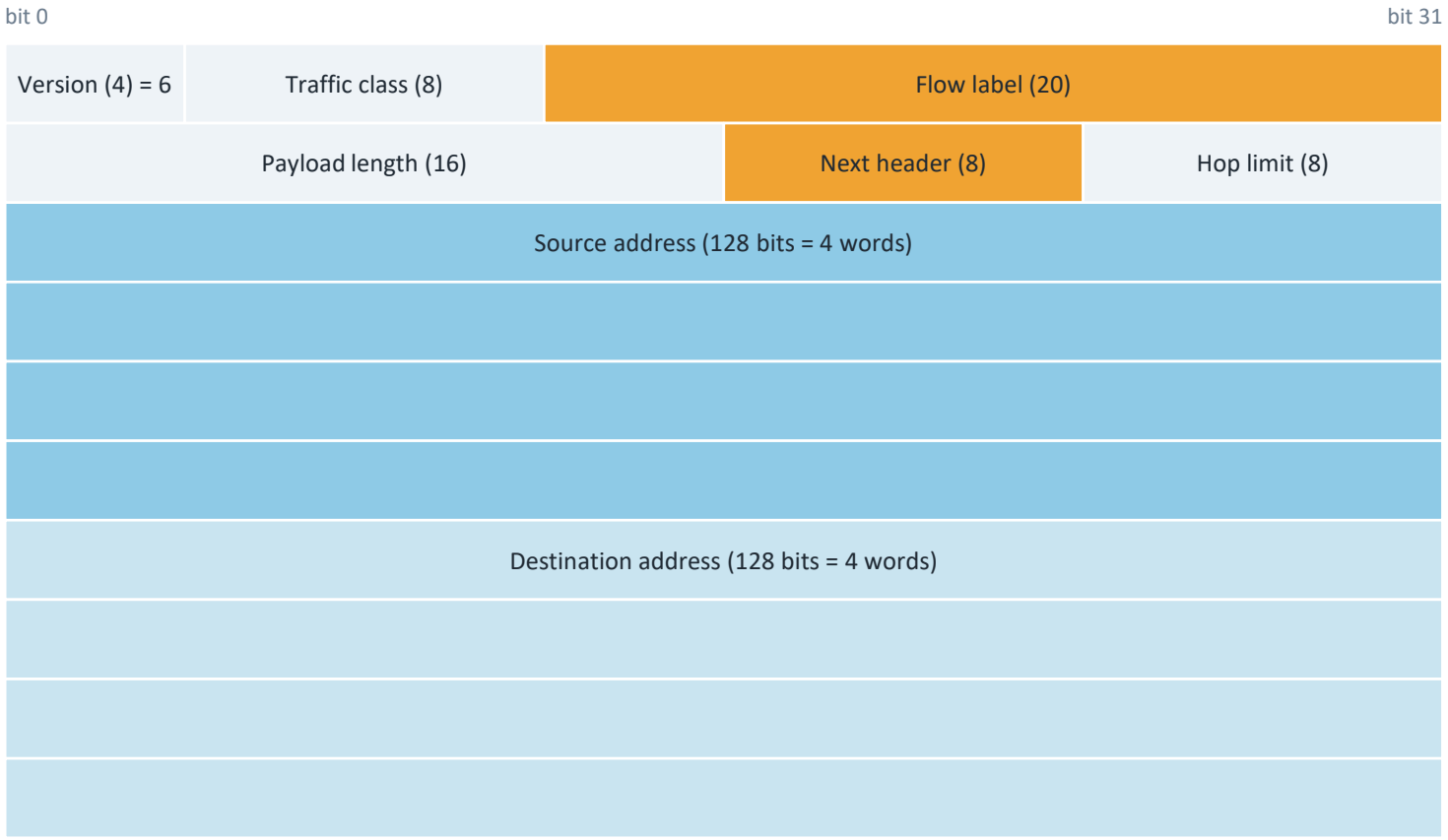
The answer

- 128 bits: 3.4 × 10^38 addresses (≈ 6.7 × 10^23 per m² of the Earth's surface)
- Simpler header, autoconfiguration, native multicast, no broadcast
- Transition still under way: dual stack, tunnels and translation (NAT64)

Main differences between IPv6 and IPv4

Aspect	IPv4 (RFC 791)	IPv6 (RFC 8200)
Address	32 bits; 4.3 × 10^9	128 bits; 3.4 × 10^38; allocation hierarchy; native anycast
Header	20 to 60 bytes, 13 fields, options in the header, checksum	Fixed 40 bytes, 8 fields, no checksum; options in extension headers
Fragmentation	In routers and at the source	Only at the source (Path MTU Discovery); minimum MTU 1,280 bytes
Address resolution	ARP (broadcast)	Neighbor Discovery, ICMPv6 (solicited-node multicast)
Broadcast	Yes (limited and directed)	Does not exist: multicast ff02::1 (all nodes) plays the role
Configuration	Manual or DHCP	SLAAC (RA), DHCPv6 (stateful or stateless), manual; several addresses per interface
QoS / flows	ToS → DSCP/ECN	Traffic Class (DSCP/ECN) + 20-bit Flow Label
Security	IPsec optional (added later)	IPsec built in through extension headers (AH, ESP); support recommended (SHOULD), not mandatory (RFC 8504)
NAT	Ubiquitous (scarcity)	Unnecessary: end-to-end model; stateful firewall instead of NAT

The IPv6 header



Fixed 40 bytes; no IHL, no checksum, no fragmentation fields, no options in the header

Field	Function
Traffic class	DSCP + ECN (RFC 2474, 3168)
Flow label	Identifies flows; ECMP hash (RFC 6437/6438)
Payload length	Bytes after the 40-byte header
Next header	6 TCP, 17 UDP, 58 ICMPv6, 0/43/44/50/51/60 extensions
Hop limit	Former TTL; 0 → discard + ICMPv6

History: RFC 1883 (1995) had a 4-bit 'Prio' and a 24-bit Flow Label, as in the 1999 slide; RFC 2460 (1998) settled on 8 + 20 bits.

Extension headers



Extension	Value	Who processes it / use
Hop-by-Hop Options	0	Every router; jumbograms, Router Alert (MLD, RSVP)
Destination Options	60	Destination (and intermediate destinations if before Routing); Mobile IPv6
Routing	43	Type 0 deprecated (RFC 5095); type 2 MIPv6; type 4 SRv6 (RFC 8754)
Fragment	44	Only the source fragments; reassembly at the destination
AH / ESP	51 / 50	IPsec: authentication; confidentiality + integrity
No next header	59	Nothing follows

Rules (RFC 8200)

- Recommended order: the chain above; each extension at most once (Dest. Options up to twice)
- Routers do not insert or remove extension headers in transit
- Only Hop-by-Hop is examined by every hop
- In practice: many networks drop packets with extension headers (RFC 7872); use sparingly

IPv6 addressing: 128 bits, three types, no broadcast

Notation (RFC 4291, canonical form RFC 5952)

```
2001:0db8:0000:0000:0000:0000:0000:0001  →  2001:db8:0:0:0:0:0:1  →  2001:db8::1
```

Eight 16-bit groups in hexadecimal; leading zeros omitted; a single run of zeros becomes '::'; lowercase; CIDR prefix (/64)

Unicast

- Identifies one interface
- Global, unique local (ULA), link-local

Anycast

- Same address on several interfaces; delivered to the 'nearest' one (IGP/BGP metric)
- Syntactically identical to unicast; DNS root, CDNs

Multicast

- ff00::/8; delivered to all members of the group
- Replaces broadcast: ff02::1 = all nodes on the link

Typical structure of a global unicast address (GUA)



Subnets are /64 by convention (SLAAC requires it); organizations receive a /48 (65,536 subnets) or a /56; ISPs, a /32 or larger. 2^128 ≈ 3.4 × 10^38 addresses.

IPv6 address types and scopes

Type	Prefix	Scope and use	Reference
Global unicast (GUA)	2000::/3	Routable on the Internet; allocated by IANA → RIR → ISP → organization (/48, /56)	RFC 4291
Unique local (ULA)	fc00::/7 (fd00::/8)	Private, not routable on the Internet; 40 random bits avoid collisions in mergers; no NAT	RFC 4193
Link-local	fe80::/10	Automatic on every interface; ND, next hop for OSPFv3/BGP; never routed	RFC 4291
Loopback / unspecified	::1 / ::	Equivalent to IPv4's 127.0.0.1 and 0.0.0.0	RFC 4291
IPv4-mapped	::ffff:0:0/96	Represents IPv4 in dual-stack sockets (::ffff:203.0.113.5)	RFC 4291
NAT64 well-known	64:ff9b::/96	IPv6 → IPv4 translation (DNS64 synthesizes AAAA)	RFC 6052
Documentation	2001:db8::/32	Examples and courses (used in this deck)	RFC 3849
Multicast	ff00::/8	ff02:: link, ff05:: site, ff0e:: global; ff02::1 all nodes, ff02::2 routers, ff02::1:ffxx:xxxx solicited-node	RFC 4291

'Site-local' addresses (fec0::/10), planned in the drafts of the 1990s, were deprecated by RFC 3879 (2004) and replaced by ULAs.

Neighbor Discovery (ND) and autoconfiguration (SLAAC)

ICMPv6 message	Type	Function
Router Solicitation	133	Host asks for an RA at startup (ff02::2)
Router Advertisement	134	Prefixes, default router, MTU, M/O/A flags, RDNSS; periodic and on request
Neighbor Solicitation	135	Resolves an address (= ARP Request); DAD; NUD
Neighbor Advertisement	136	Reply (= ARP Reply); unsolicited announcement
Redirect	137	Router points to a better next hop on the link

Equivalences: NS/NA = ARP (but over solicited-node multicast ff02::1:ffxx:xxxx, without interrupting every host); RA = default gateway + prefix + (optionally) DNS, without DHCP; DAD prevents duplicate addresses; NUD tracks the state of each neighbor (reachable, stale, probe).

1 **Link-local**
fe80:: + interface ID; duplicate address detection (DAD) with an NS for its own address.

2 **RS → RA**
Host sends an RS to ff02::2; the router replies with an RA: prefix 2001:db8:10::/64 (A flag), default router, MTU, DNS.

3 **Global address**
Prefix + interface ID (stable random, RFC 7217; EUI-64 in legacy systems) + temporary addresses (RFC 8981); new DAD.

4 **DHCPv6 if M/O**
M = 1: addresses from DHCPv6 (stateful); O = 1: options only (DNS, NTP); ff02::1:2, UDP 546/547 (RFC 9915).

Switch protections: RA Guard (RFC 6105), DHCPv6 Guard, ND inspection; SEND (RFC 3971) exists but is rarely used.

IPv6 routing

IGP / EGP	IPv6	Notes
RIP	RIPng, RFC 2080	UDP 521, ff02::9; same limitations
OSPF	OSPFv3, RFC 5340	Per link, link-local; AF for IPv4 (RFC 5838)
IS-IS	RFC 5308 / MT RFC 5120	One protocol for both families; multiple topologies
EIGRP	EIGRP for IPv6, RFC 7868	ff02::a; same DUAL and metric
BGP	MP-BGP, RFC 4760 (AFI 2)	Global IPv6 table > 200,000 prefixes; sessions over IPv6 or IPv4
Static	ipv6 route 2001:db8::/32 ...	Next hop may be link-local + interface

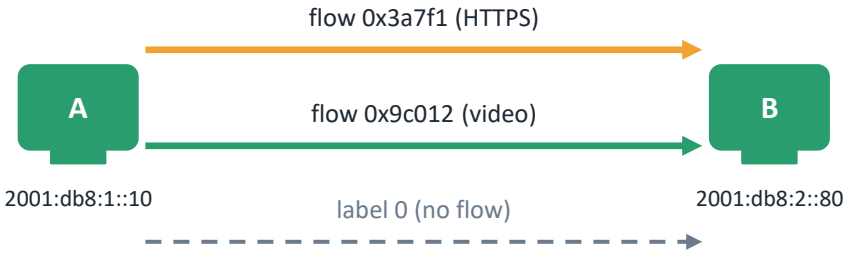
Recommendations: /64 subnets on every link (even point-to-point, with /127 optional, RFC 6164); /128 loopbacks; filter ULA and link-local at the borders; handle ICMPv6 with care (RFC 4890).

The 'new features' of 1999, seen from 2026

- **Provider selection** (policy, cost): exists through BGP multihoming with PI/PA prefixes and through traffic engineering; not through IPv6 fields
- **Host mobility** (Mobile IPv6, RFC 6275): standardized, little used; mobility was solved above (QUIC, apps) and in the mobile networks
- **Automatic re-addressing**: works; prefixes with preferred/valid lifetimes in the RA allow renumbering (RFC 4192)
- Typical practice: dual stack with OSPFv3 or IS-IS + MP-BGP; IPv6-only with NAT64 in mobile carriers and data centers

IPv6 flows and traffic class

- Several **flows** active between the same source and destination; traffic without a flow carries label **0**
- Flow = **source address + flow label ≠ 0** (20 bits), assigned pseudorandomly by the source (RFC 6437); routers do not change it
- Real use: input to the **ECMP / LAG hash** without reading the transport header (RFC 6438), useful with extension headers or ESP
- Per-flow resource reservation (RSVP/IntServ) never took off; differentiation through the **Traffic Class** (DSCP + ECN), as in IPv4



Routers doing ECMP use (source, destination, label) to pick the path: each flow always follows the same path.



Security in IPv6

IPsec built in through extension headers

- **AH** (51, RFC 4302): authentication and integrity, no confidentiality; little used today
- **ESP** (50, RFC 4303): confidentiality + integrity (AES-GCM); transport and tunnel modes; IKEv2 (RFC 7296)

Mandatory? Not anymore

- RFC 4294 (2006, since replaced): IPv6 nodes **must** implement IPsec
- RFC 6434 (2011) → **SHOULD**; kept in RFC 8504 (2019)
- In practice, IPsec in IPv6 is as optional as in IPv4

Specific risks

- No NAT: an explicit stateful firewall is mandatory
- ND as vulnerable as ARP: rogue RA, forged NS/NA, DAD flooding
- Reconnaissance by scanning is unfeasible (/64), but DNS and ND caches leak hosts

Mitigations

- RA Guard (RFC 6105), DHCPv6 Guard, ND inspection, SEND (RFC 3971, rare)
- Temporary addresses and stable random IDs (RFC 8981, RFC 7217)
- Careful ICMPv6 filtering (RFC 4890); filter ULA/link-local at the borders; RPKI for prefixes

UPDATED

IPv4 / IPv6 transition and coexistence

Dual stack (RFC 4213)

- Both stacks on hosts and routers
- DNS: AAAA preferred; Happy Eyeballs (RFC 8305) picks the fastest connection
- Basis of today's Internet; cost: running two networks

Tunnels

- IPv6 over IPv4: 6in4 (proto 41), GRE, 6rd (RFC 5969)
- IPv4 over IPv6: DS-Lite (RFC 6333), MAP-T/MAP-E
- Where one stack does not arrive natively; reduced MTU

Translation

- NAT64 (RFC 6146) + DNS64 (RFC 6147): an IPv6-only network reaches IPv4 servers
- 464XLAT (RFC 6877): IPv4-only apps on IPv6-only phones
- Standard at large mobile carriers

Direction in 2026: IPv6-only networks with 'IPv4 as a service' (NAT64/464XLAT), including campuses and data centers; DHCP option 108 (RFC 8925) lets hosts that do not need IPv4 turn it off.

Data: native IPv6 access to Google > 50% since March 28, 2026 (Internet Society Pulse / Google); Brazil ≈ 59% IPv6-capable (APNIC Labs, October 5, 2026). Resource in Portuguese: ipv6.br (NIC.br).

MODULE 2 | PART 8

IP multicast

Concept, addresses and mapping to MAC, IGMP and MLD, dense and sparse modes, multicast routing algorithms and protocols (DVMRP, MOSPF, PIM), RSVP

Multicast: one sender, many receivers

Unicast

- **1 → 1**
- N copies for N receivers

Broadcast

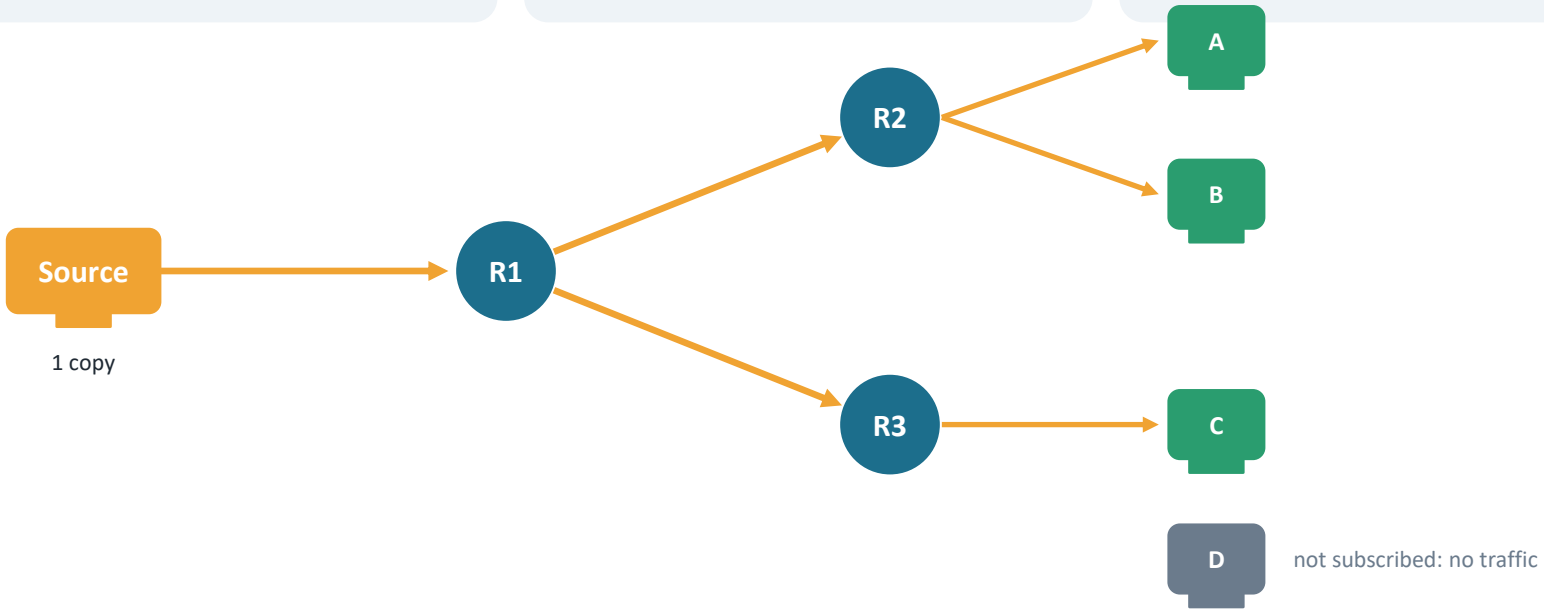
- **1 → everyone on the link**
- Interrupts those who do not want it; does not cross routers

Multicast

- **1 → group (subscribers)**
- One copy; the network replicates at the branches

Anycast

- **1 → the nearest member of the group**
- A single receiver; DNS root, CDN



Applications in 2026

- Carrier IPTV; stock market feeds
- Video and signaling on campuses; service discovery (mDNS, SSDP)
- Routing protocols (OSPF, EIGRP, VRRP)
- Public Internet: replaced by unicast CDNs

IPv4 (and IPv6) multicast addresses

1110	Multicast group identifier (28 bits)

Class D: 224.0.0.0 to 239.255.255.255 (224.0.0.0/4)

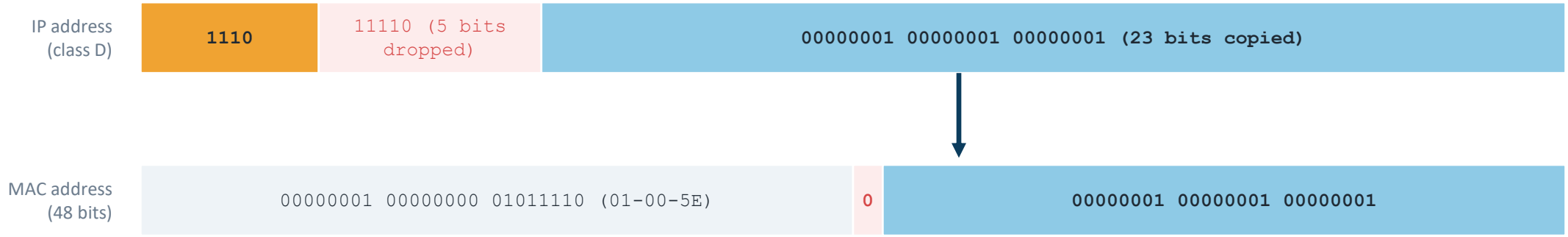
IPv4 range	Use	Ref.
224.0.0.0/24	Link-local control (TTL 1): .1 hosts, .2 routers, .5/.6 OSPF, .9 RIPv2, .10 EIGRP, .13 PIM, .18 VRRP, .22 IGMPv3, .251 mDNS	RFC 5771
224.0.1.0/24	Internetwork control: 224.0.1.1 NTP, .39/.40 Auto-RP	RFC 5771
232.0.0.0/8	Source-Specific Multicast (SSM)	RFC 4607
233.0.0.0/8	GLOP: 233.X.Y.0/24 per 16-bit ASN	RFC 3180
239.0.0.0/8	Administratively scoped (private): organization, site	RFC 2365

IPv6: ff00::/8

- ff | flags | scope | group ID (112 bits)
- **Scopes:** 1 interface, 2 link, 4 admin, 5 site, 8 organization, e global
- **ff02::1** all nodes, **ff02::2** all routers, ff02::5/6 OSPFv3, ff02::9 RIPng, ff02::a EIGRP, ff02::d PIM, ff02::12 VRRP, ff02::1:2 DHCPv6
- **ff02::1:ffxx:xxxx** solicited-node (ND)
- **ff3x::/32** SSM; ff05::/16 site; ff0e::/16 global
- Registry: IANA IPv6 Multicast Address Space Registry

Mapping IP multicast addresses to MAC addresses

IPv4 (RFC 1112): 239.1.1.1 → 01-00-5E-01-01-01



01-00-5E-01-01-01: IANA OUI (01-00-5E) + bit 0 + 23 bits of the group

32-to-1 ambiguity

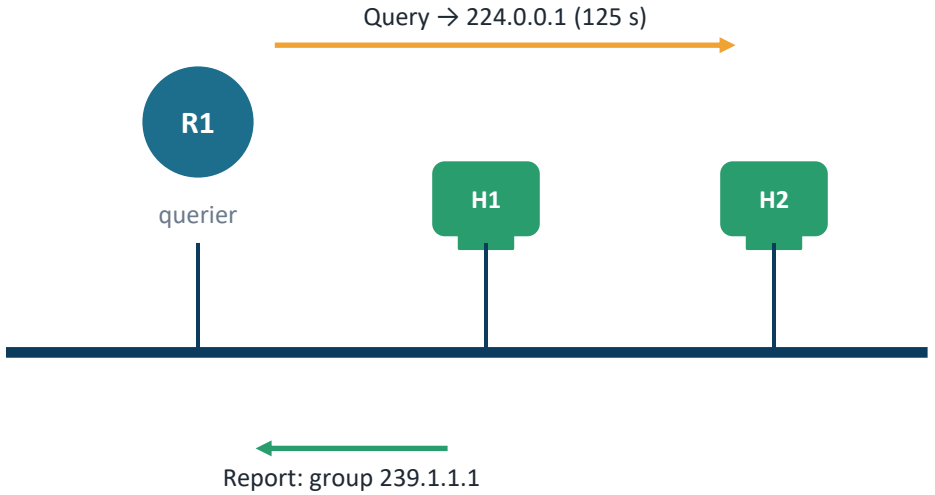
- 5 bits of the group do not fit in the MAC: 2^5 = 32 groups share the same MAC
- 224.1.1.1, 224.129.1.1, 225.1.1.1 … 239.129.1.1 → all 01-00-5E-01-01-01
- The NIC accepts; IP discards the groups not joined (CPU cost)

IPv6 (RFC 2464)

- 33-33 + the 32 least significant bits of the group
- ff02::1 → 33-33-00-00-00-01; ff02::1:ff12:3456 (solicited-node) → 33-33-ff-12-34-56
- Switches with MLD snooping filter per group

IGMP and MLD: hosts say what they want to receive

Version	RFC	What's new
IGMPv1	1112 (1989)	Query and Report; leave by timeout
IGMPv2	2236 (1997)	Leave; group-specific query; querier election (lowest IP); 224.0.0.2
IGMPv3	9776 (2025)	INCLUDE/EXCLUDE source filters (SSM); reports to 224.0.0.22
MLDv1	2710 (1999)	IGMPv2 for IPv6, over ICMPv6 (types 130 to 132)
MLDv2	9777 (2025)	IGMPv3 for IPv6; reports to ff02::16 (type 143)



IGMP / MLD snooping in switches (RFC 4541)

- The switch listens to Reports and Leaves and builds a group → ports table
- Multicast is forwarded only to the ports with members (and to router ports)
- Without snooping, each group floods the whole VLAN as if it were broadcast
- Requires a querier on the VLAN (a router or a 'querier' in the switch itself)

Timers (IGMPv2)

- Query every 125 s; answer within 10 s
- A member expires after 260 s without a report
- Leave → group-specific query (2 × 1 s) → fast removal

Dense mode vs. sparse mode

Dense mode

- Assumes most users belong to the group (receivers everywhere)
- **Flood and prune:** forwards out every port; branches without members send a Prune; the prune expires (3 min) and the flooding starts over
- (S,G) state in every router, even without receivers
- Protocols: **DVMRP, MOSPF, PIM-DM** (RFC 3973)
- Today: small networks, labs

Sparse mode

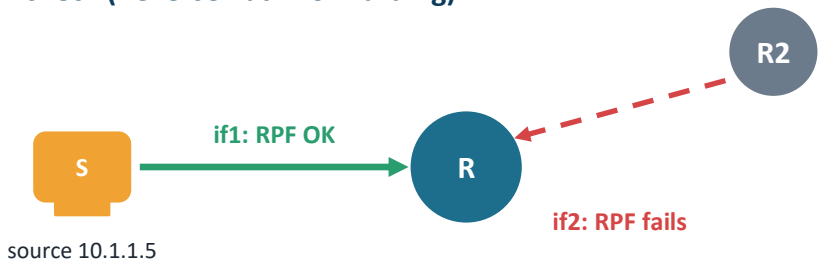
- Assumes few receivers, scattered across the network
- **Explicit join:** routers with members join the tree; nothing flows without a request
- Shared tree around a **rendezvous point (RP)** or core; optionally a per-source tree (SPT)
- Protocols: **CBT** (RFC 2201, historical), **PIM-SM** (RFC 7761), SSM (RFC 4607), Bidir-PIM (RFC 5015)
- Today: the industry standard

Criterion	Dense	Sparse
Initial traffic	Flooded to everyone	Only after a join
State in routers	In all of them	Only along the receivers' path
Scale	Small networks	Large networks and carriers

Multicast routing algorithms and the RPF check

Algorithm	Idea	Status
Flooding	Resends out every interface; memory avoids duplicates	Historical
Spanning tree	One tree for all groups	Historical
RPB	Reverse Path Broadcasting: RPF check + resend out the other interfaces (MBone)	Conceptual basis
TRPB	RPB without sending to leaves without members (IGMP)	Conceptual basis
RPM	TRPB + prunes traveling up the tree; DVMRP, PIM-DM	In use (PIM-DM)
Steiner trees	Minimum total cost tree; NP-hard, unstable	Theoretical
CBT (Core Based Trees)	Shared tree per group around a core	Evolved into PIM-SM

RPF check (Reverse Path Forwarding)



R's unicast table	Via
10.1.1.0/24 (source)	if1

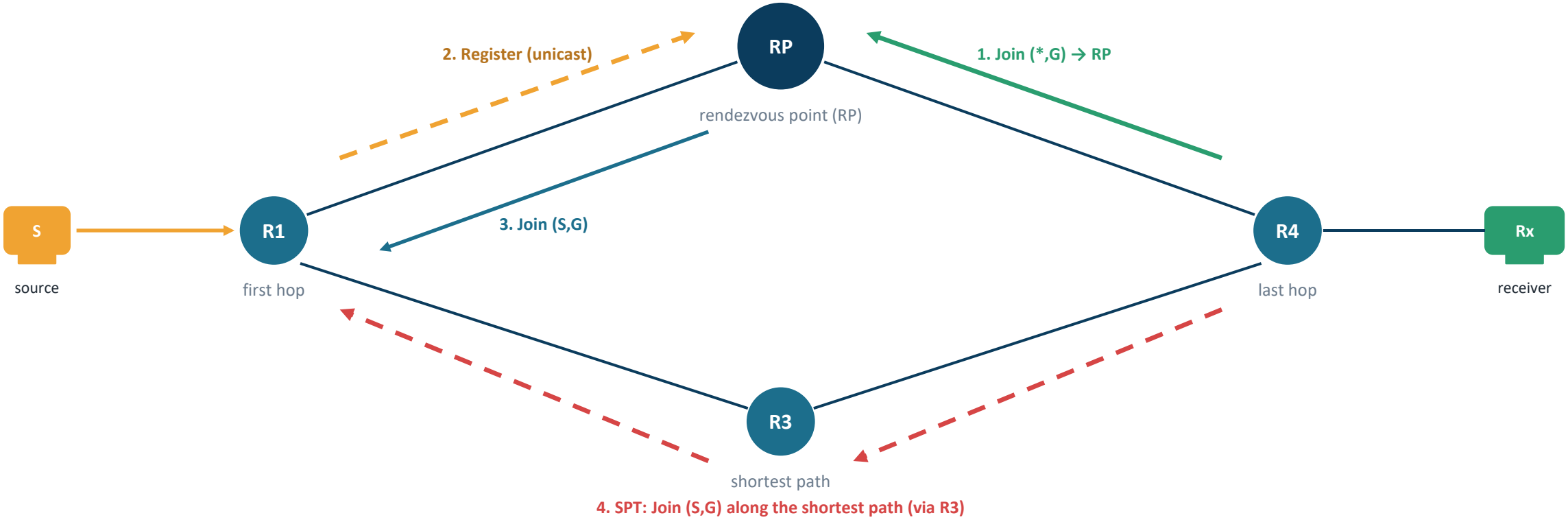
Rule: accept the packet from S only if it arrived on the interface the unicast table uses to reach S (if1); copies arriving on another interface (if2) are discarded. Prevents loops without a TTL and is used by DVMRP, PIM-DM and PIM-SM (with the RP in place of the source on the shared tree).

Multicast routing protocols

Protocol	RFC	Model	How it works	Status in 2026
DVMRP	1075 (1988)	Dense	Its own distance vector (RIP-like) for RPF; RPM; tunnels between islands; the MBone protocol	Historical
MOSPF	1584 (1994)	Dense	OSPF extension: group membership LSAs; (S,G) tree computed in the LSDB	Historical
PIM-DM	3973 (2005)	Dense	Flood and prune; RPF from the existing unicast table; no RP	Small networks
PIM-SM	7761 (2016)	Sparse	Explicit join; shared tree (*,G) at the RP; switchover to the SPT (S,G)	Industry standard
PIM-SSM	4607 (2006)	Sparse	Only (S,G); the receiver names the source (IGMPv3/MLDv2); 232/8, ff3x::/32; no RP	IPTV, market data
Bidir-PIM	5015 (2007)	Sparse	Bidirectional shared tree; many-to-many; no (S,G) state	Niche (finance)
CBT	2201 (1997)	Sparse	Shared tree around a core	Historical

MBone (1992 to ≈ 2000): a virtual network overlaid on the Internet, with DVMRP tunnels between multicast islands; it carried IETF meetings and NASA missions and reached thousands of subnets in dozens of countries. It disappeared with native multicast and, later, with unicast CDNs.

PIM-SM in operation: RP, shared tree and SPT



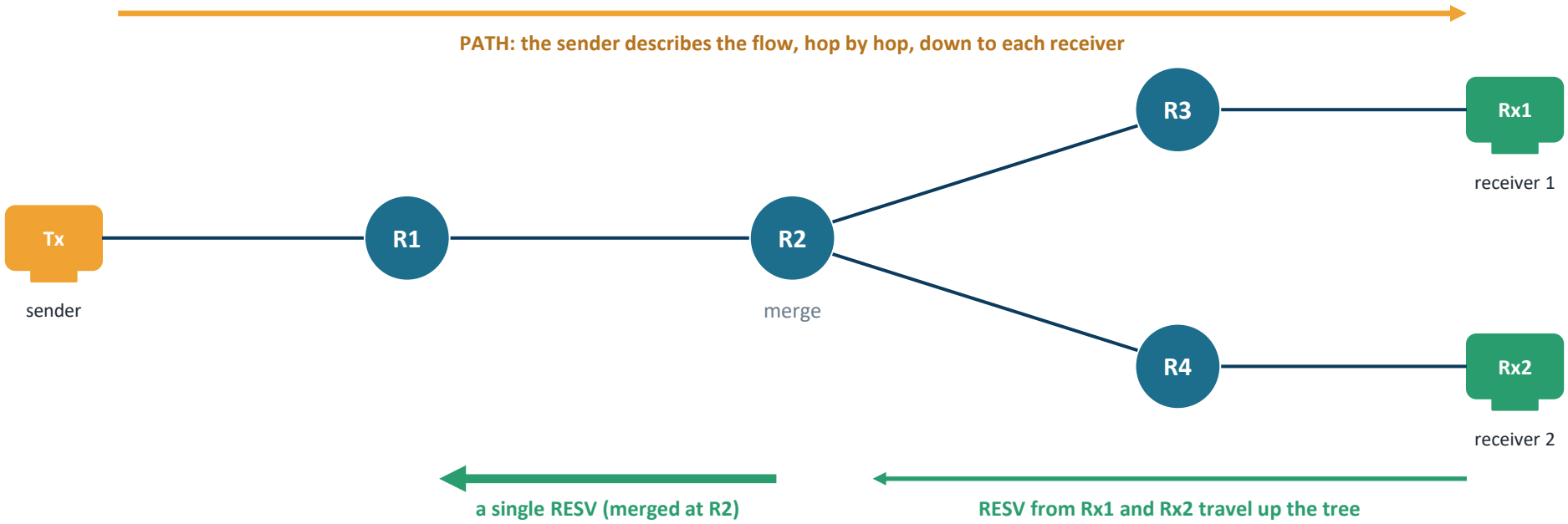
1. **Join on the shared tree**
   The receiver sends an IGMP Report; R4 sends a PIM Join (*,G) hop by hop up to the RP.
2. **Source registration**
   R1 encapsulates the first packets from S in unicast PIM Register messages to the RP, which delivers them down the (*,G) tree.
3. **Native path to the RP**
   The RP sends a Join (S,G) toward S and then a Register-Stop; traffic now flows natively.
4. **Switchover to the SPT**
   R4 sends a Join (S,G) along the shortest path to S (via R3) and prunes the shared tree (Prune (S,G,rpt)).

RP discovery

- Static; Auto-RP (Cisco); BSR (RFC 5059); Anycast-RP (RFC 4610) for redundancy
- PIM: IP protocol 103; Hello on 224.0.0.13 / ff02::d

LEGACY

RSVP: resource reservation (and what is left of it)



Original RSVP (RFC 2205, IntServ)

- **Receiver-oriented** reservations: PATH goes down, RESV goes up and **merges** at the first reserved branch
- Soft per-flow state in every router: did not scale on the Internet
- Real QoS went to DiffServ (DSCP) and overprovisioning

What survived: RSVP-TE (RFC 3209)

- Signals MPLS traffic-engineering tunnels: explicit path, bandwidth, priority
- Fast Reroute protection (RFC 4090)
- Losing ground to Segment Routing (no signaling)

MODULE 2 | PART 9

MPLS and VRRP

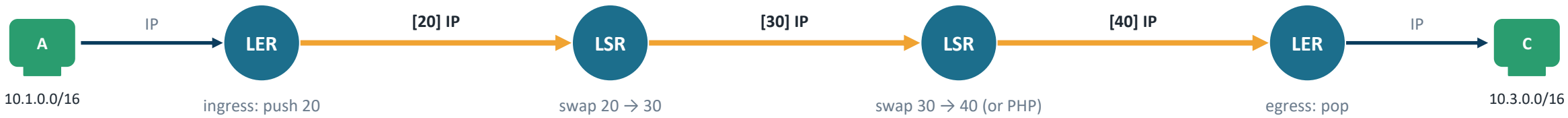
Label switching: concepts, terminology, a LIB/LFIB example and current applications; gateway redundancy with VRRP

MPLS: label switching

MPLS header ('shim', RFC 3032): 32 bits between the link layer and IP



Labels 0 to 15 are reserved (0 IPv4 explicit null, 1 router alert, 2 IPv6 explicit null, 3 implicit null = PHP); S = bottom of stack; the label stack allows nested tunnels



LSP (Label Switched Path) for FEC '10.3.0.0/16' from the ingress LER to the egress LER

Operations

- **Push** (ingress): classifies into the FEC and inserts a label
- **Swap** (core): replaces the label using the LFIB, without reading IP
- **Pop** (egress or penultimate hop, PHP): removes it

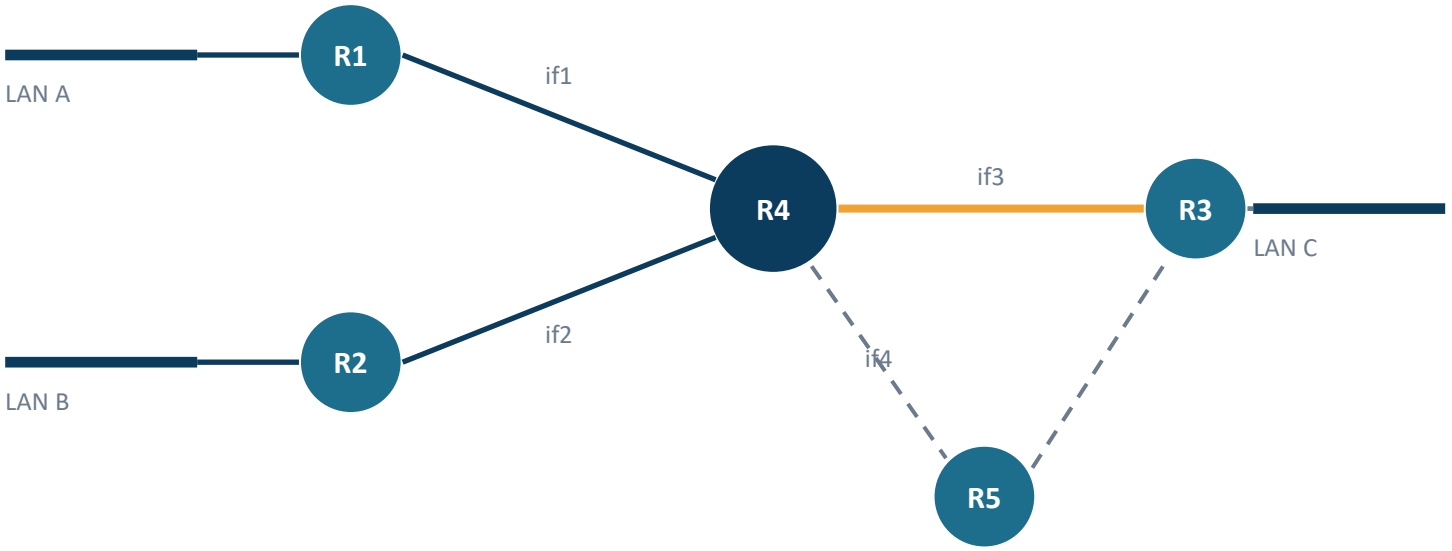
Why it still exists in 2026

- No longer for speed (ASICs route IP at line rate)
- For what the label makes possible: L3/L2 VPNs, traffic engineering, multiprotocol transport, Segment Routing

MPLS terminology

Term	Definition	Note
FEC	Forwarding Equivalence Class: packets forwarded the same way (e.g., same destination prefix, or prefix + class of service)	One label per FEC on each link
FIB	Forwarding Information Base: IP table with one row per FEC; at the ingress, says which label to push	Derived from the RIB
LSP	Label Switched Path: label-switched path established for each route (or tunnel) through the network	Unidirectional
LDP	Label Distribution Protocol (RFC 5036): distributes labels between neighbors (TCP/UDP 646), following the IGP	Alternatives: RSVP-TE, BGP, Segment Routing
LIB	Label Information Base: every label received from every neighbor, per FEC	Control plane
LFIB	Label Forwarding Information Base: incoming label → action (swap/pop) + outgoing label and interface	Data plane
LER / LSR	Label Edge Router (ingress/egress: push/pop) / Label Switch Router (core: swap)	'PE' and 'P' in VPNs
PHP	Penultimate Hop Popping: the penultimate hop removes the label (implicit null, 3)	Saves a lookup at the egress

Example: FIB, LIB and LFIB of router R4



R4's FIB (IP → FEC → outgoing label)

Prefix (FEC)	Next hop	Interface	Outgoing label
10.3.0.0/16	R3	if3	30
10.1.0.0/16	R1	if1	10
10.2.0.0/16	R2	if2	11

R4's LIB (every label learned)

FEC	Neighbor	Its label
10.3.0.0/16	R3 (if3)	30
10.3.0.0/16	R5 (if4)	55
10.1.0.0/16	R1 (if1)	10
10.2.0.0/16	R2 (if2)	11

R4's LFIB (data plane)

Label in	If in	Action	Label out	If out
20	if1	swap	**30**	if3
21	if2	swap	**30**	if3
22	if3	swap	10	if1
23	if3	swap	11	if2

LIB = everything LDP learned; LFIB = only the path chosen by the IGP.
A → C: arrives [20], leaves [30]; B → C: arrives [21], leaves [30].

UPDATED

MPLS applications in 2026

Layer 3 VPN (RFC 4364)

- One VRF per customer on the PEs; MP-BGP VPNv4/VPNv6 with route targets
- Two-label stack: transport (IGP/LDP/SR) + VPN
- The carriers' 'premium' corporate WAN

Layer 2 VPN and EVPN

- Pseudowires (RFC 8077), VPLS (RFC 4762)
- EVPN (RFC 7432): BGP control plane for MACs/IPs; also over VXLAN in data centers

Traffic engineering

- RSVP-TE (RFC 3209): explicit paths, bandwidth, priorities
- Fast Reroute (RFC 4090): protection within tens of ms
- QoS across the core through the TC field (3 bits)

Segment Routing

- No LDP or RSVP: segments advertised by the IGP (OSPF/IS-IS)
- SR-MPLS (RFC 8660): label stack at the source; SRv6 (RFC 8986): list of IPv6 addresses
- SR Policy with a controller (PCE); TI-LFA for protection

UPDATED

VRRP: default gateway redundancy

The problem

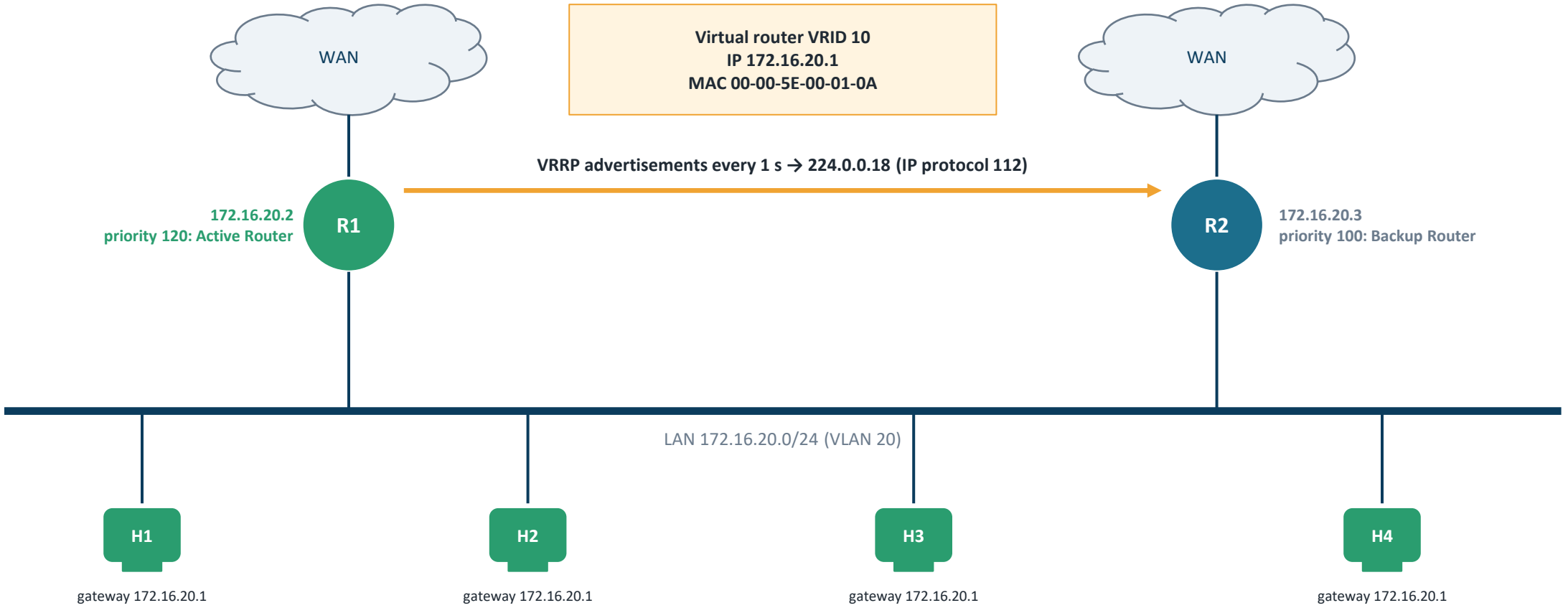
- Hosts have **one** default gateway: if the router goes down, the LAN is cut off, even with another router available
- Old solutions: Proxy ARP, ICMP Router Discovery, RIP on the hosts (all slow or fragile)

The solution: a virtual router

- **VRID** (1 to 255) + **virtual IP** + **virtual MAC** 00-00-5E-00-01-{VRID} (IPv6: 00-00-5E-00-02-{VRID})
- The hosts use the virtual IP as their gateway; whoever answers is the current **Active Router** (Master in the predecessor RFC 5798 and in vendor CLIs)

Parameter	Value	Parameter	Value
Priority	1 to 254 (255 = IP owner); default 100; highest wins	Advertisements	Active Router → 224.0.0.18 / ff02::12, IP protocol 112, every 1 s (v3: 100 cs)
Failover	≈ 3 intervals without advertisements + skew → backup takes over (≈ 3 s; v3 sub-second)	After taking over	Gratuitous ARP / unsolicited NA with the virtual MAC: switches relearn the port
Preemption	Enabled by default (the highest-priority router takes back the role)	Versions	RFC 2338 (1998) → RFC 3768 v2 (2004) → RFC 5798 v3 (2010, IPv4 + IPv6) → RFC 9568 v3 (2024), the current standard
Proprietary	HSRP (Cisco, RFC 2281), GLBP (Cisco, load balancing)	Tracking	Object tracking lowers the priority if the Active Router's WAN goes down

VRRP in operation: a failover example



R1 fails

R2 gets no advertisements for ≈ 3 s → becomes the Active Router → answers for the virtual MAC → gratuitous ARP moves the MAC in the switches. Hosts notice nothing.

Best practices

Track the WAN (tracking); alternate the Active Router per VLAN to use both routers; v3 for IPv6 and centisecond timers.

Module 2 summary

TCP/IP and addressing

- 4-layer stack; 20-byte IPv4 header; protocol and port demultiplex
- CIDR/VLSM replaced the classes; special addresses (RFC 6890)
- Comparison under the mask and longest prefix match

Host behavior

- Route determination (local/remote) + ARP/ND resolution
- ARP for the destination if local, for the gateway if remote; Proxy ARP
- MACs change at every hop; IPs do not; TTL decreases

Dynamic routing

- RIP (small), EIGRP (DUAL), OSPF (areas, DR, LSAs), IS-IS
- Administrative distance decides between sources; redistribute with care
- BGP-4: path vector and policies between ASes

Control and operations

- Ordered ACLs with an implicit deny; stateful firewalls
- DHCP relay; directed broadcast turned off (RFC 2644)
- Secondary addresses and unnumbered interfaces

IPv6

- 128 bits, 40-byte header, extension headers
- ND and SLAAC instead of ARP and DHCP; no broadcast
- Dual stack, tunnels, NAT64; > 50% at Google (2026)

Multicast, MPLS, VRRP

- IGMP/MLD + PIM-SM/SSM; RPF; 01-00-5E mapping
- MPLS: labels, LSP, LDP/SR; VPNs and TE
- VRRPv3: virtual IP and MAC; transparent failover

References: RFCs and sources by topic (1 of 2)

Topic	Main references
IPv4, TCP, UDP, ICMP	RFC 791 (IPv4), RFC 9293 (TCP, 2022), RFC 768 (UDP), RFC 792 (ICMP), RFC 1122 (host requirements), RFC 2474 and 3168 (DSCP/ECN), RFC 1191 (PMTUD)
IPv4 addressing	RFC 950 (subnets), RFC 4632 (CIDR), RFC 1918 (private), RFC 6598 (CGNAT), RFC 5737 (documentation), RFC 6890 (special), RFC 3021 (/31), RFC 1878
ARP, DHCP, DNS	RFC 826 (ARP), RFC 1027 (Proxy ARP), RFC 2131 (DHCP), RFC 1542 and 3046 (relay, option 82), RFC 2644 (directed broadcast), RFC 1034/1035 (DNS)
Ports and numbers	RFC 6335 (port ranges) and the IANA port and protocol number registries (successors of RFC 1060 / 1700)
RIP, EIGRP, AD	RFC 1058 (RIPv1), RFC 2453 (RIPv2), RFC 2080 (RIPng), RFC 4822 (authentication), RFC 7868 (EIGRP), RFC 2991/2992 (ECMP)
OSPF, IS-IS	RFC 2328 (OSPFv2), RFC 5340 (OSPFv3), RFC 5838 (AF), RFC 5709 and 7166 (authentication), RFC 5880 (BFD), ISO 10589 and RFC 1195/5308 (IS-IS)
BGP	RFC 4271 (BGP-4), RFC 4760 (MP-BGP), RFC 4456 (route reflectors), RFC 6793 (32-bit ASN), RFC 6996 (private ASN), RFC 1997/8092 (communities), RFC 6811 (ROV), RFC 7938 (BGP in data centers)

References: RFCs and sources by topic (2 of 2)

Topic	Main references
IPv6	RFC 8200 (IPv6), RFC 4291 (addressing), RFC 5952 (notation), RFC 4193 (ULA), RFC 4861 (ND), RFC 4862 (SLAAC), RFC 7217 and 8981 (interface IDs), RFC 9915 (DHCPv6), RFC 8106 (RDNSS), RFC 4443 (ICMPv6), RFC 6437 (flow label), RFC 8504 (node requirements)
IPv6: security and transition	RFC 4301 to 4303 (IPsec), RFC 6105 (RA Guard), RFC 4890 (ICMPv6 filtering), RFC 4213 (dual stack), RFC 6146/6147 (NAT64/DNS64), RFC 6877 (464XLAT), RFC 8305 (Happy Eyeballs), RFC 8925
Multicast	RFC 1112, RFC 5771 (addresses), RFC 2365 (admin scope), RFC 2236 and 9776 (IGMP), RFC 2710 and 9777 (MLD), RFC 4541 (snooping), RFC 3973 (PIM-DM), RFC 7761 (PIM-SM), RFC 4607 (SSM), RFC 5015 (Bidir), RFC 5059 (BSR), RFC 1075 and 1584 (DVMRP, MOSPF, historical)
RSVP and MPLS	RFC 2205 (RSVP), RFC 3209 (RSVP-TE), RFC 4090 (FRR), RFC 3031/3032 (MPLS), RFC 5462 (TC field), RFC 5036 (LDP), RFC 4364 (L3VPN), RFC 7432 (EVPN), RFC 8660 (SR-MPLS), RFC 8986 (SRv6)
VRRP and redundancy	RFC 9568 (VRRPv3, current), RFC 5798 (VRRPv3, predecessor), RFC 3768 (VRRPv2), RFC 2338 (original), RFC 2281 (HSRP, informational)
Layer 2 and history	IEEE 802.1D/802.1Q/802.1w (bridging, VLANs, RSTP), RFC 9542 (documentation MACs), RFC 904 (EGP), RFC 1930 (AS), RFC 1001/1002 (NetBIOS over TCP), RFC 2166 (DLSw)
2026 data	Internet Society Pulse: '18 years later, IPv6 reaches majority' (Google: 50.10% on March 28, 2026); APNIC Labs IPv6 Capability Metrics, Brazil ≈ 59% (October 5, 2026); bgp.potaroo.net (global table)

Thank you!

Rodolfo Nützmann | ronutz.com

TCP/IP: Concepts and IP Routing, 2026 edition (US English), based on the March 1999 course